

[HBM Roadmap ver 1.98 by KAIST Teralab] Overview of Next Generation HBM Architectures

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- Basics of HBM
- Overview of HBM Roadmap
- Next-Generation HBM Roadmap by KAIST TERALAB (ver 1.3)
 - ✓ HBM4 2026
 - ✓ HBM5 2029
 - ✓ HBM6 2032
 - ✓ HBM7 2035
 - ✓ HBM8 2038

Part1: Basics HBM

DALL·E 2: OpenAI's Image Generation AI System

Text Prompt

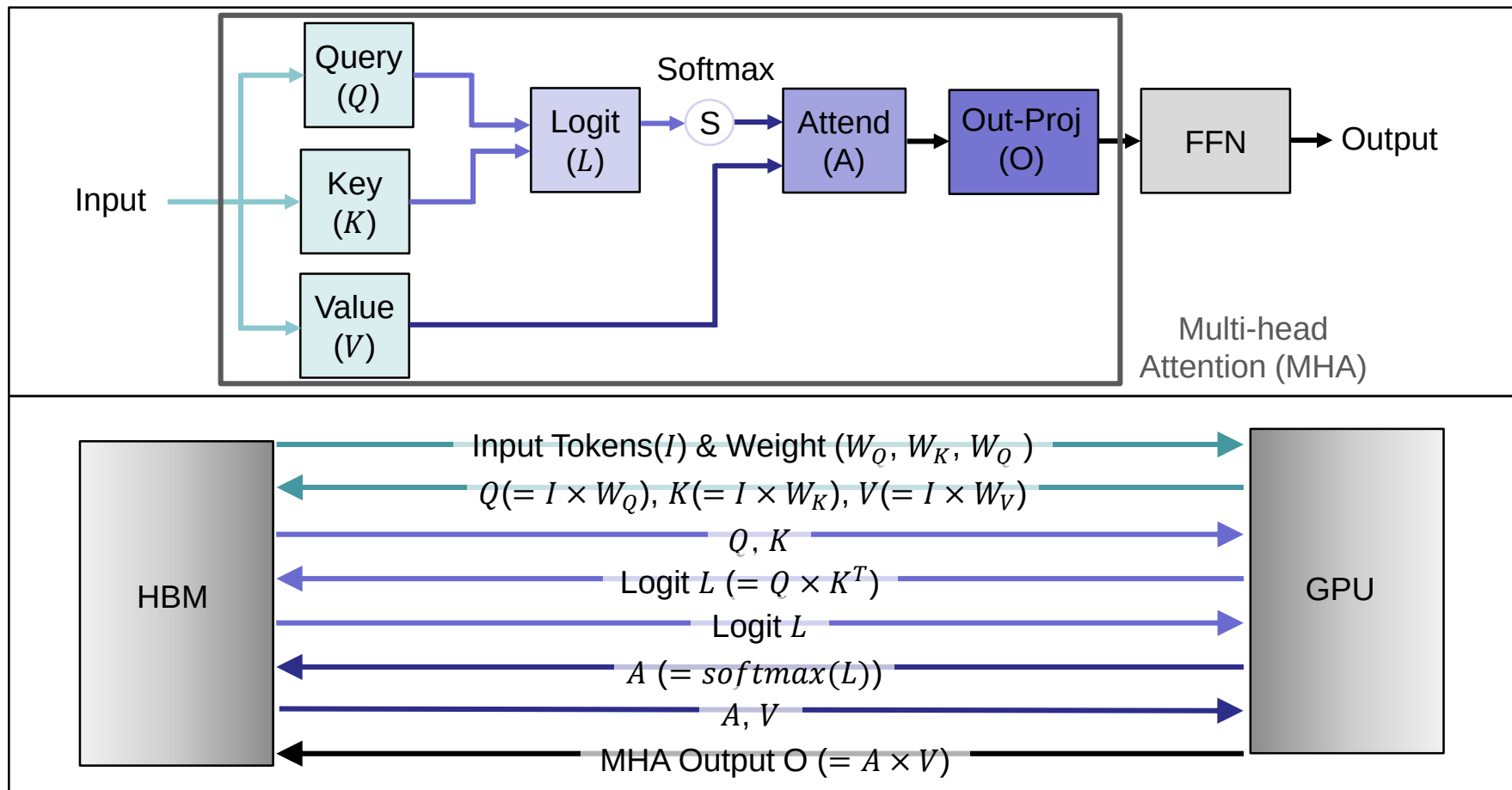
Vibrant portrait painting of Salvador Dalí with a robotic half face.



< Salvador Dalí image generated by DALL·E 2 >

- DALL·E 2 is an AI system that can create realistic images and art from a description in natural language.
- OpenAI released DALL·E 2 on April 6, 2022.
- The parameter of DALL·E 2 is 3.5 billion, which is 1/4 of the previous version.

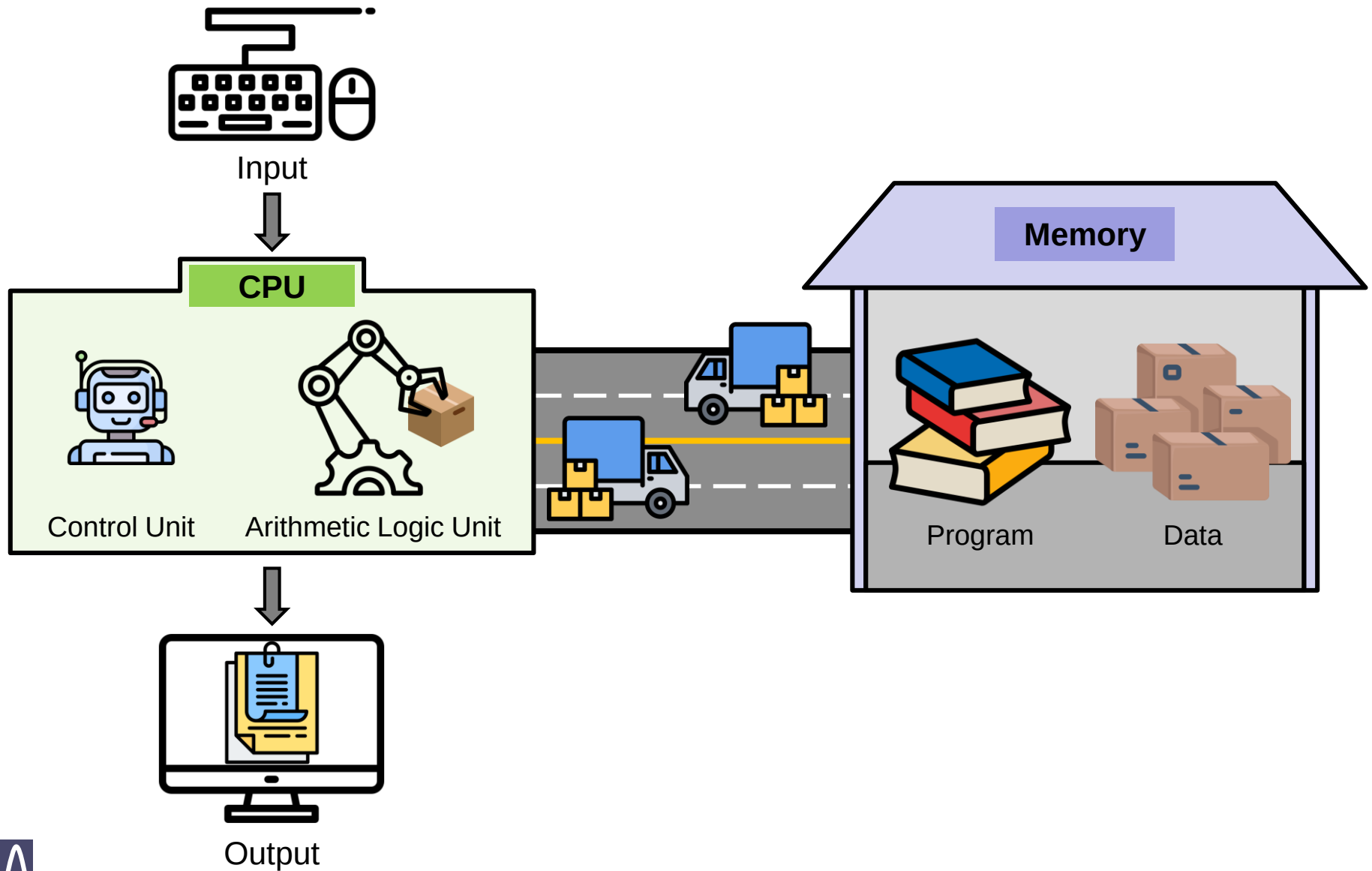
Massive Data Movement between GPU and HBM during Transformer-based AI Computing



< Structure of MHA and FFN in Transformer and Conceptual Data Movement between GPU and HBM >

- Transformer model consists of an encoder and decoder, which are composed of two main components: multi-head attention (MHA) and feed forward network (FFN).
- During these processes, significant data movement occurs between GPU and HBM.

Von Neumann Architecture

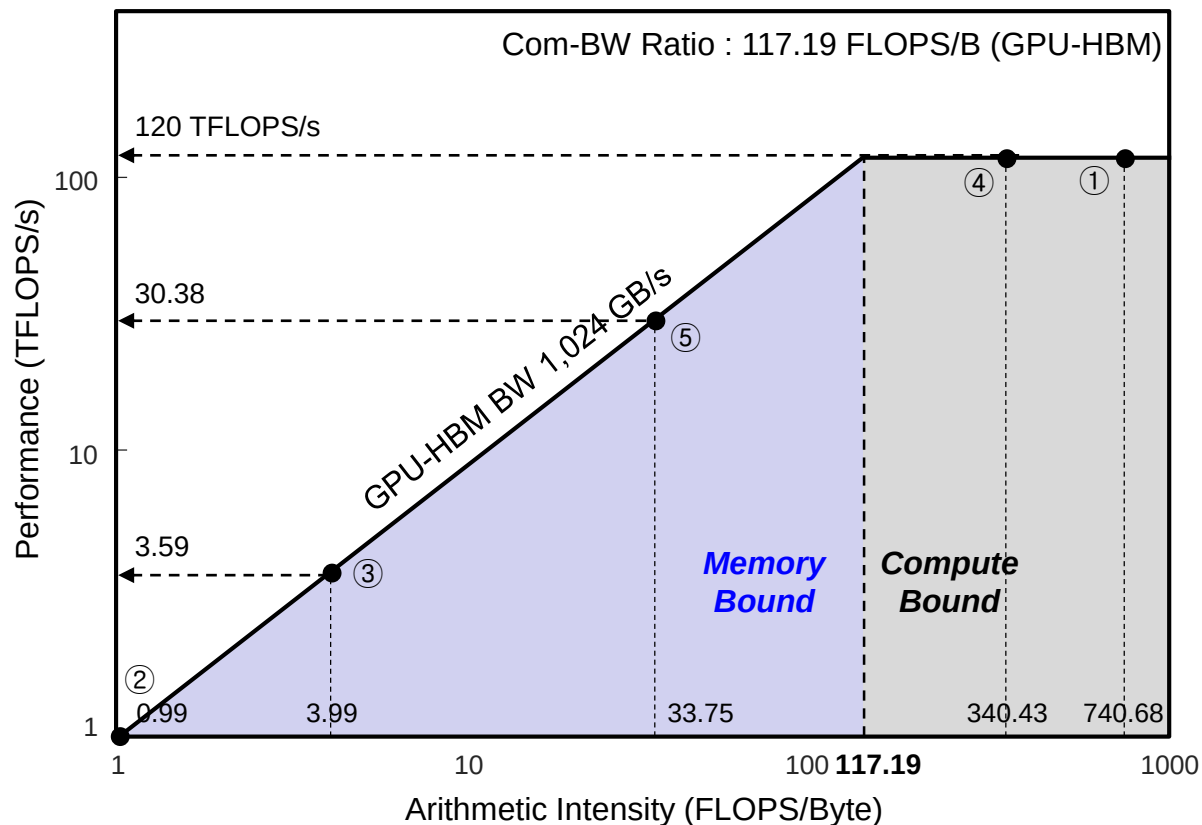


Performance Evaluation of GEMM Workloads with GPU-HBM Roofline Model

$$\text{Arithmetic Intensity} = \frac{M \cdot N \cdot K}{M \cdot K + N \cdot K + M \cdot N}$$

$$\text{Com-BW Ratio} = \frac{\text{Peak Performance}}{\text{Bandwidth}}$$

GEMM	Arithmetic Intensity [FLOPS/Byte]	Achieved Performance [TFLOPS/s]
①	740.68	120
②	0.99	0.89
③	3.99	3.59
④	340.43	120
⑤	33.75	30.38



< Roofline Model of Simulated GEMM Workloads >

- Assuming the GPU-HBM architecture meets the system's peak computation capabilities, the roofline model of GPU-HBM and GEMM workloads was plotted based on V100.
 - ✓ GEMM ①, ④ : Arithmetic Intensity > 117.19 FLOPS/B → **Compute-bound**
 - ✓ GEMM ②, ③, ⑤ : Arithmetic Intensity < 117.19 FLOPS/B → **Memory-bound**

영상 생성시 필요한 Token size 계산 과정



< Video generation by continuous frame >

- 해상도 : 원본 비디오 프레임 가로/세로 픽셀 수 : $H \times W$
- 패치 크기 (P) : 한 토큰으로 묶는 공간적 블록 크기
- 프레임 수 (F) : 영상 길이 (t) x 초당 프레임 (fps)
- codebook 수 (C) : 하나의 패치마다 생성하는 토큰 수를 결정하는 인코더의 깊이

$$\text{Total number of token} = \left(\frac{H}{P} \times \frac{W}{P}\right) \times (t \times fps) \times C$$

- 따라서 256x256 해상도, 16x16 패치, 30fps, 10초, 단일 VAE기반 ($C=1$)의 영상을 만든다면 $(256/16 \times 256/16) \times (10 \times 30) = \mathbf{76,800}$ 개의 토큰이 필요함.
- 이 때 해상도 4배 증가, 저/고해상도 두 단계 토큰화 ($C=2$), 1분 영상을 만들 시 총 **460,800개**의 토큰 필요함 → 현재 GPT 계열은 128k 토큰까지 지원하므로 앞으로 수십 ~ 수백만 토큰을 소화할 LLM이 필수가 됨.

Diffusion Transformer (DiT)로 동영상 생성 시 필요한 메모리

Key & Value matrix size **w/o VAE** : $\left(\frac{W}{P_w}\right) \times \left(\frac{H}{P_h}\right) \times \left(\frac{f \times t}{P_t}\right) \times (\# \text{ of layer}) \times \text{hidden_dim} \times 2 \text{ Bytes}$

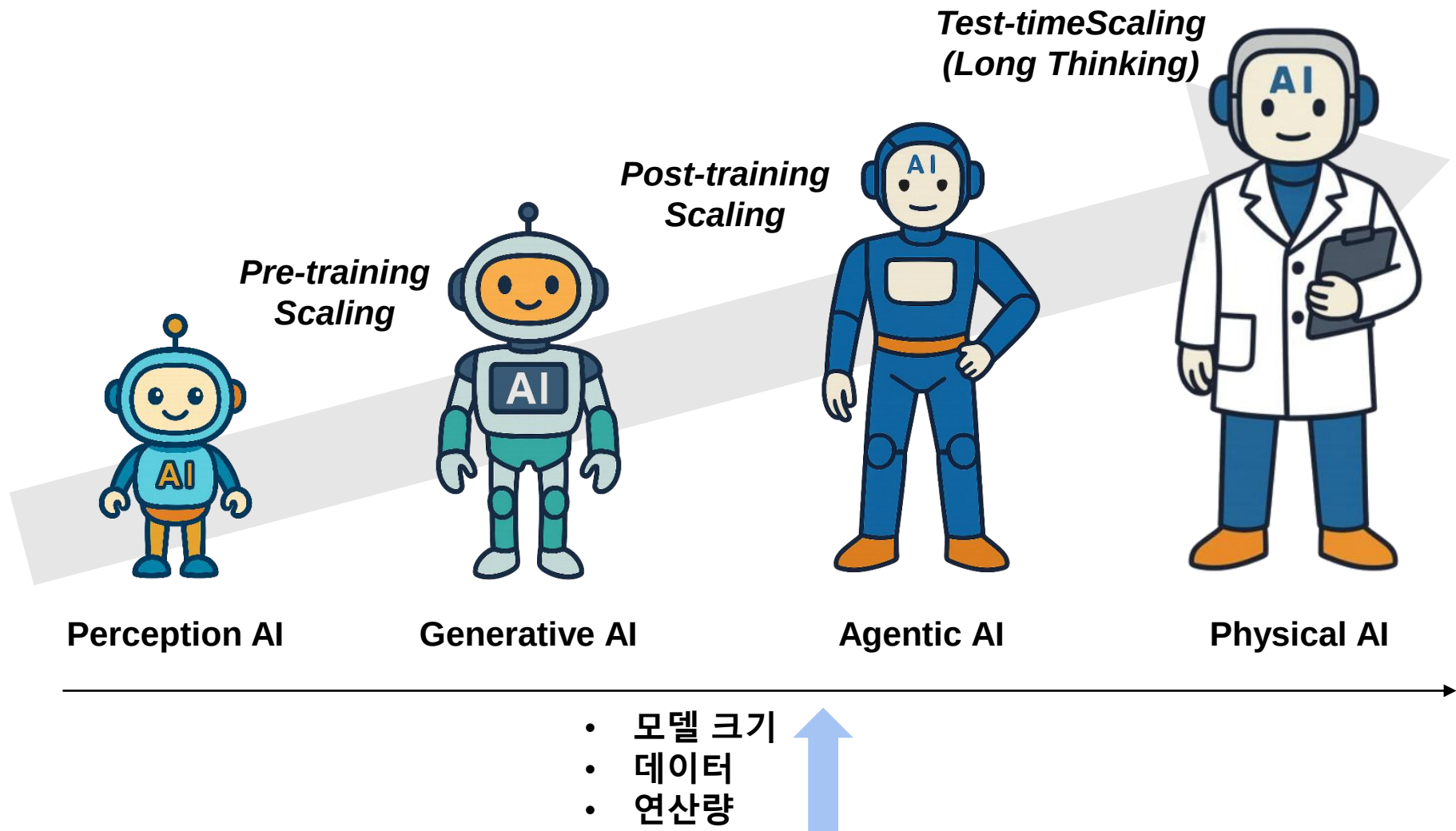
Key & Value matrix size **w/ VAE** : $\left(\frac{W}{C_s P_w}\right) \times \left(\frac{H}{C_s P_h}\right) \times \left(\frac{f \times t}{C_t P_t}\right) \times (\# \text{ of layer}) \times \text{hidden_dim} \times 2 \text{ Bytes}$

W, H, f, t : 동영상 가로 및 세로 해상도, 프레임, 시간
 P_w, P_h, P_t : Patch 크기
 C_s, C_t : VAE 압축률

< DiT 모델로 동영상 생성시 필요한 메모리 계산 식 >

- DiT 모델은 patch 단위로 이미지와 동영상을 이해함.
→ Transformer는 근본적으로 언어처리 모델이기 때문에 patch를 통해 동영상을 분할하고 시퀀스화 시켜서 Transformer의 input으로 사용함.
- Variable Auto Encoder (VAE)를 사용해 입력 시퀀스를 효과적으로 압축할 수 있음.
- FP16 정밀도의 4,096 hidden dimension을 갖는 layer 40개를 사용하는 DiT로 1,920x1,080 해상도의 30 frame 10초 길이의 영상을 만든다고 가정했을 때,
→ **w/o VAE** & 16x16x2 patch : **370.79 GB**
→ **w/ VAE** & 2x2x2 patch : **185.39 GB** (VAE 압축률 C_s : 8, C_t : 2)
- 매 de-noising step마다 몇 백 GB의 matrix를 메모리에 load하고 erase해야함.
→ HBM capacity와 bandwidth가 bottleneck임.

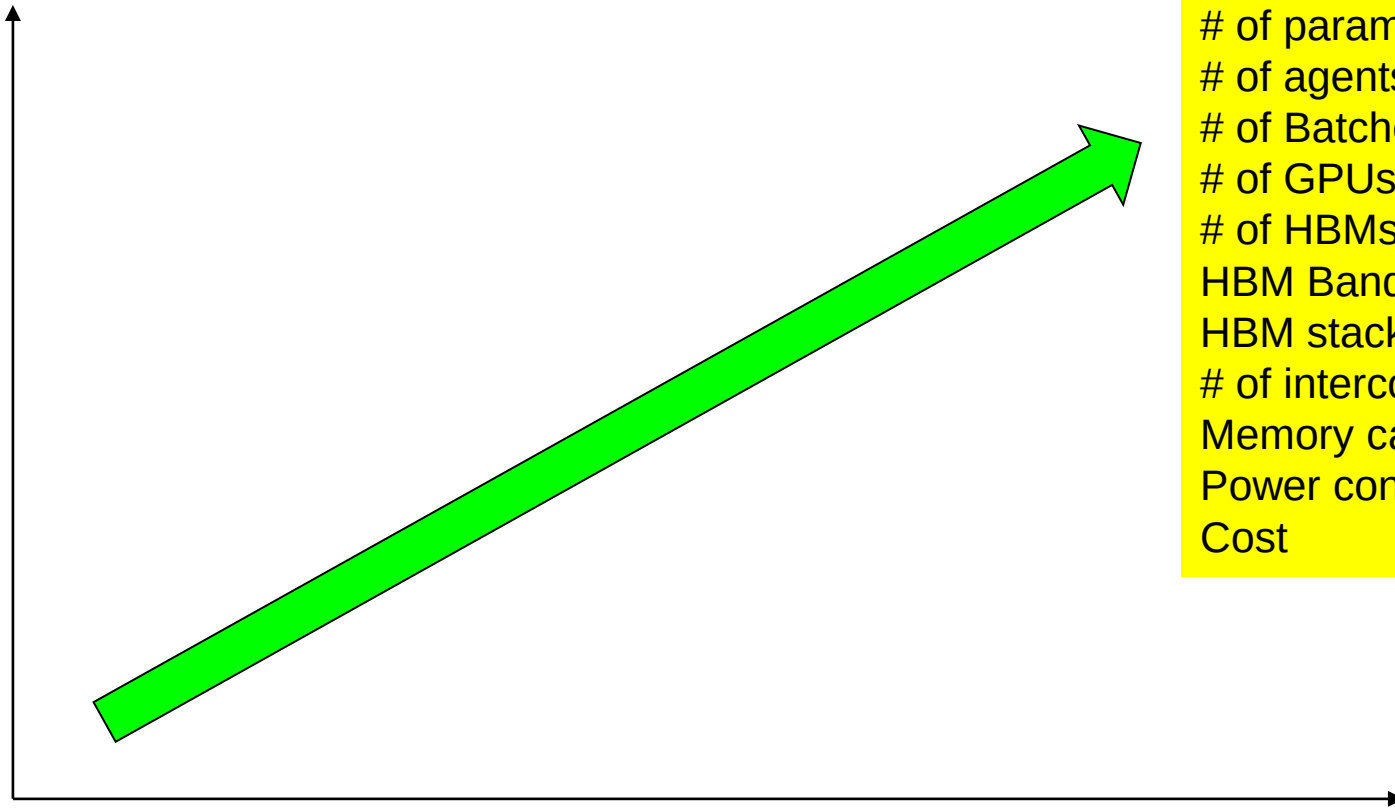
9. AI Scale Up 되는 이유



- AI가 더 똑똑해지려면 모델 크기, 데이터, 연산량에서 계속 Scale Up 필요

Scaling 법칙은 계속된다

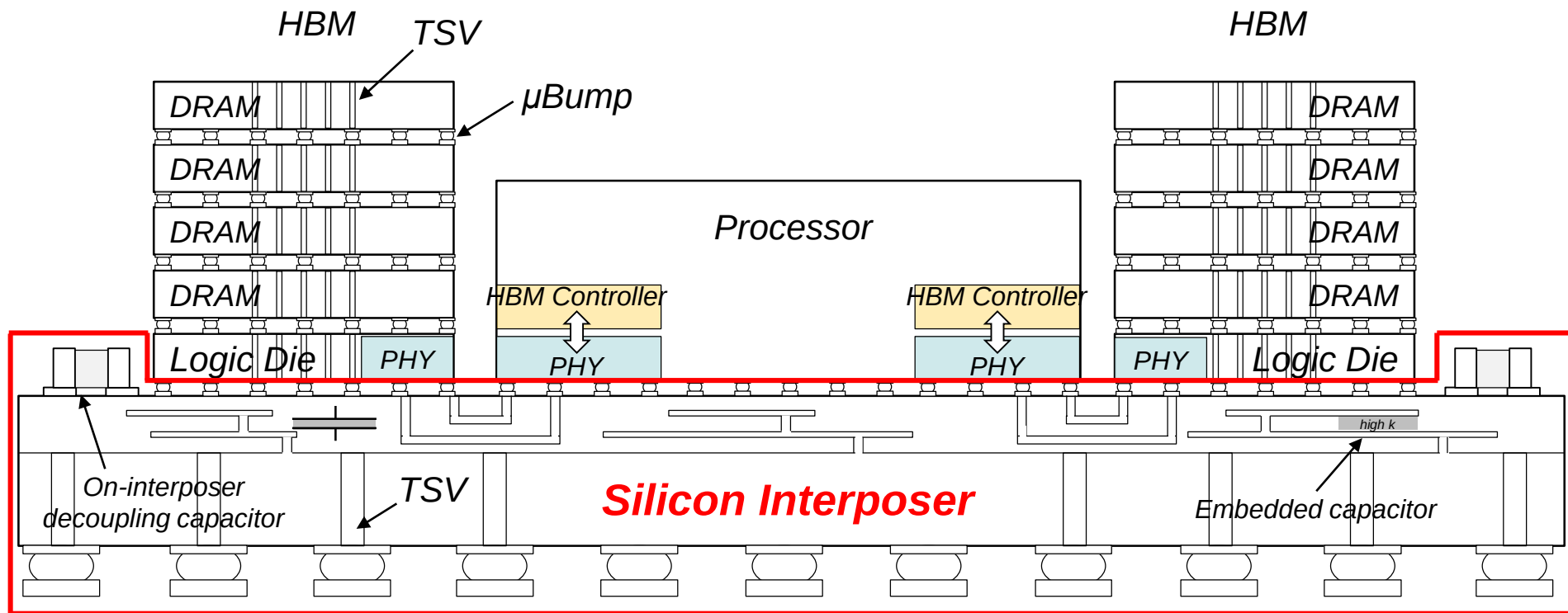
Size



Model Size
of parameters
of agents
of Batches
of GPUs
of HBMs
HBM Bandwidth
HBM stacks
of interconnections
Memory capacity
Power consumptions
Cost

Year

HBM Interposer Design: Cross-sectional View

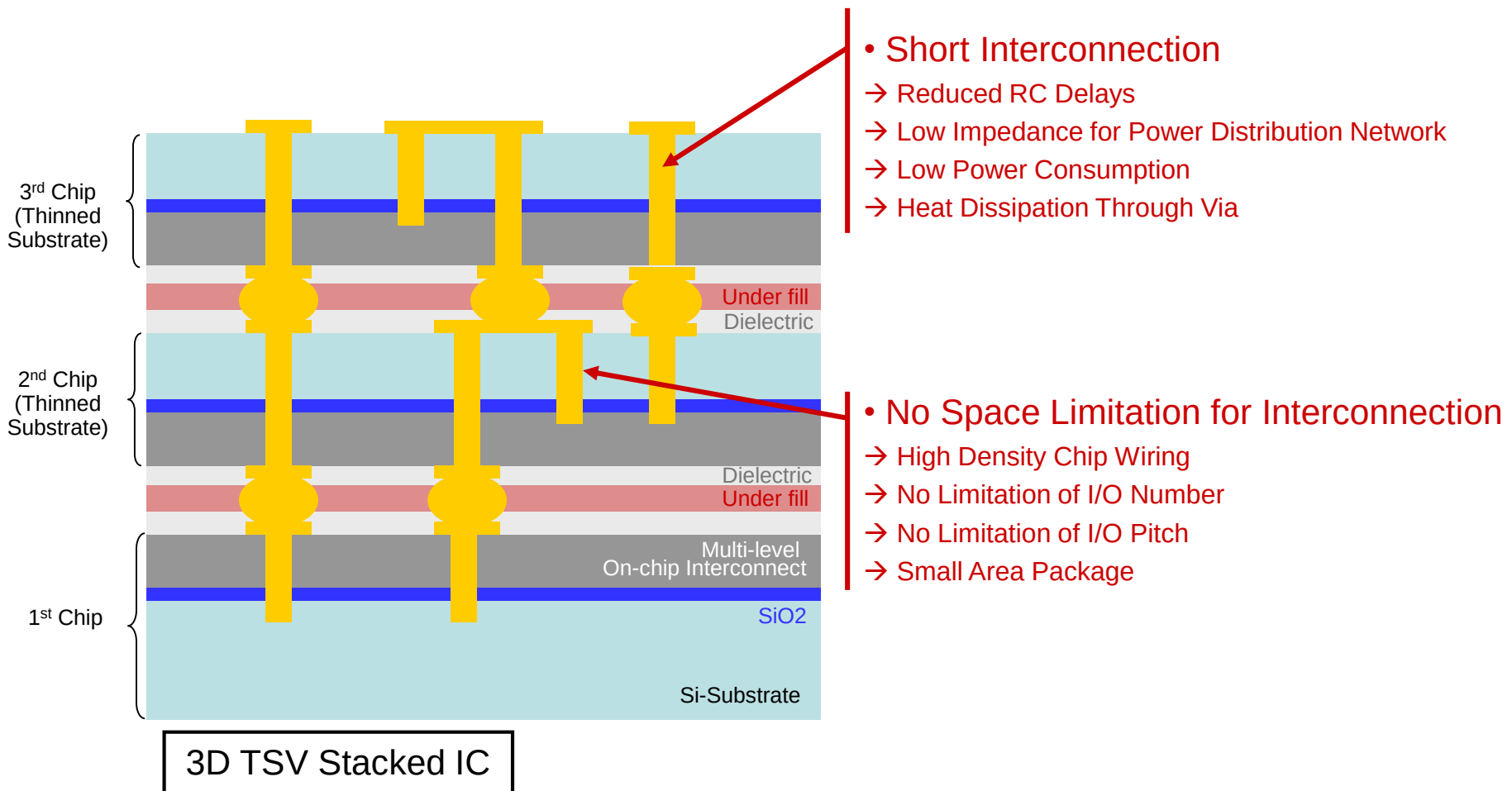


< Cross-sectional View of Silicon Interposer based HBM Module >

주상복합 건물



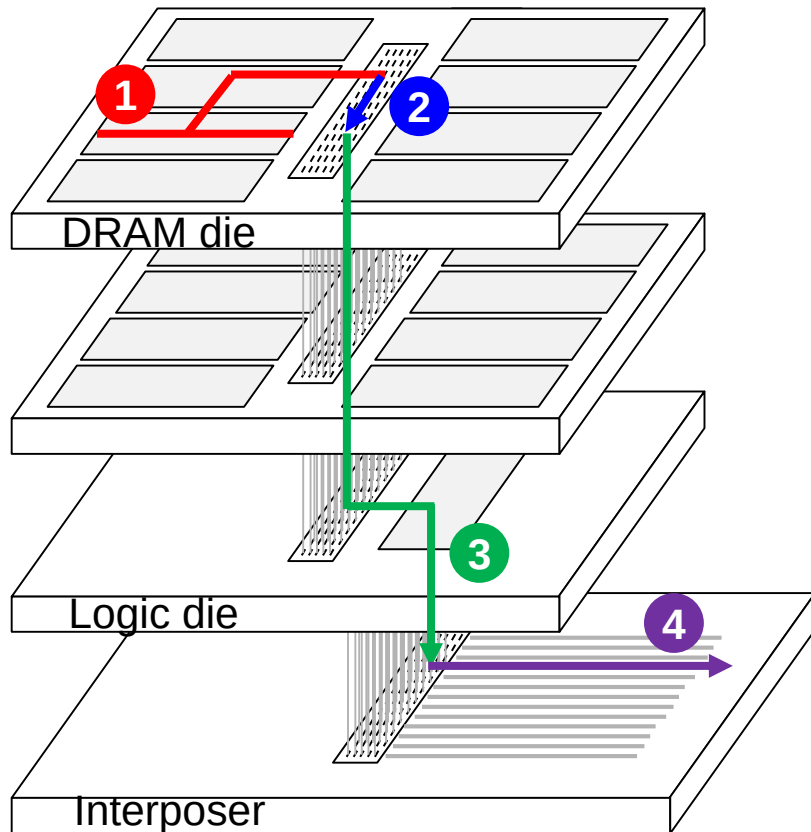
Key Technology : TSV (Through Silicon Via)



TSV Elevator



DRAM Bandwidth Teardown in HBM3



< Data path in HBM3 >

Data path	Configuration	Bandwidth
Bank group (on-chip)	16384 I/O x 0.4375 Gbps (4nCK)	896 GB/s
Global (on-chip)	8192 I/O x 0.875 Gbps (2nCK)	896 GB/s
Die-to-die (TSV)	4096 I/O x 1.75 Gbps (1nCK)	896 GB/s
Logic die (on-chip)	4096 I/O x 1.75 Gbps (1nCK)	896 GB/s
Interface (interposer)	1024 I/O x 7 Gbps (0.25 nCK)	896 GB/s

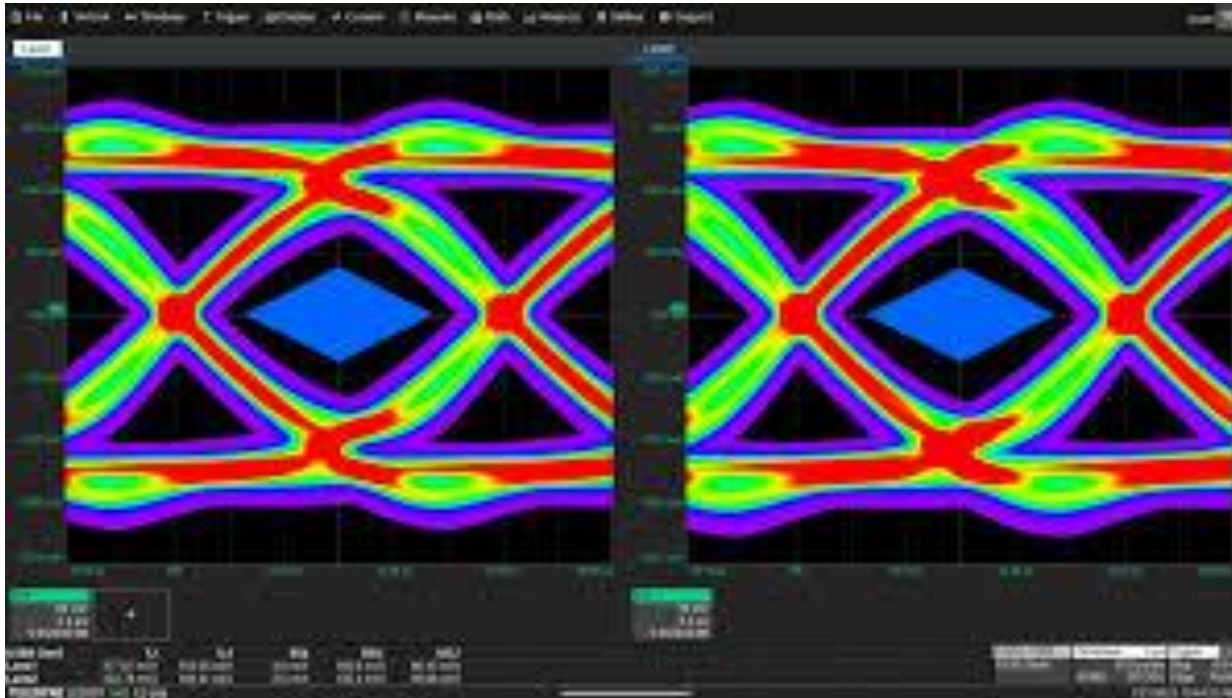
< DRAM bandwidth of each data path in HBM3 >

- From bank group I/O to interface I/O, the number of I/O and data rate are hierarchical.
- I/O bus window is extended by time-division multiplexing.

인천 공항



시간의 불확실성: ultimate bandwidth limit



Deterministic Jitter
Random jitter
ISI jitter
PDN jitter
PSIJ

Interconnection Future(10): Design Innovations: 5 Is

Design and Analysis for High-performance System

Co-Simul. Co-Design

System on Chip
(On-Chip Level)

System in Package
(On-Package Level)

PCB, Cable, Module
(System Level)



Signal **I**ntegrity
(SI)

Power
Integrity
(PI)

Thermal
Integrity
(TI)

5I
for High-
Performance
System

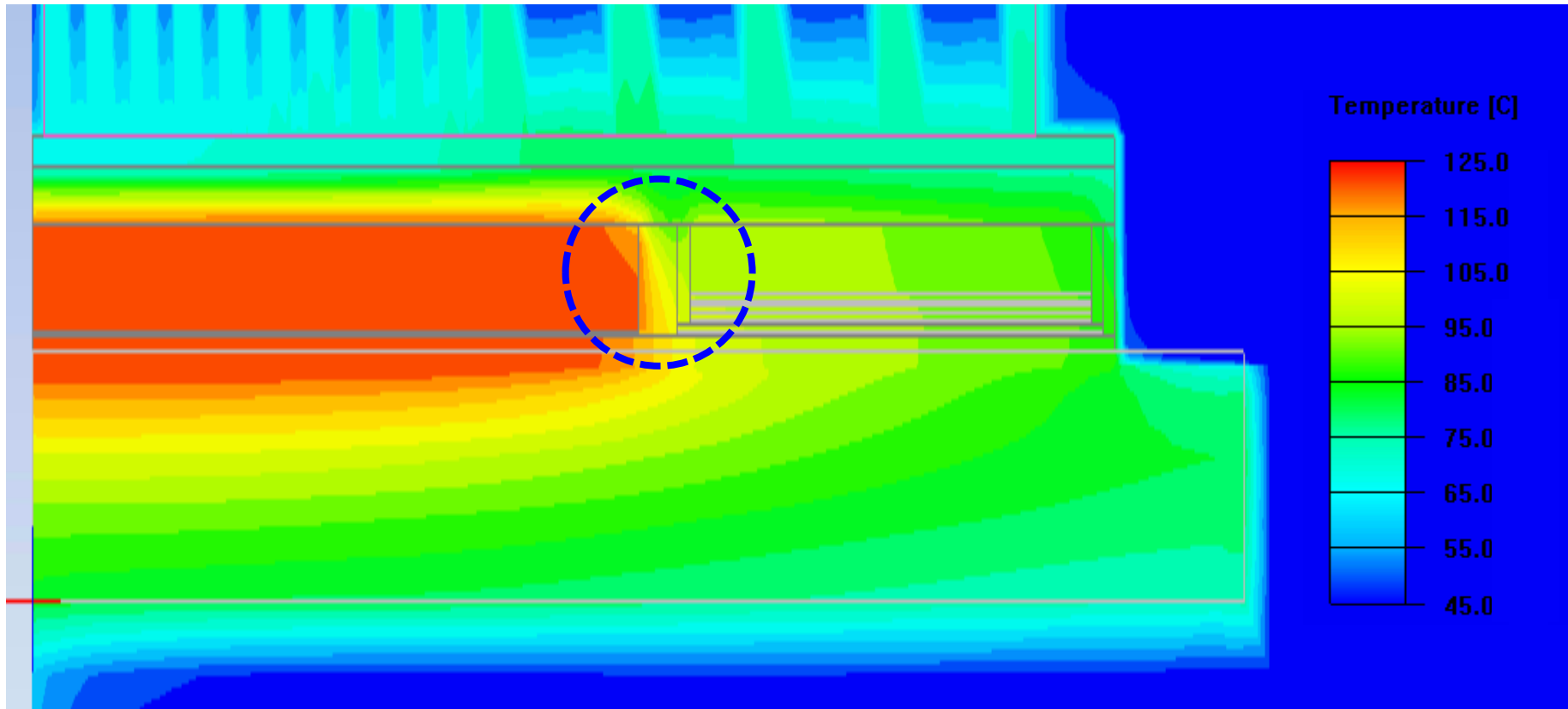
Electromagnetic
Interference
(EMI)

Artificial
Intelligence
(AI)



Improvement of System Integrity, Performance, Reliability, Cost, ...

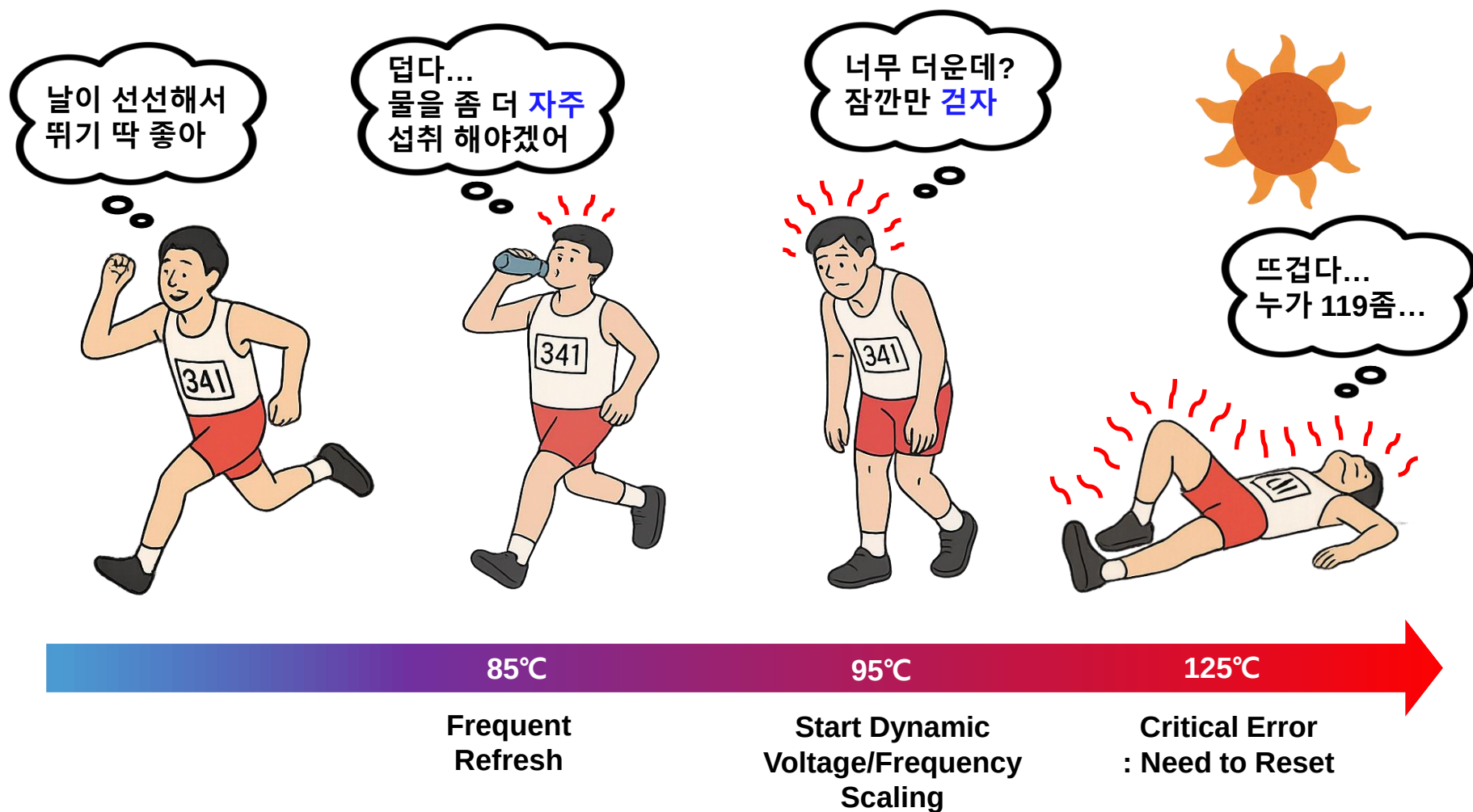
Thermal Coupling on Operating HBM System



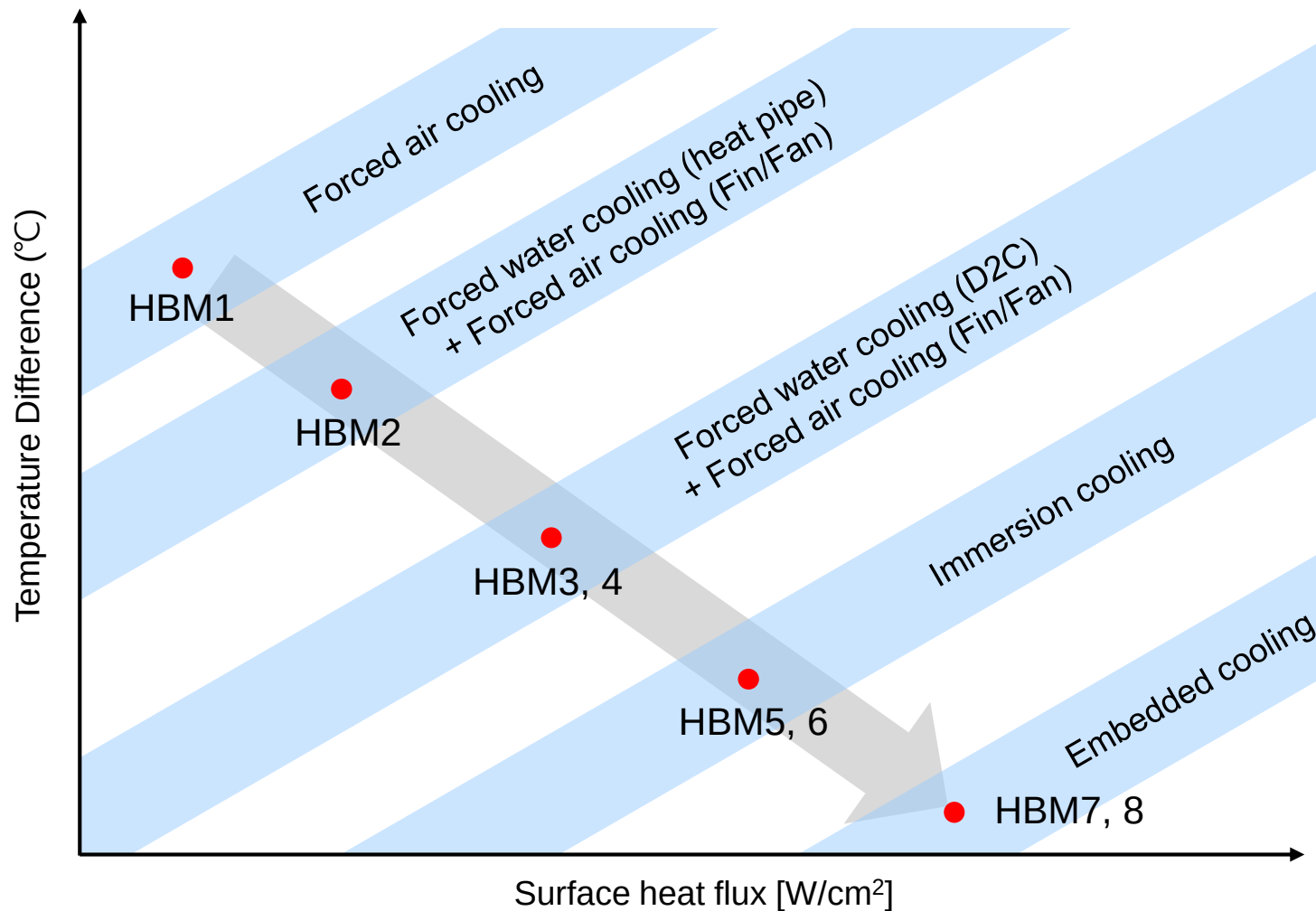
<Thermal distributions of operating HBM system>

- GPU's heat thermal coupling to HBM through molding compound, heat sink, interposer and etc.
- GPU's heat is dominantly affected to HBM's thermal gradient.
- Only molding compound part can be customized for reducing thermal coupling effects.

12. HBM 온도가 올라가면 생기는 문제 – Refresh, DVFS, Data Flip



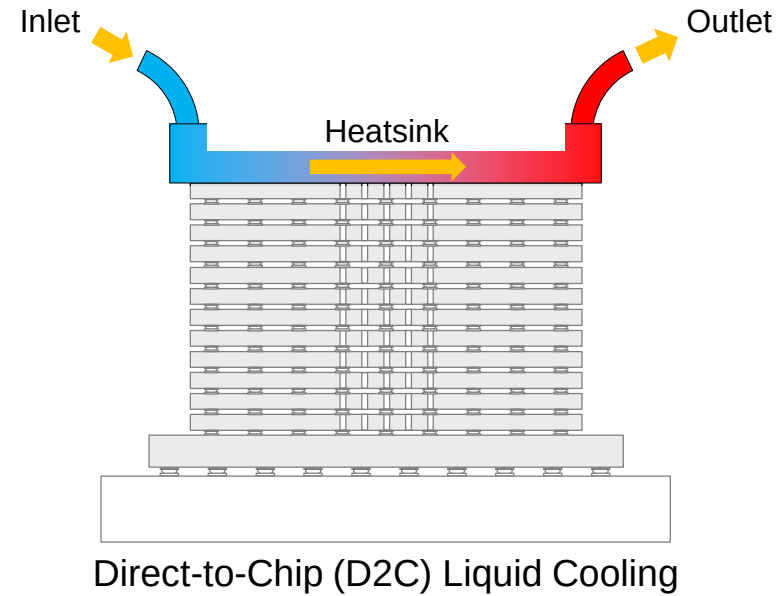
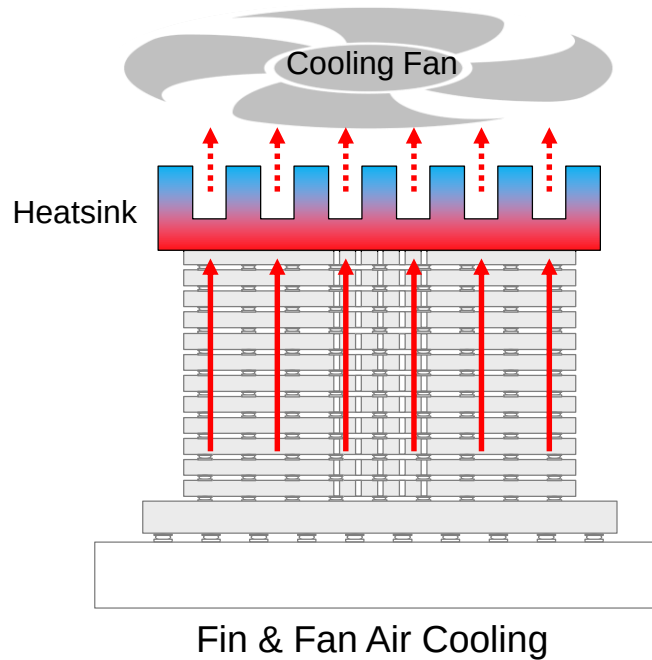
< HBM 과열 시 동작을 마라토너에 비유 >



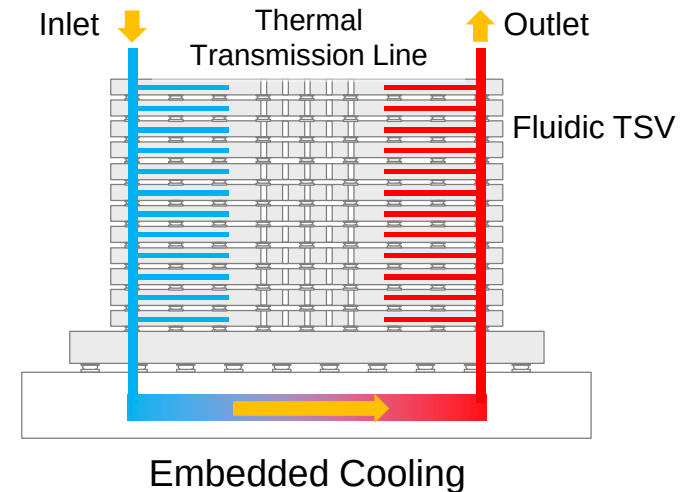
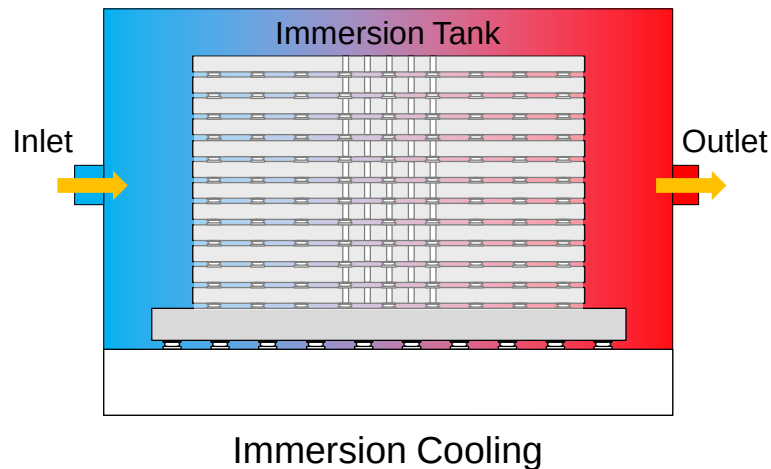
< Thermal Management & Cooling Methods for Next-Generation HBM >

Next-Generation HBM Roadmap : Thermal Management & Cooling Method

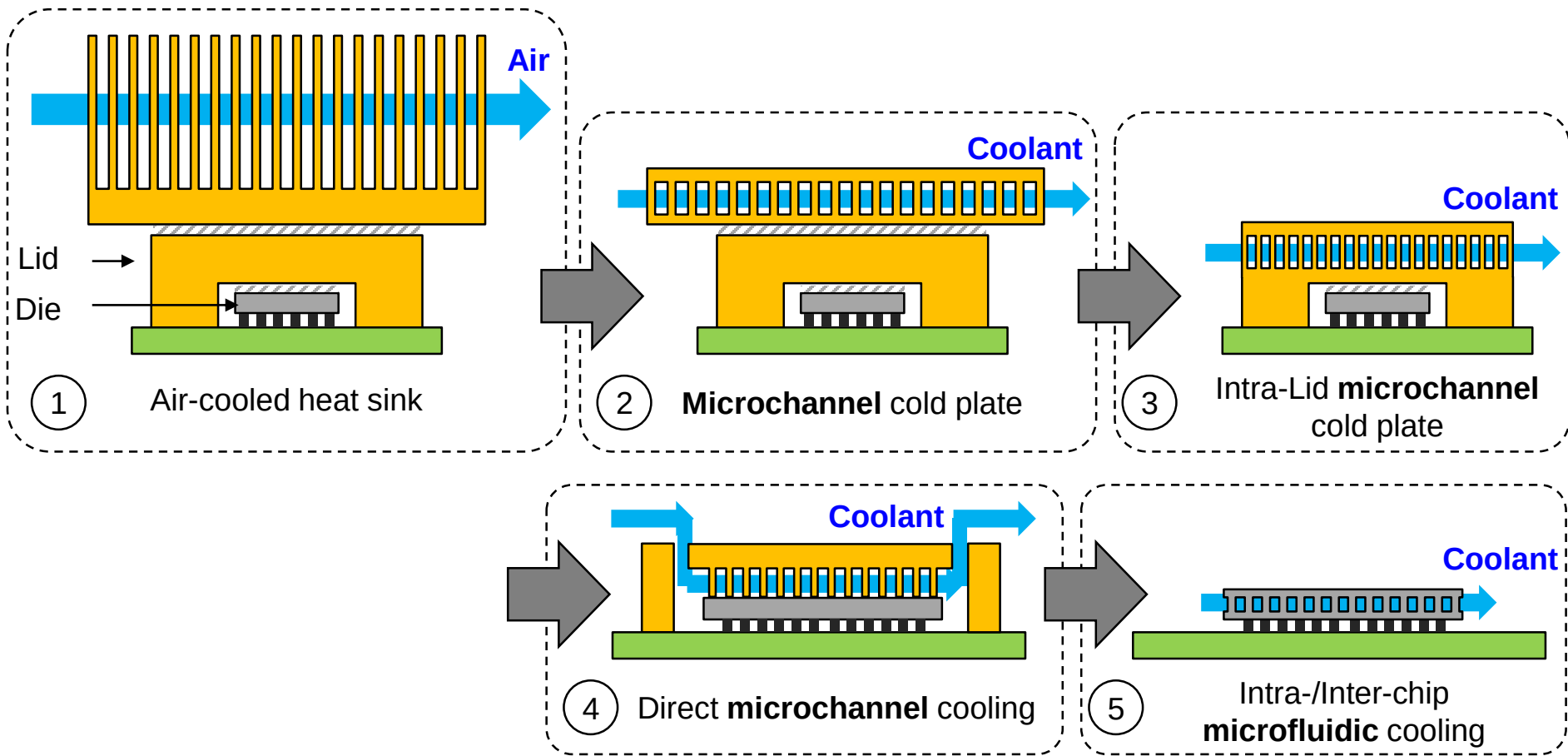
**(Present)
Conventional
HBM Cooling
Method**



**Next-Generation
HBM Cooling
Method**



Microchannel Cooling 기술 발전 동향

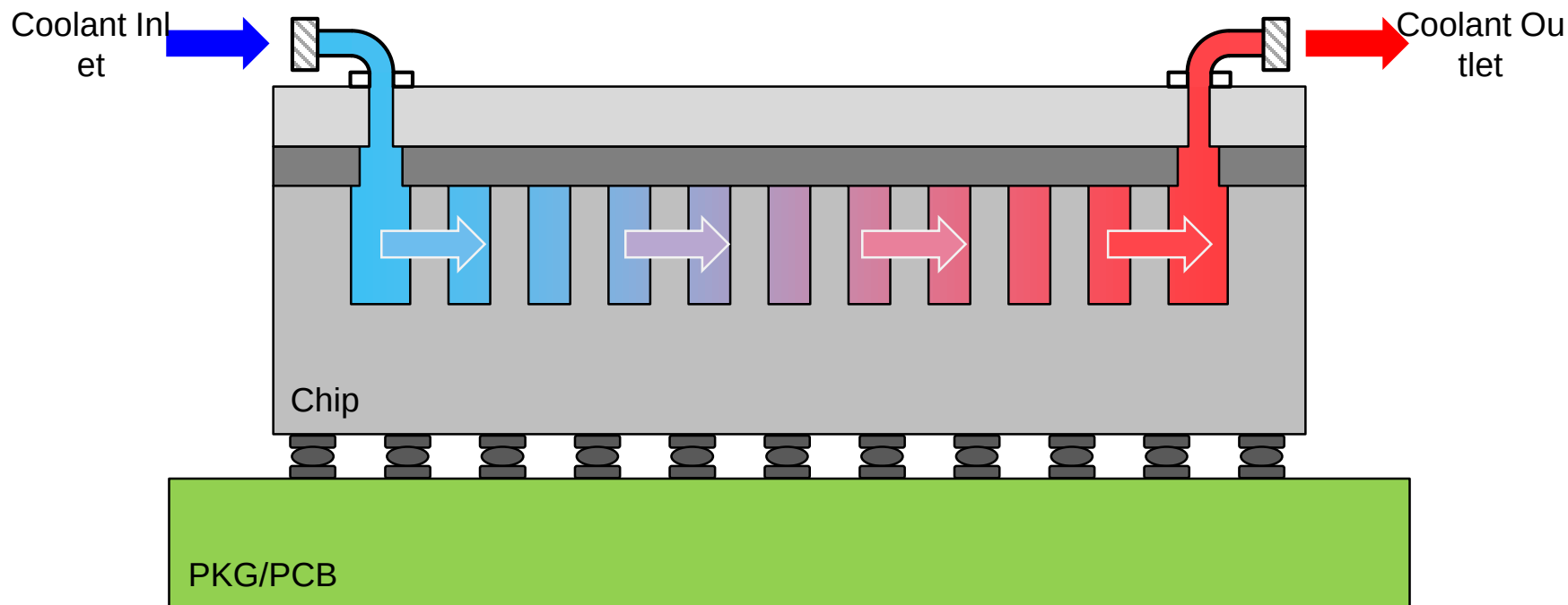


< Microchannel Cooling 기술 발전 동향 >

- Microchannel cooling은 칩 외부에 부착되던 기존의 공랭식 냉각 장치와 비슷한 구조부터 기술이 개발됨.
- 기술이 발전함에 따라 점차 칩에 물리적으로 가까워 지는 방향으로 microchannel cooling 구조가 변형되고 성능이 개선됨.
- 최종적으로는 칩 내부에 microchannel 을 형성하여, 칩에서 발생하는 열을 direct 로 냉각시킬 수 있는 Microfluidic cooling 구조로 발전됨.

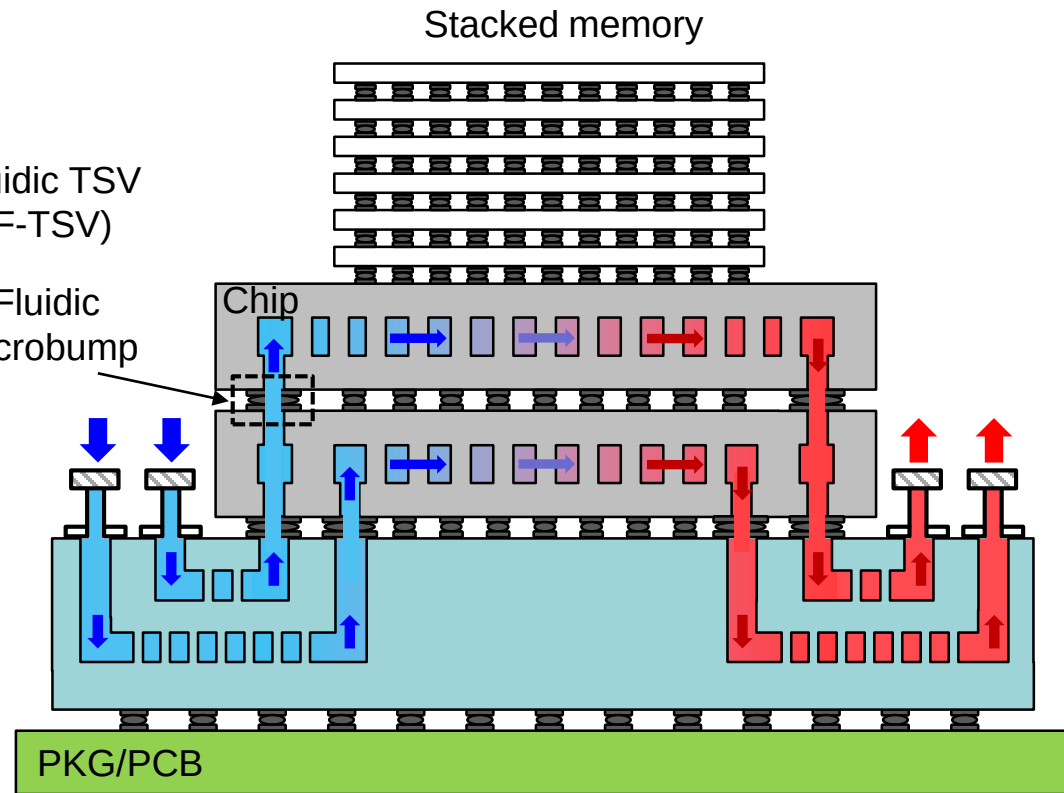
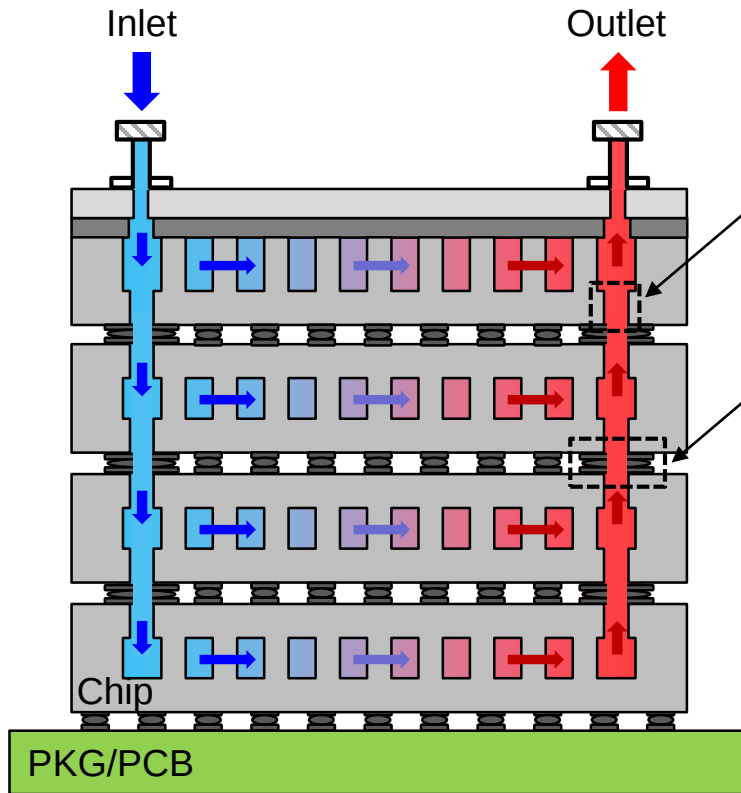
Inter-Chip Cooling Method: Microfluidic Cooling

– Advancement of Microchannel Cooling



- Microfluidic cooling 방법은 microchannel cooling 방법의 발전형이며, Chip 내부에 microchannel 을 가공하여 direct cooling 하는 방법임.
- Chip 의 윗부분에서 coolant 를 inlet 하여, chip 내부에 직접 coolant 를 유동시킨 뒤, Chip의 열을 흡수한 coolant 를 다시 Chip 의 윗부분에서 outlet 하는 방식임.
- Chip 내부에서 direct cooling 하는 방법이기때문에, 다른 냉각 방법 대비 보다 높은 냉각 성능을 가짐.

Microfluidic Cooling for 3D ICs



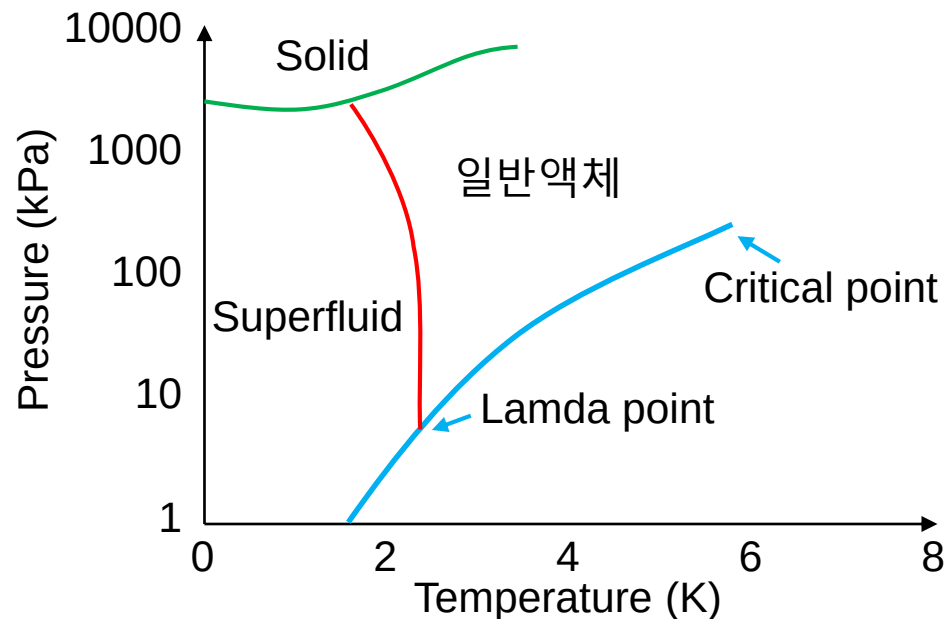
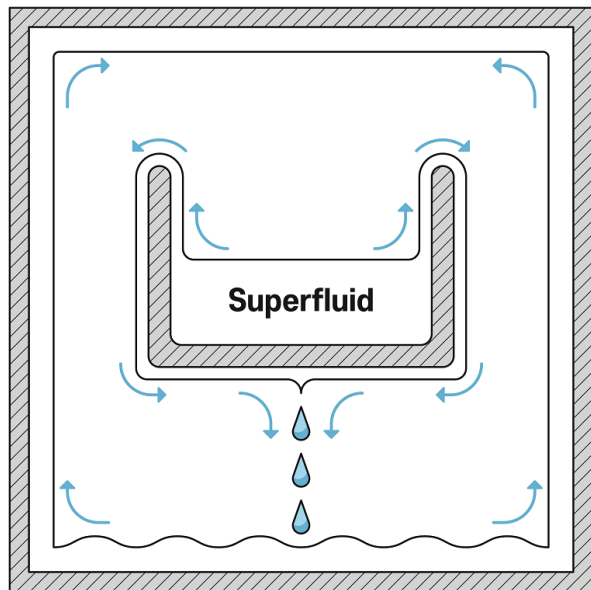
< Inter-chip microfluidic cooling based 3D ICs >

< Inter-chip/interposer microfluidic cooling based 3D ICs >

- 3D ICs 향 microfluidic cooling 구조는 inter-chip, 그리고 inter-interposer 구조를 가짐.
- Stacked 구조에서 microfluidic cooling 을 위해 fluidic TSV 와 fluidic microbump 가 요구됨.
- Microfluidic cooling 구조의 기구적 안정성 및 공정을 고려해 칩의 두께가 수백 μm 인 경우에 대해서만 제조 가능.

(HBM 의 경우 die thickness 가 50 μm 미만이기때문에 microchannel 공정 어려움)

초유동체의 정의와 임계 조건



< 초유동 상태 조건 및 온도와 압력에 따른 상태 >

- 물리적 정의
 - 완전 유동성: 점성 $\eta = 0$
 - 표면장력 소실: 표면장력 $\sigma = 0$, 벽을 타고 무한정 상승
- 3대 임계조건
 1. 임계 온도 (Critical Temperature, T_λ): 초유동 전이가 일어나는 온도
 2. 임계 속도 (Landau Critical Velocity, v_c): 초유동성을 파괴하는 최소 상대속도
 3. 임계 압력 (Critical Pressure, P_c): 초유동 상태를 유지할 수 있는 압력 범위

⇒ 초유동 유지 조건: $T < T_\lambda$, $v < v_c$, $P < P_c$

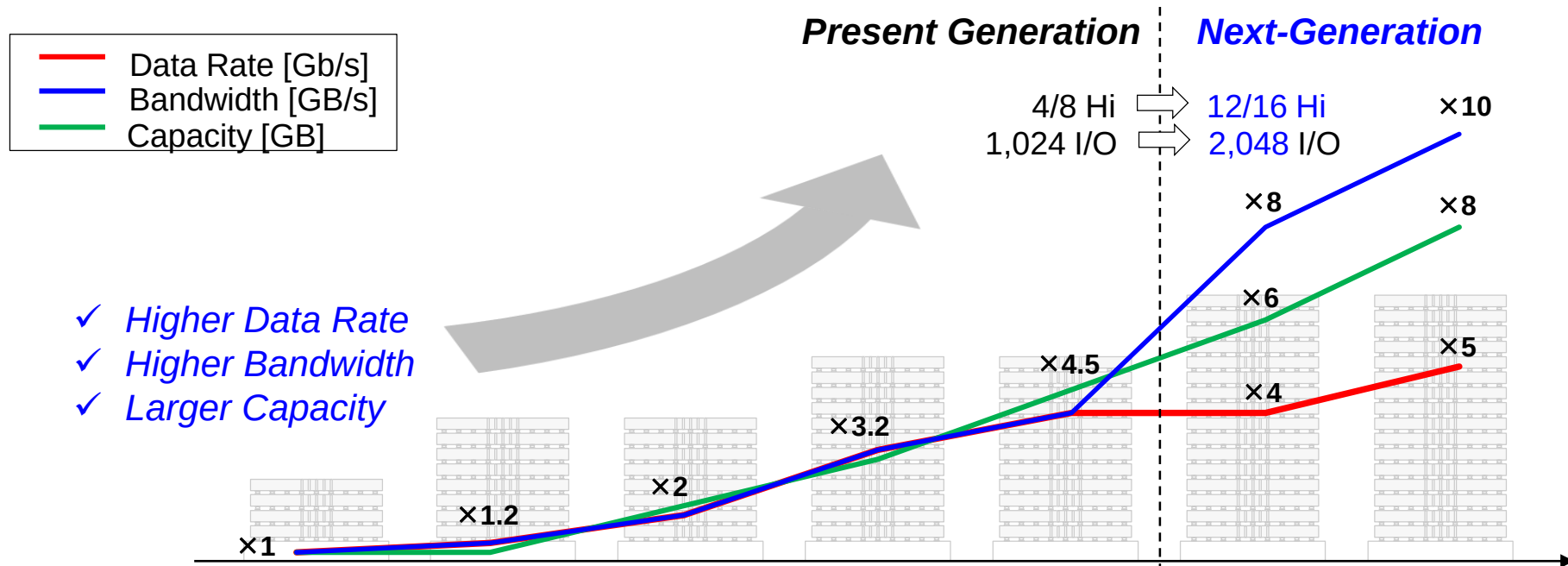
Part2: Overview of HBM Roadmap

- Higher Bandwidth
 - ✓ Larger Number of Interconnects
 - ✓ Higher Gbps/line
 - ✓ Larger Number of TSV
 - ✓ Hybrid bonding, Narrow pitch,
- Higher Memory Capacity
 - ✓ Integration with LPDDR-HBM, HBF
 - ✓ Hierarchical Memory Architecture
 - ✓ Integrated Memory network, CXL
- Computing in HBM
 - ✓ Near Memory Computing
 - ✓ Steaming Multi-processor
 - ✓ Data Compression
 - ✓ Error Corrections

- 3D Integration
 - ✓ Stacked Cash
 - ✓ Active/Embedded Interposer
 - ✓ Hybrid interposer (Si/Glass)
- Innovative cooling architecture
 - ✓ Liquid cooling
 - ✓ Immersion Cooling
 - ✓ Embedded cooling
- HBM Centric Computing
 - ✓ Full 3D Integration
 - ✓ CPU, GPU, HBM, HBF integration
 - ✓ Instruction set, programming

Technical Trend of AI-Specialized HBM in AI Semiconductor Industry

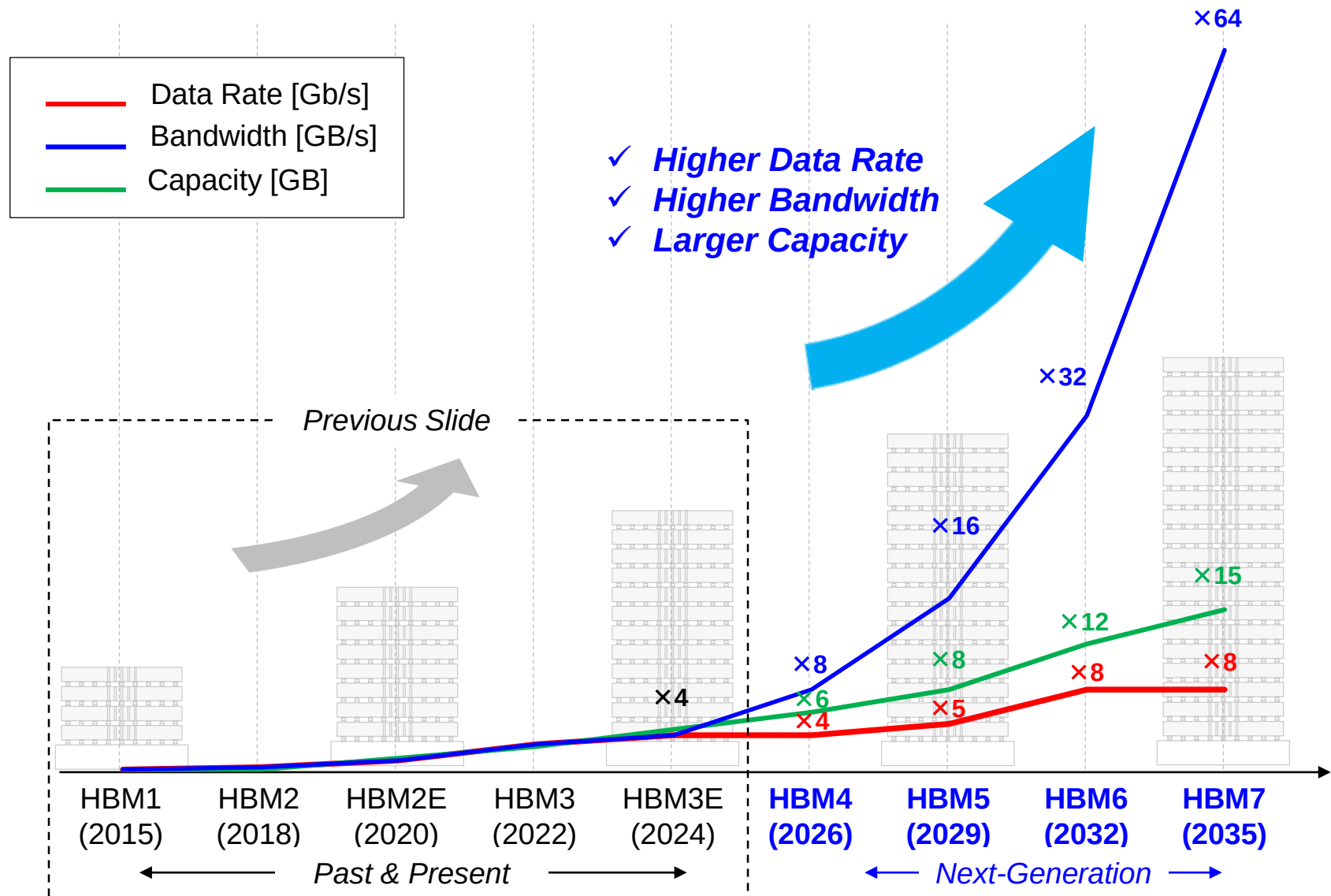
: Increasing Bandwidth, Data Rate, Capacity



	HBM (2015)	HBM2 (2018)	HBM2E (2020)	HBM3 (2022)	HBM3E (2024)	HBM4 (2026)	HBM4E (*2028)
Data Rate	2 Gbps	2.4 Gbps	3.6 Gbps	6.4 Gbps	8 Gbps	8 Gbps	10 Gbps
# of I/O	1,024					2,048	
Bandwidth	256 GB/s	307 GB/s	461 GB/s	819 GB/s	1.0 TB/s	2.0 TB/s	2.5 TB/s
Capacity/die	8 Gb		16 Gb		24 Gb		32 Gb
# of stack die	4/8-Hi			8/12-Hi		12/16-Hi	
Capacity/HBM	4/8 GB		8/16 GB	16/24 GB	24/36 GB	36/48 GB	48/64 GB
Power/HBM	4 W	10 W	19 W	25 W	32 W	43 / 75 W	48 / 80 W
Cooling Method	Thermo-Electric Cooling (TEC) w/ Heatsink			Direct-to-Chip (D2C) Liquid Cooling			

Technical Trend of AI-Specialized HBM in AI Semiconductor Industry

: Next Generation HBM Technology Trend by KAIST TERALAB



Next-Generation HBM Roadmap by KAIST TERALAB

 HBM Spec.  Packaging/Cooling
 Architecture  AI Design Agent

Ver 1.2 / updated.250521

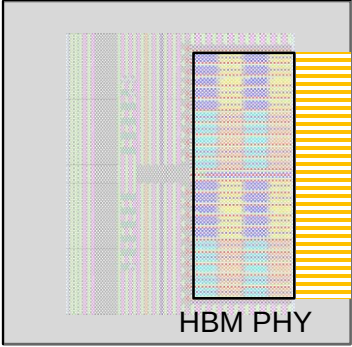
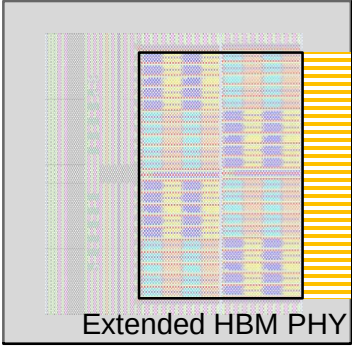
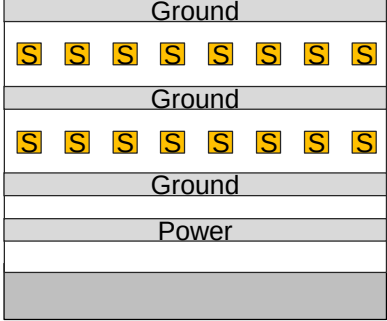
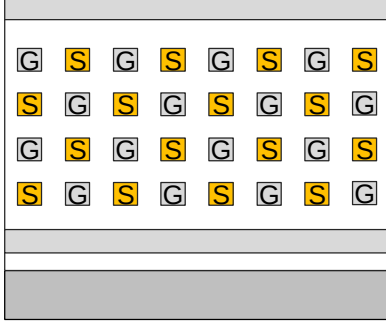
	HBM4 (2026)	HBM5 (2029)	HBM6 (2032)	HBM7 (2035)	HBM8 (2038)
Data Rate	8 Gbps	8 Gbps	16 Gbps	24 Gbps	32 Gbps
# of I/O	2,048	4,096	4,096	8,192	16,384
Bandwidth	2.0 TB/s	4 TB/s	8 TB/s	24 TB/s	64 TB/s
Capacity/die	24 Gb	40 Gb	48 Gb	64 Gb	80 Gb
# of die stack	12/16-Hi	16-Hi	16/20-Hi	20/24-Hi	20/24-Hi
Capacity /HBM	36/48 GB	80 GB	96/120 GB	160/192 GB	200/240 GB
Power/HBM	75 W	100 W	120 W	160 W	180 W
Die stacking	Microbump (MR-MUF)		Bump-less Cu-Cu Direct bonding		
Cooling Method	Direct-to-Chip (D2C) Liquid Cooling	Immersion Cooling		Embedded Cooling	
HBM Architecture	Custom HBM Base Die HBM-LPDDR	3D NMC-HBM & stacked cache / decap	Multi-tower HBM Active / Hybrid Interposer	Hybrid HBM Architecture HBM-HBF HBM-3D LPDDR	Full-3D / HBM Centric Computing Architecture
Additional Features (Patent)	NMC processor + LPDDR Ctrl	+ Cache + CXL + on-die/stacked decap + HBM shielding	+ Network switch + Bridge die + Asymmetric TSV	+ HBF/LPDDR Ctrl + Storage network	+ HBM Centric Interposer + Double side Cooling + Edge-expand Stack
AI Design Agent	ubump & TSV-array Decap placement Optimization	I/O Interface Optimization considering PSIJ	Hybrid Equalizer + Generative AI based SI/PI Metric Estimation	LLM based Human Interactive AI Design Agent	

Next-Generation GPU-HBM Roadmap

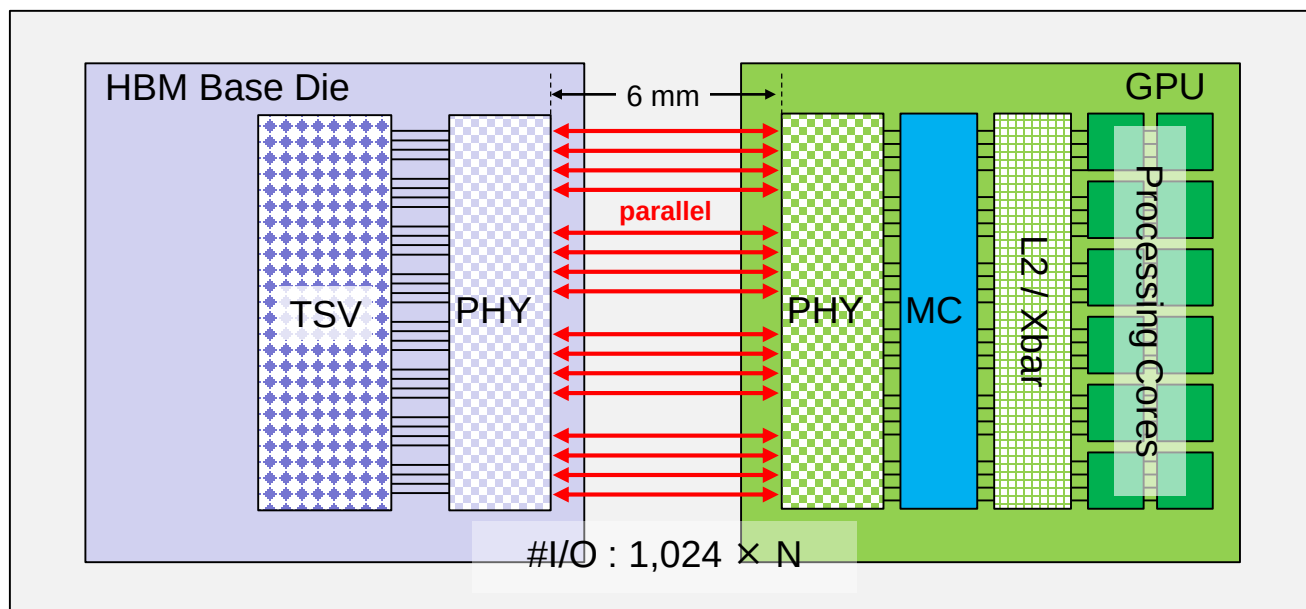
: More GPU & HBM Integrated Above Interposer

GPU Architecture	Rubin (2026)	Feynman (2029)	Post Feynman (2032)	Next-Gen Architecture (2035)
GPU Die Size	728 mm ²	750 mm ²	700 mm ²	600 mm ²
GPU Power	800 W	900 W	1,000 W	1,200 W
GPU-HBM Module	R200	F400	Post Feynman GPU-HBM Module	Next-Gen GPU-HBM Module
Interposer Size				
# of GPU Dies	×2	×4	×4	×8
# of HBM Stack	HBM4×8	HBM5×8	HBM6×16	HBM7×32
Interposer Die Size	2,194 mm ² (46.2 mm x 48.5 mm)	4,788 mm ² (85.2 mm x 56.2 mm)	6,014 mm ² (102.8 mm x 58.5 mm)	9,245 mm ² (96.4 mm x 95.9 mm)
Total Bandwidth	16 / 32 TB/s	48 TB/s	128/256 TB/s	1,024 TB/s
Total HBM Capacity	288/384 GB	400/500 GB	1,536/1,920 GB	5,120/6,144 GB
Total Power	2,200 W	4,400 W	5,920 W	15,360 W

: Increased I/O Channels for Bandwidth Extension

	HBM3 (2022)	HBM4 (2024)	Next-Generation HBM
I/O #	1,024	2,048	4,096 ~ 8,192
HBM PHY Ball map			
microbump pitch	55 um	45 um	40 um
Interposer Channel Cross Section			
Metal w/s	2um ~ 3um	2um	< 2um
# RDL	Microstrip + Stripline (x2)	GSG Interleaved (x4)	GSG Interleaved (x6 ~ x8)

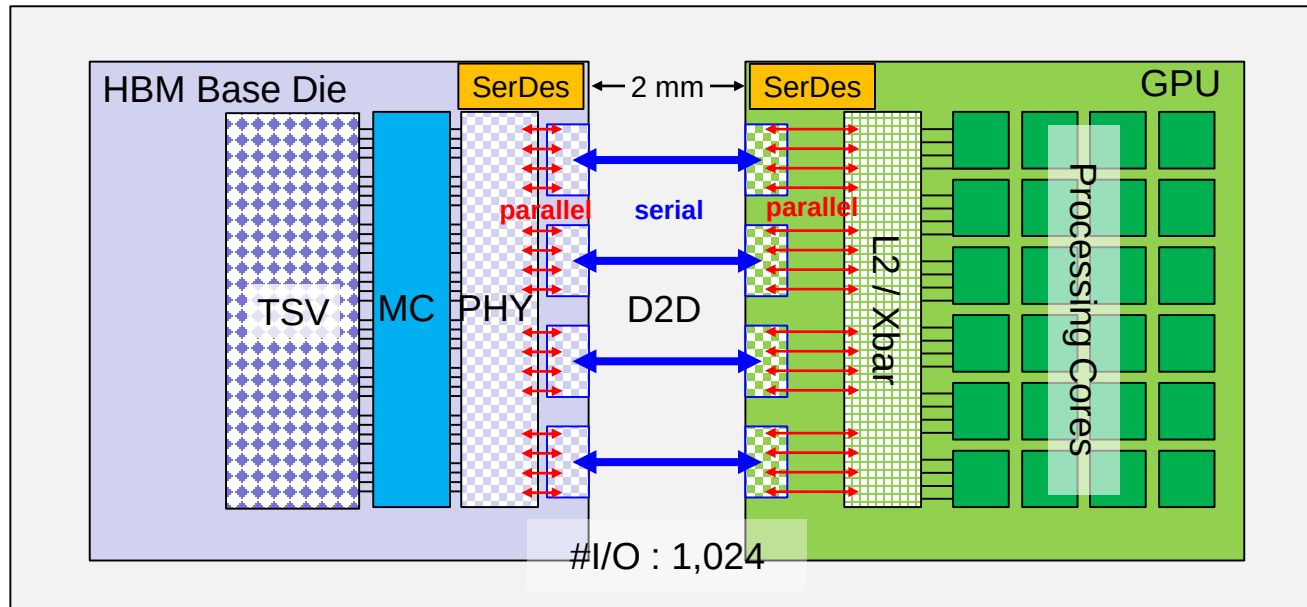
Type I. Direct Parallel GPU-HBM Interconnect



< GPU-HBM Type I : Direct Parallel Interconnect >

Advantages	Disadvantages
1. Increase Bandwidth 2. Increase # of I/Os 3. Minimized Latency	1. Narrow width/space/pitch $\rightarrow R \uparrow$ 2. Difficulty in pad/ball/bonding routing 3. Multi-layer metal RDL in interposer 4. # of Layer $\uparrow \rightarrow$ Interposer cost $\uparrow$ 5. Longer line length 6. Limited Data Speed (~ 64 Gbps)

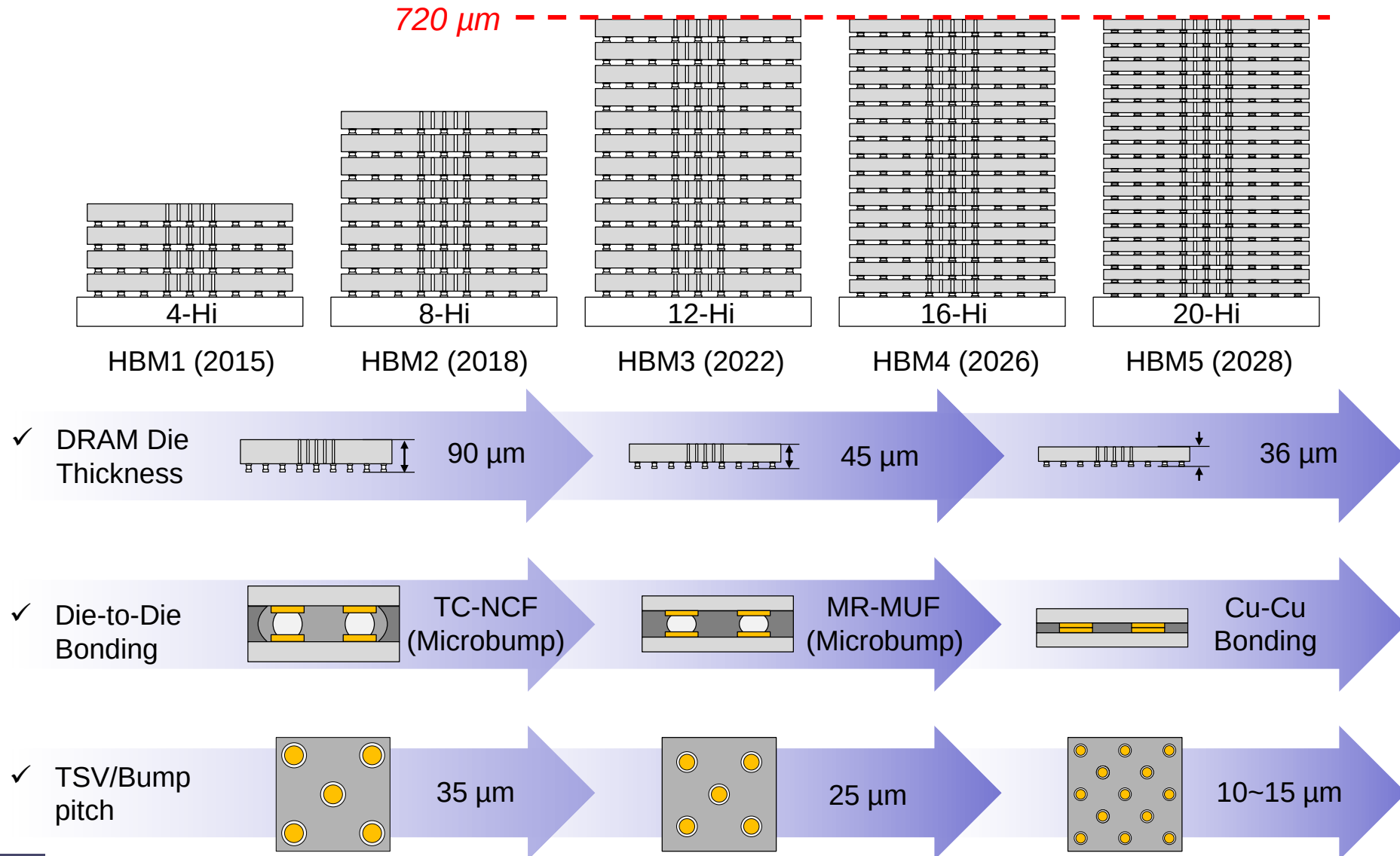
Type II. D2D Interconnect between GPU-HBM



< GPU-HBM Type II : D2D Interconnect >

Advantages	Disadvantages
1. Interconnect Congestion ↓ 2. Ease of pad/bump/bonding routing 3. Short length (< 2mm) 4. Memory controller at HBM base die (Logic IP)	1. Latency at Serdes (serial ↔ parallel) 2. High data rate (100 ~ 200 Gbps) 3. High I/O power consumption 4. Increased I/O area at die

Next-Generation HBM Roadmap : 3D Integration with Advanced Packaging Technology

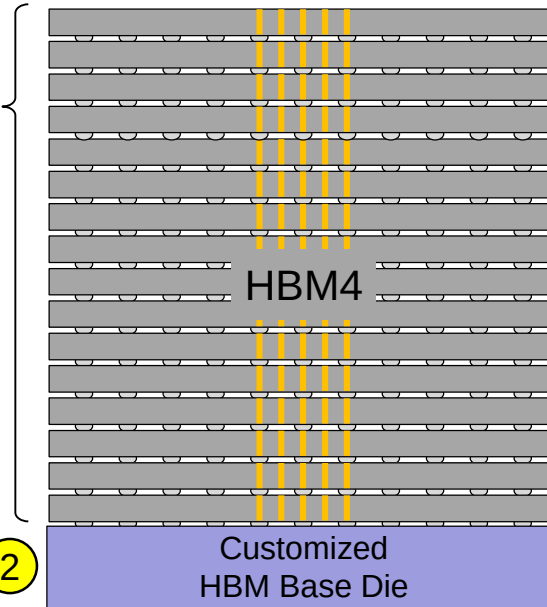


TSV(실리콘 관통전극)



① HBM 적층 수, 저장 용량 증가
8 ~ 12 단 → 12 ~ 16 단
16 ~ 24 GB → 36 ~ 48 GB

①



HBM4

GPU

② 고객 맞춤형 (Customized)
Base Die 설계
일부 GPU 계산 기능이 이동

②

Customized
HBM Base Die

실리콘 인터포저 (기판)

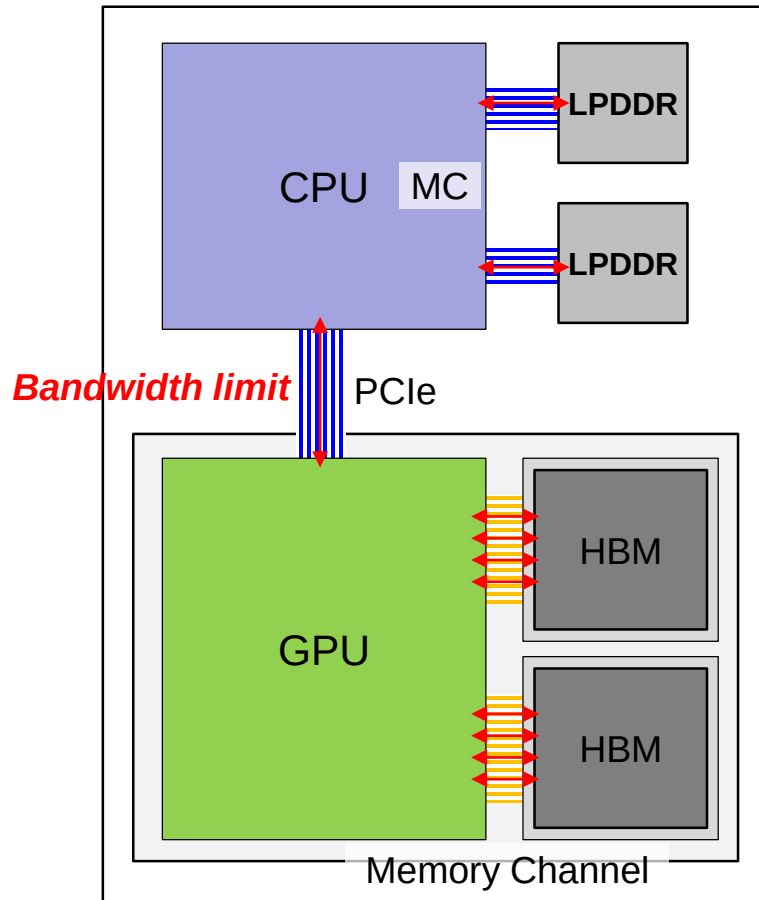


③

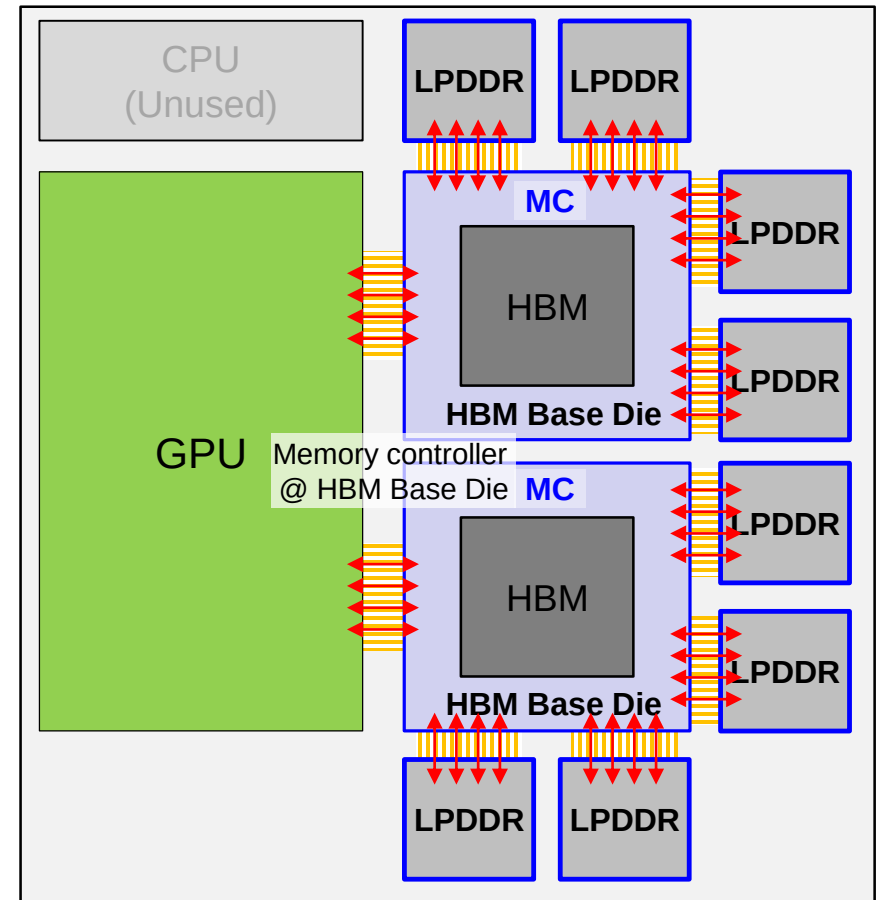
③ HBM3/3E 대비 2배 늘어난
HBM-GPU 연결선 (I/O)
2,048 개

Custom HBM Base Die Design : LPDDR Memory Channel for High Capacity & Memory Bandwidth

↔ : Low-Bandwidth ↔↔↔ : High-Bandwidth

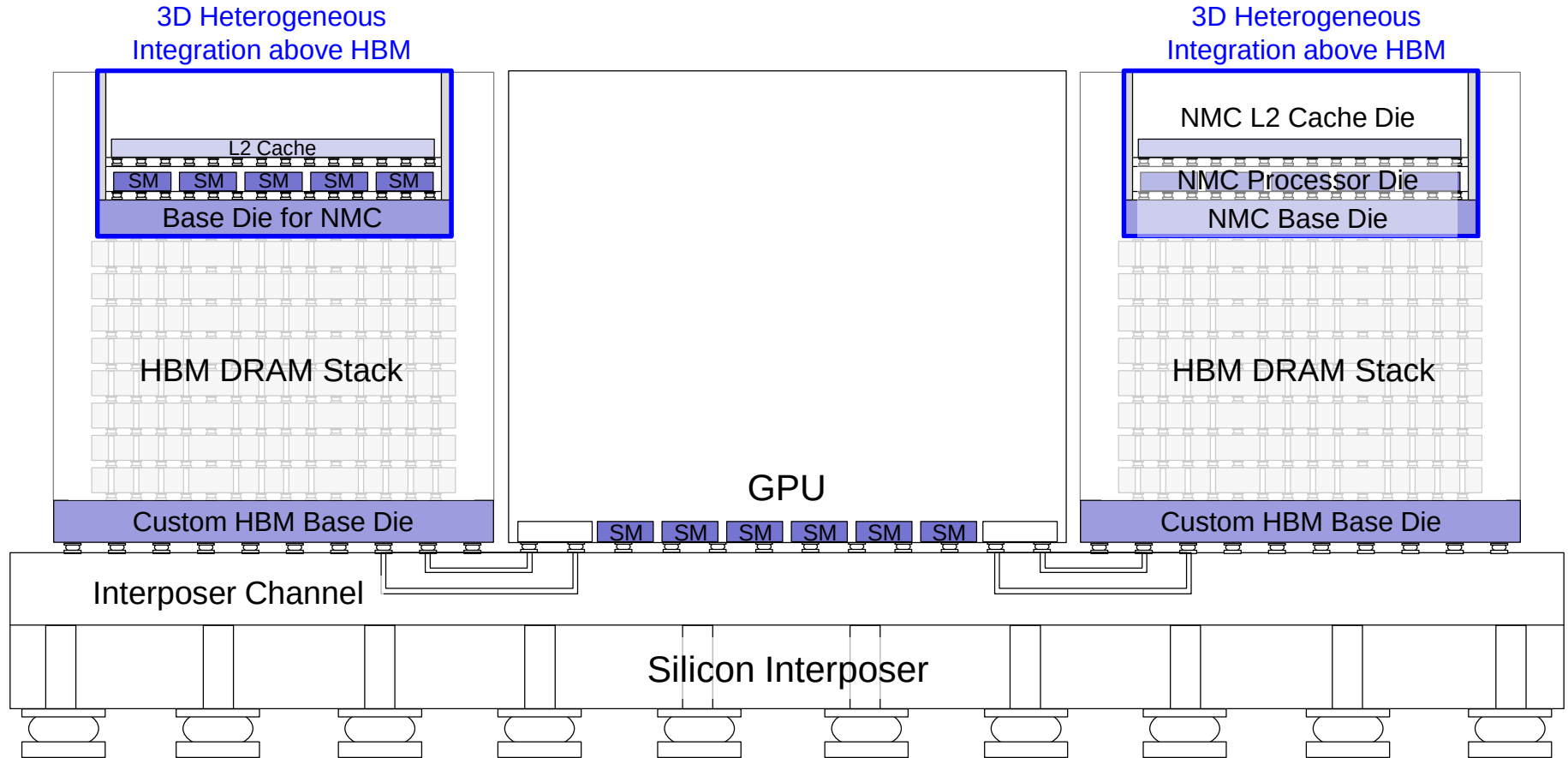


< Conventional CPU-GPU Architecture with HBM3 >



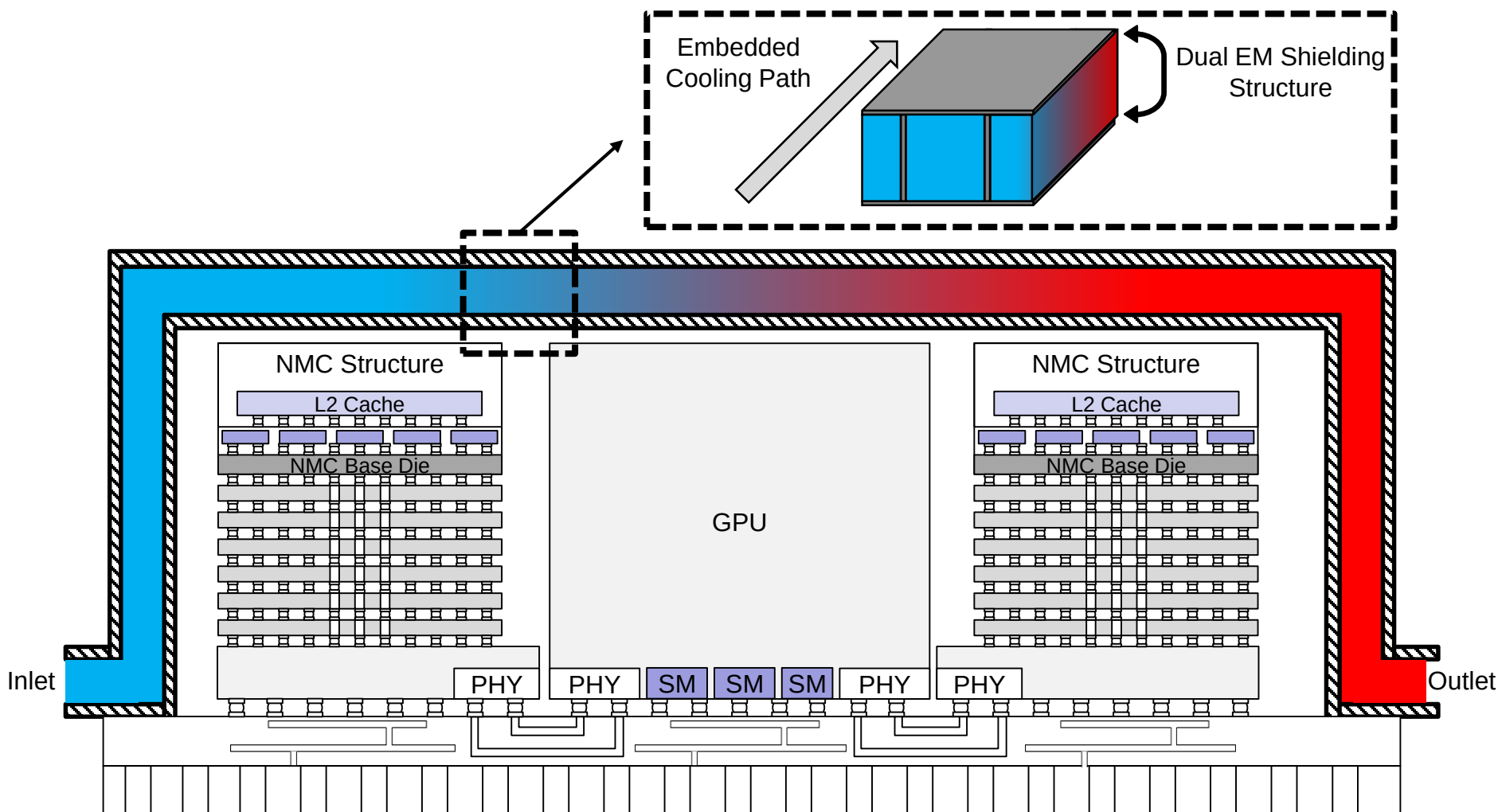
< HBM4 Architecture with LPDDR >

- The custom base die of HBM4 enables direct access to HBM and LPDDR, providing improved memory capacity without the CPU.



< 3D Integrated NMC-HBM Architecture above HBM DRAM >

- By integrating a NMC processor die and cache die above HBM, the proposed 3D NMC-HBM achieves high performance and energy efficient computing through dedicated TSV interconnection and power supply network.

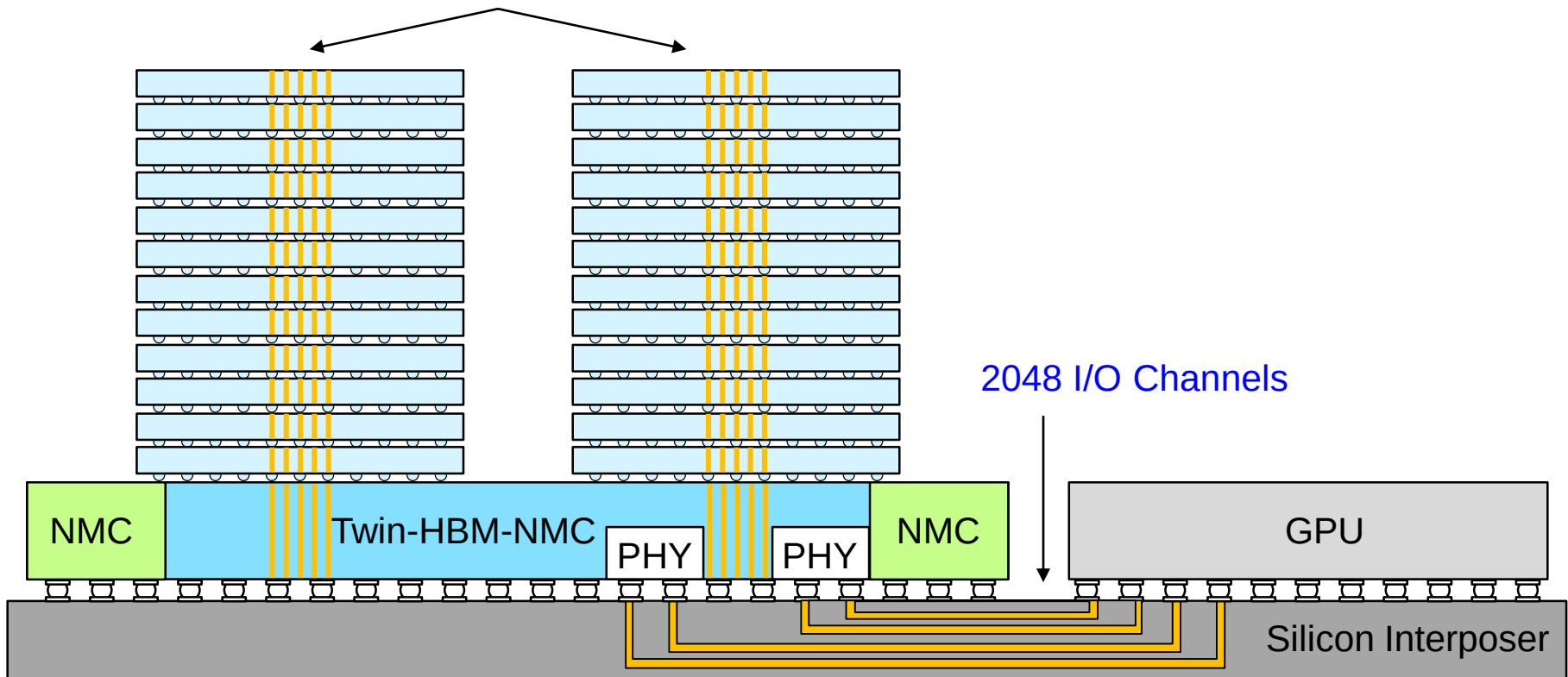


< Electromagnetic Shielding Structure with Liquid Cooling System for HBM5 Architecture >

Proposal of Twin Tower High Bandwidth Memory with Near-Memory Computing (Twin-HBM-NMC) Architecture

HBM6
Architecture

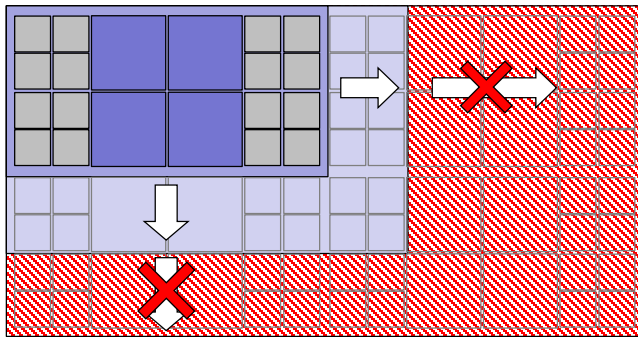
Two DRAM Stacks



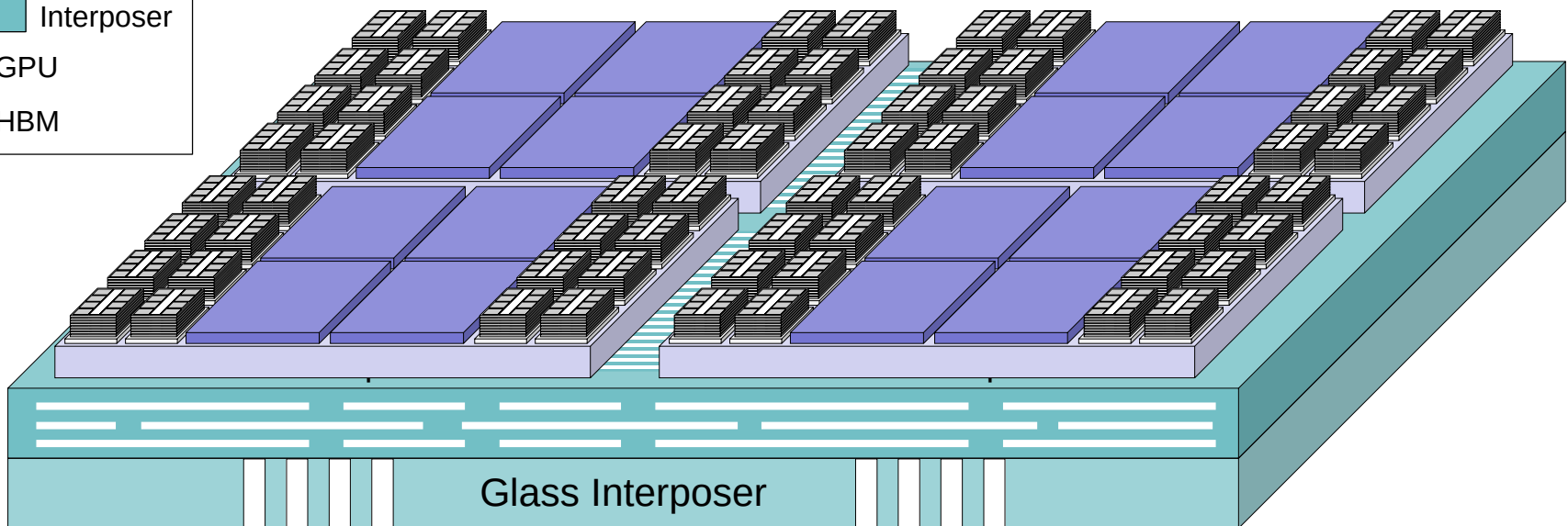
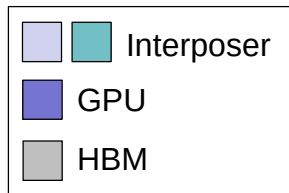
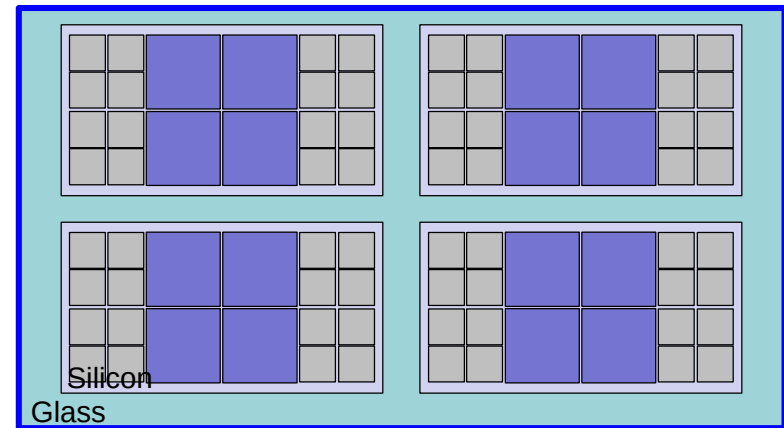
< Overview of Twin Tower HBM with NMC Architecture >

- In twin tower high bandwidth memory with near-memory-computing (Twin-HBM-NMC) architecture, two DRAM stacks are located on top of the large logic die.
- The logic die include NMC units and is connected to the GPU via 2048 interposer channels.

Limitation in increasing
silicon interposer die size

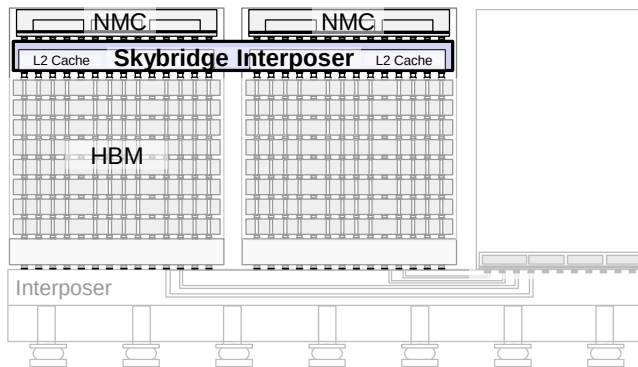


Large Scale Hybrid (Silicon+Glass) interposer

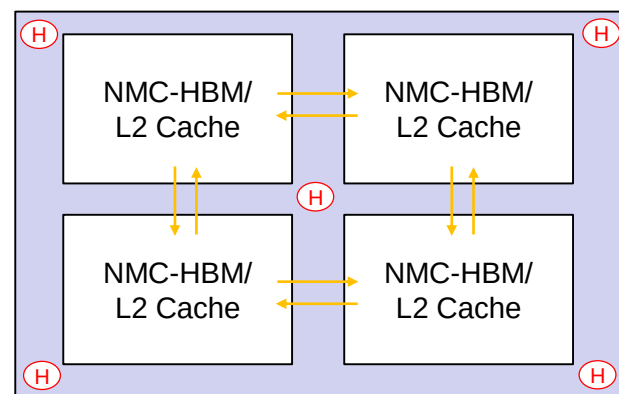


< Silicon-Glass Hybrid Interposer for Ultra Large-Scale Next Generation HBM >

Multi HBM-HBM Skybridge Interposer for Efficient Near Memory Computing Performance



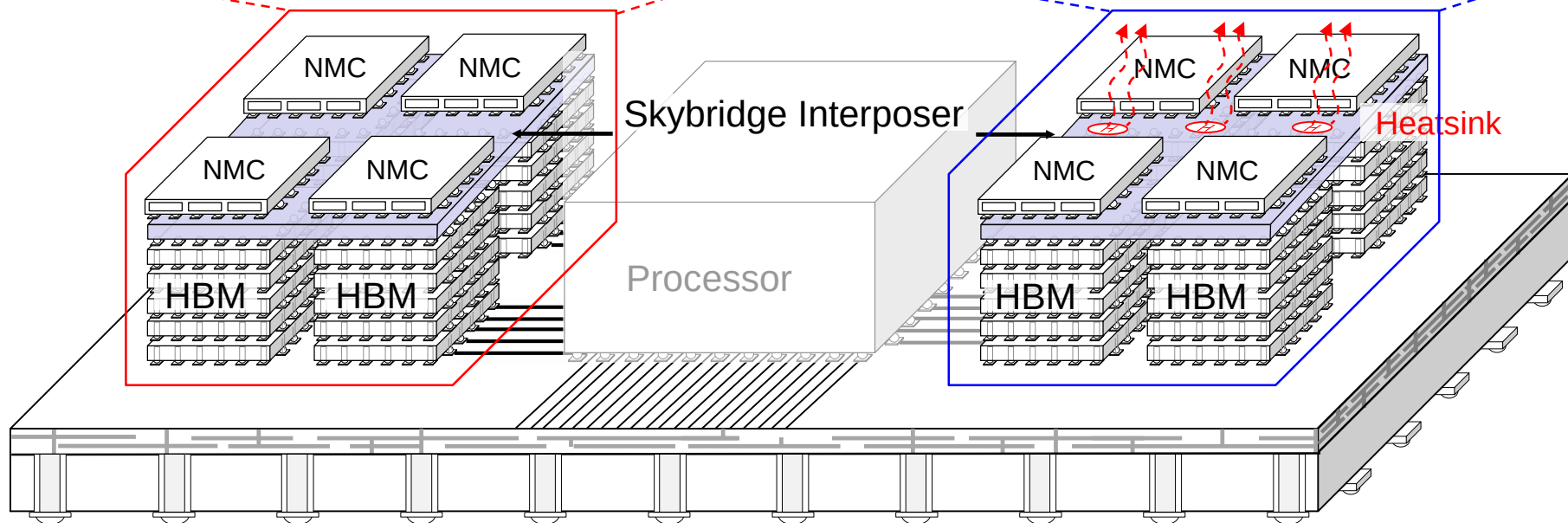
< Side view of Skybridge Interposer Architecture >

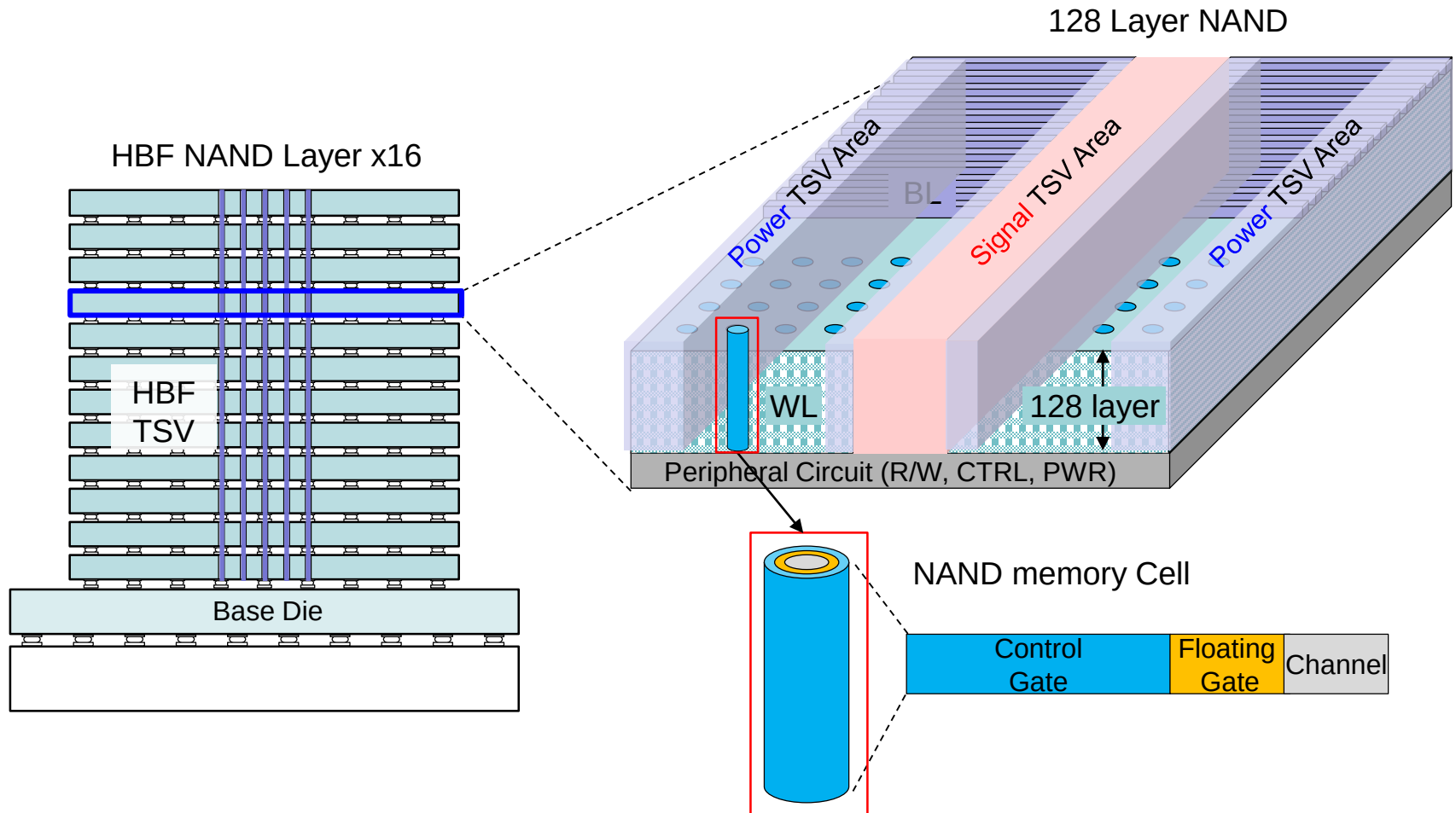


Through
Skybridge
Interposer...

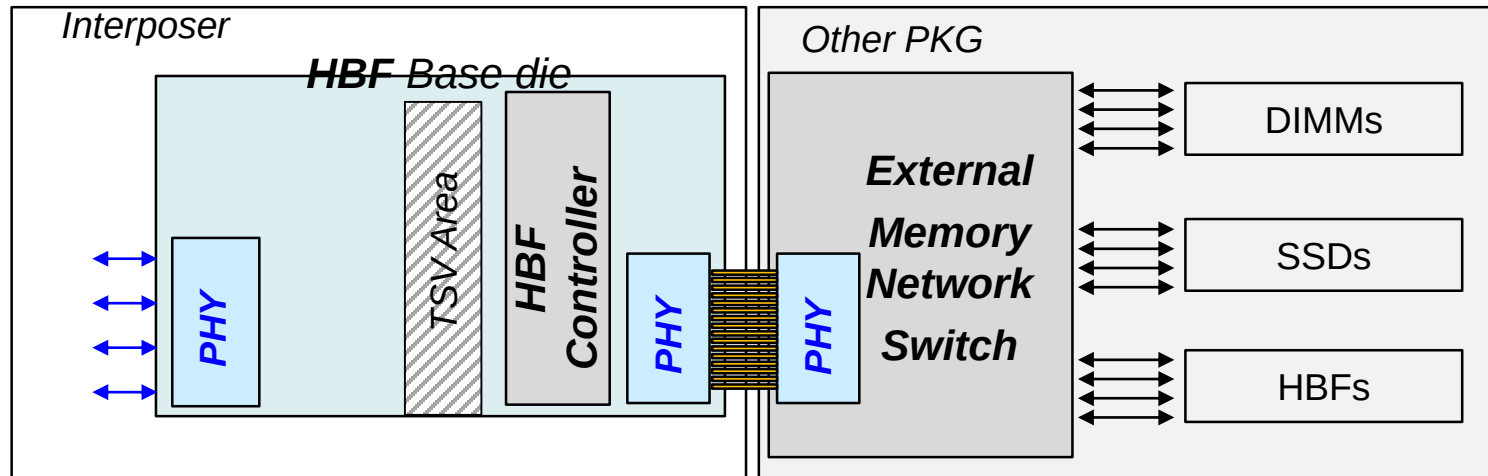
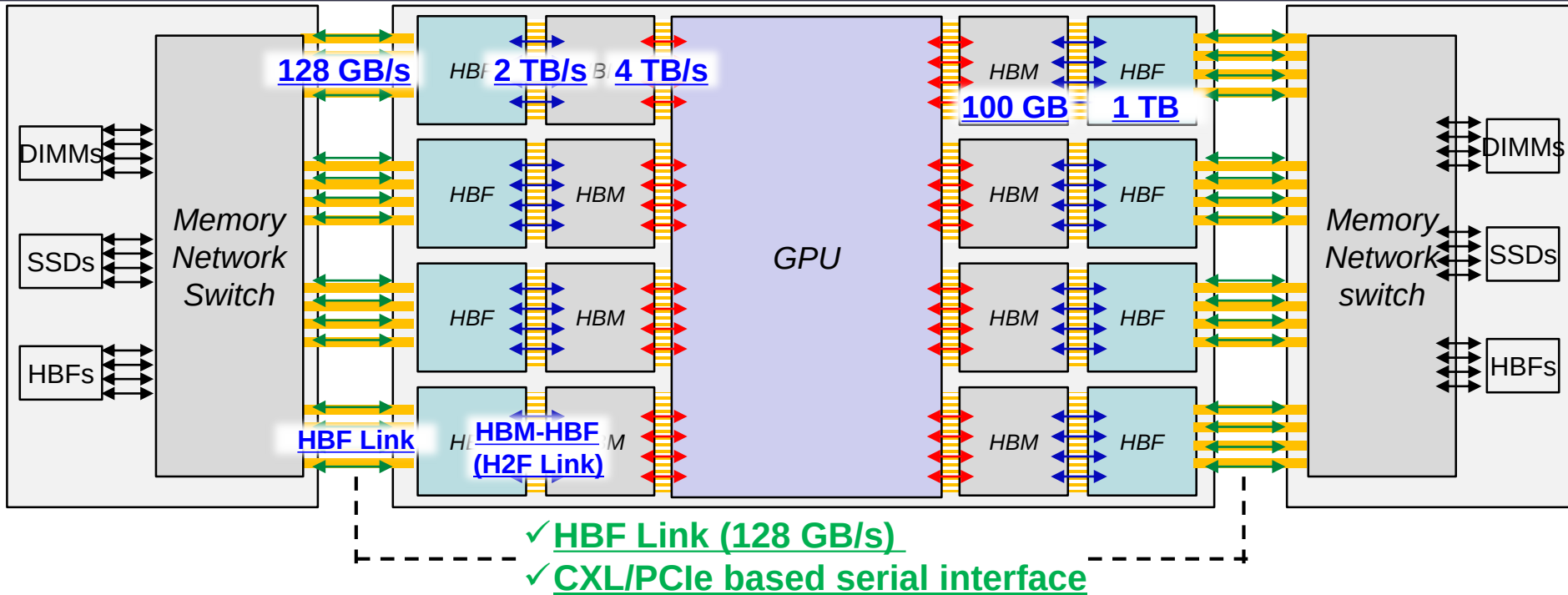
- ✓ NMC-NMC
- ✓ HBM-HBM
- ✓ NMC-HBM
- ✓ L2 cache
- ✓ Heatsink

< Top view of Skybridge Interposer >



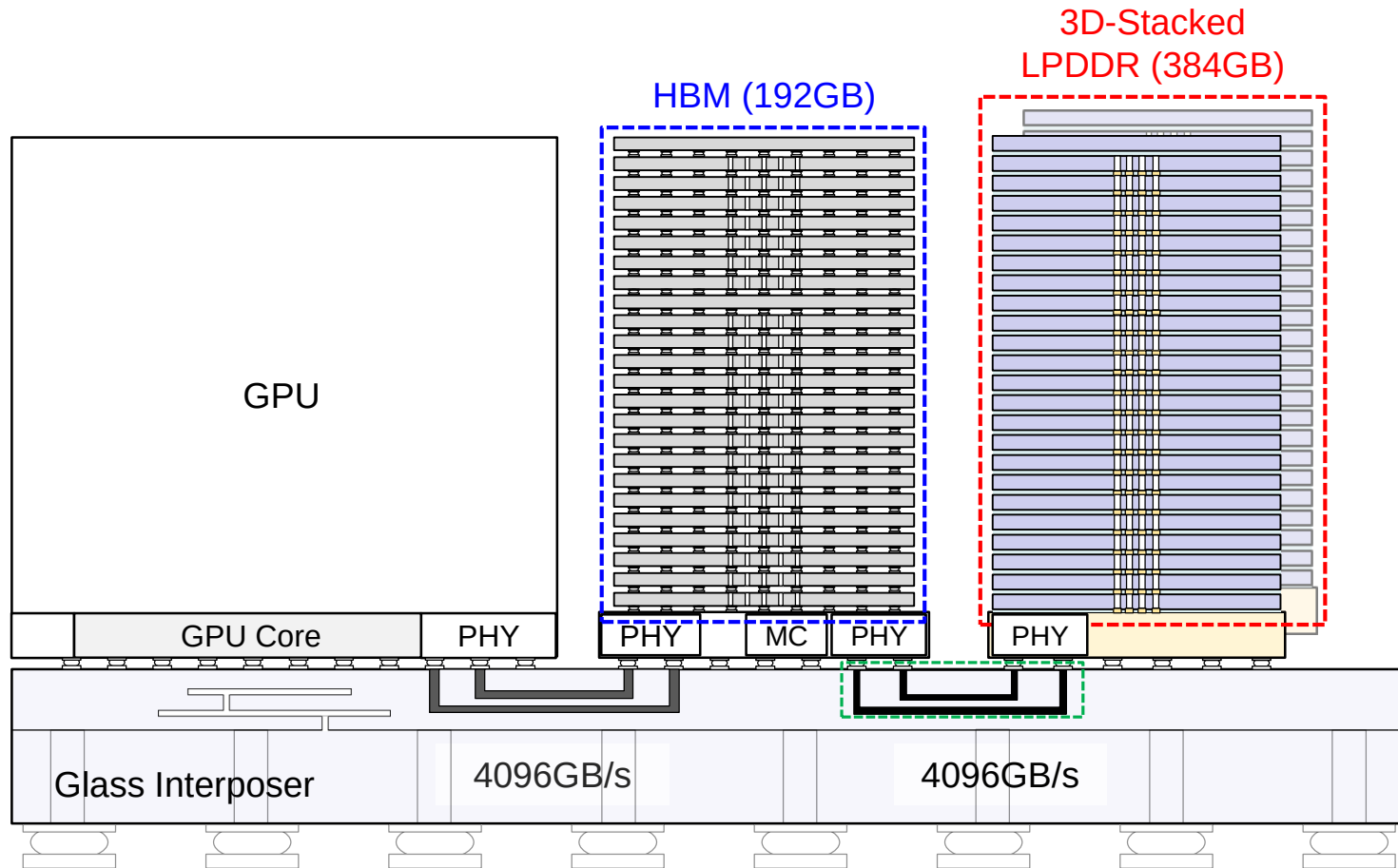


< High Bandwidth Flash (HBF) Architecture for Memory Intensive LLM Inference >



HBM7 Architecture Integrated with High-Capacity 3D-Stacked LPDDR on Glass Interposer

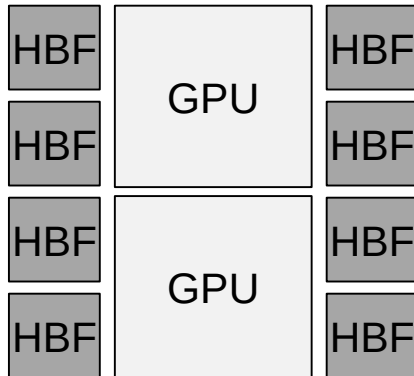
HBM7
Architecture



< HBM7 Architecture Integrated with High-Capacity 3D-Stacked LPDDR on Glass Interposer >

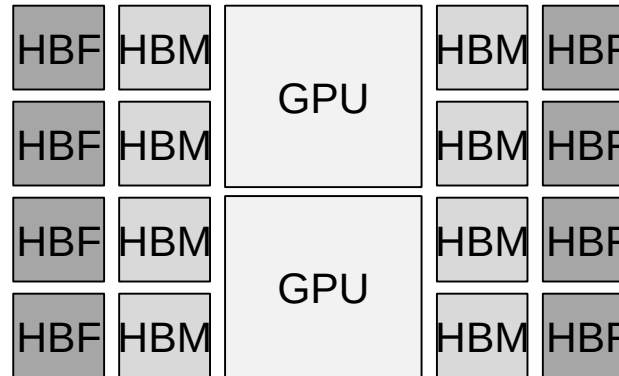
Comparison of HBM, HBF, and HBM-HBF Architecture

4TB NAND

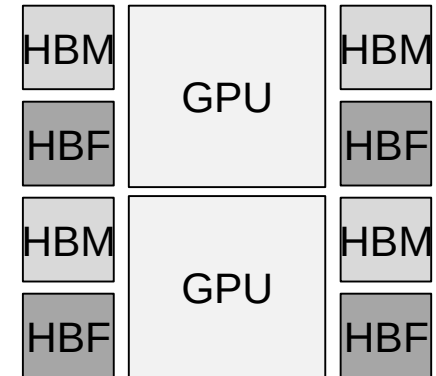


< GPU-HBF Architecture >

192GB DRAM + 4TB NAND

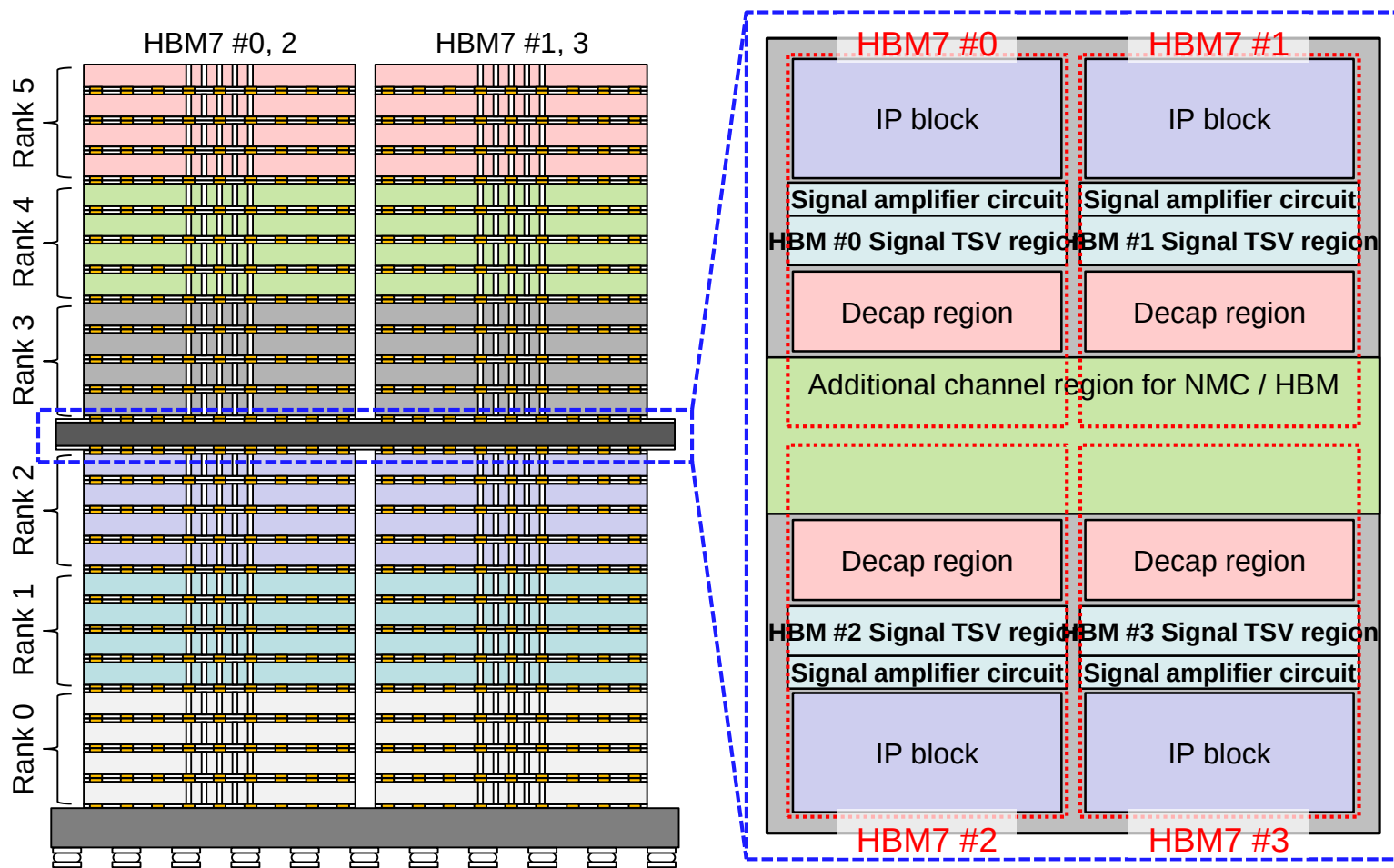


< GPU-HBM-HBF Architecture >

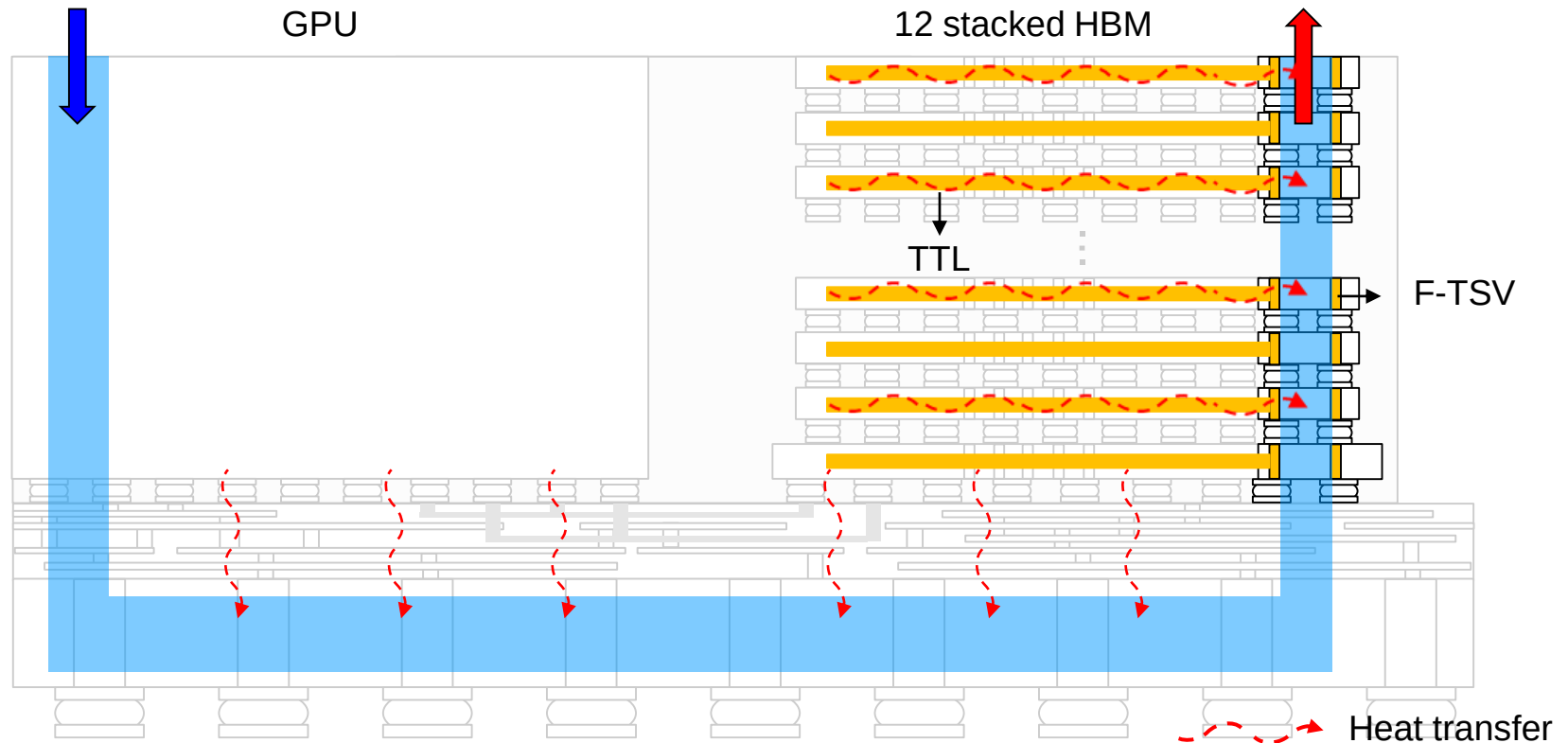


< Hybrid Architecture >

	HBM	HBF	HBM-HBF	Hybrid (1:1)
Capacity	192 GB	4 TB	4.2 TB	2.1 TB
Bandwidth	6.4 TB/s	2.7 TB/s	Mixed, Depends	Depends
Token/s	High	Medium	High	High
Cost	Medium	Low (\$ per bit is low)	High (Larger Interposer)	Medium-Low
Reliability	High	Low (HBF Write is bad)	Medium (No Write)	High



< Side and top view of proposed multi functional bridge-die >

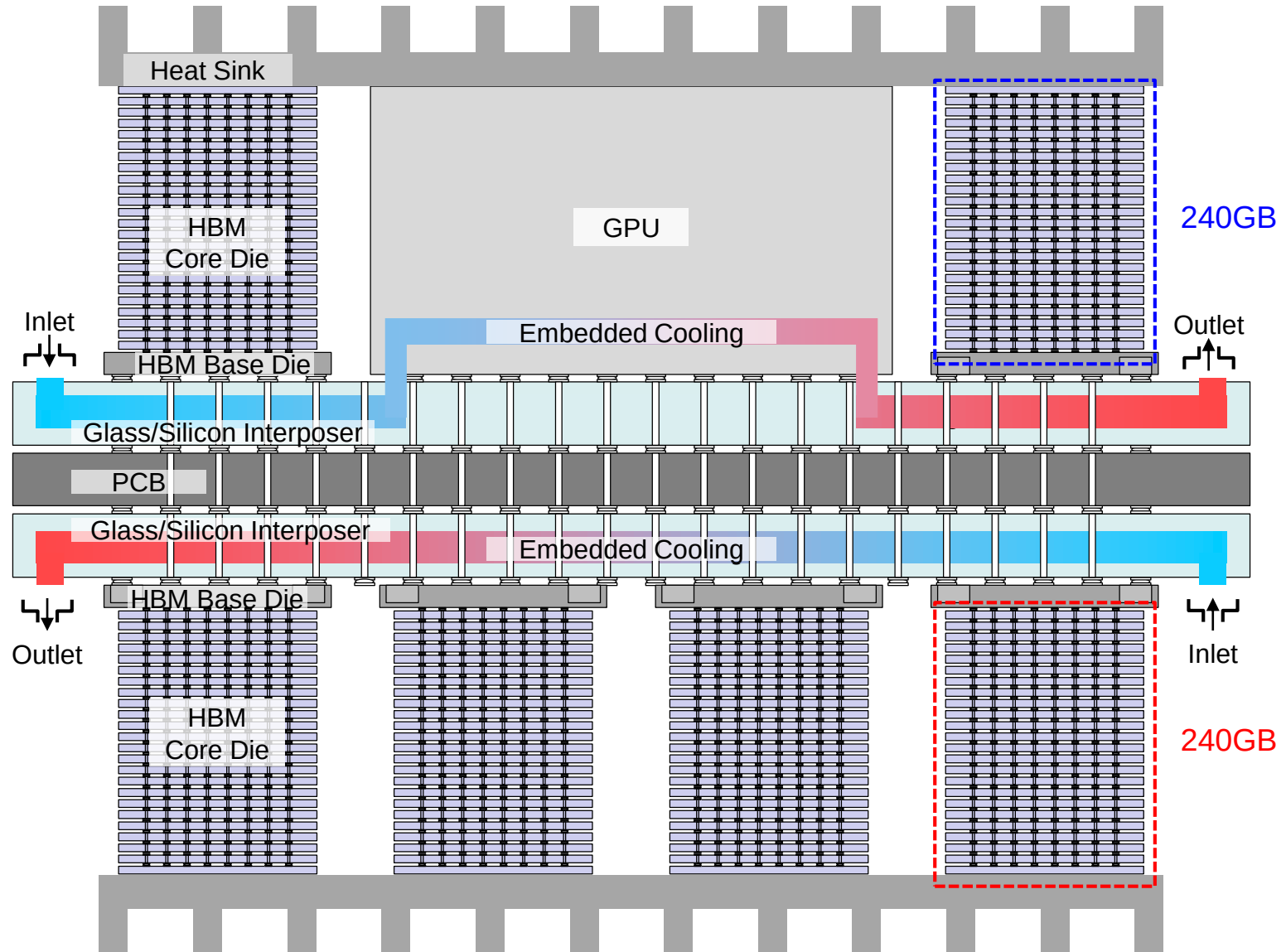


< Concept of the proposed embedded cooling structure for GPU-HBM module >

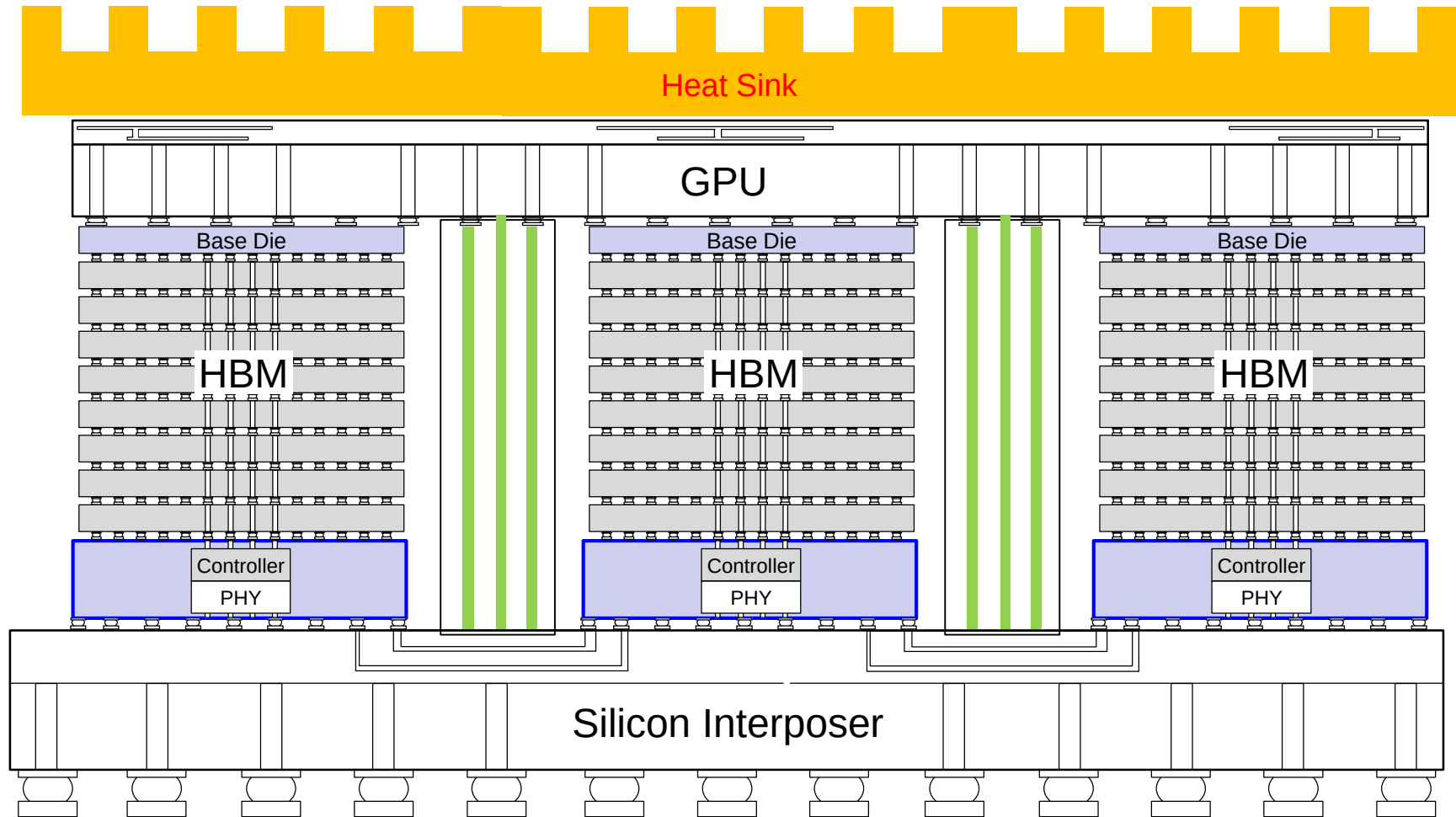
- The proposed Thermal Transmission Line (TTL) and Fluidic TSV (F-TSV) can cool the HBM module efficiently by circulating cooling fluid through the GPU to the interposer and HBM.
- The proposed TTL transfers the internal heat within HBM die to the fluid flowing in F-TSV

3D Memory Expansion Architecture with Embedded Cooling Structure for HBM8 with Double-Sided Interposer [1/3]: GPU-HBM-HBM

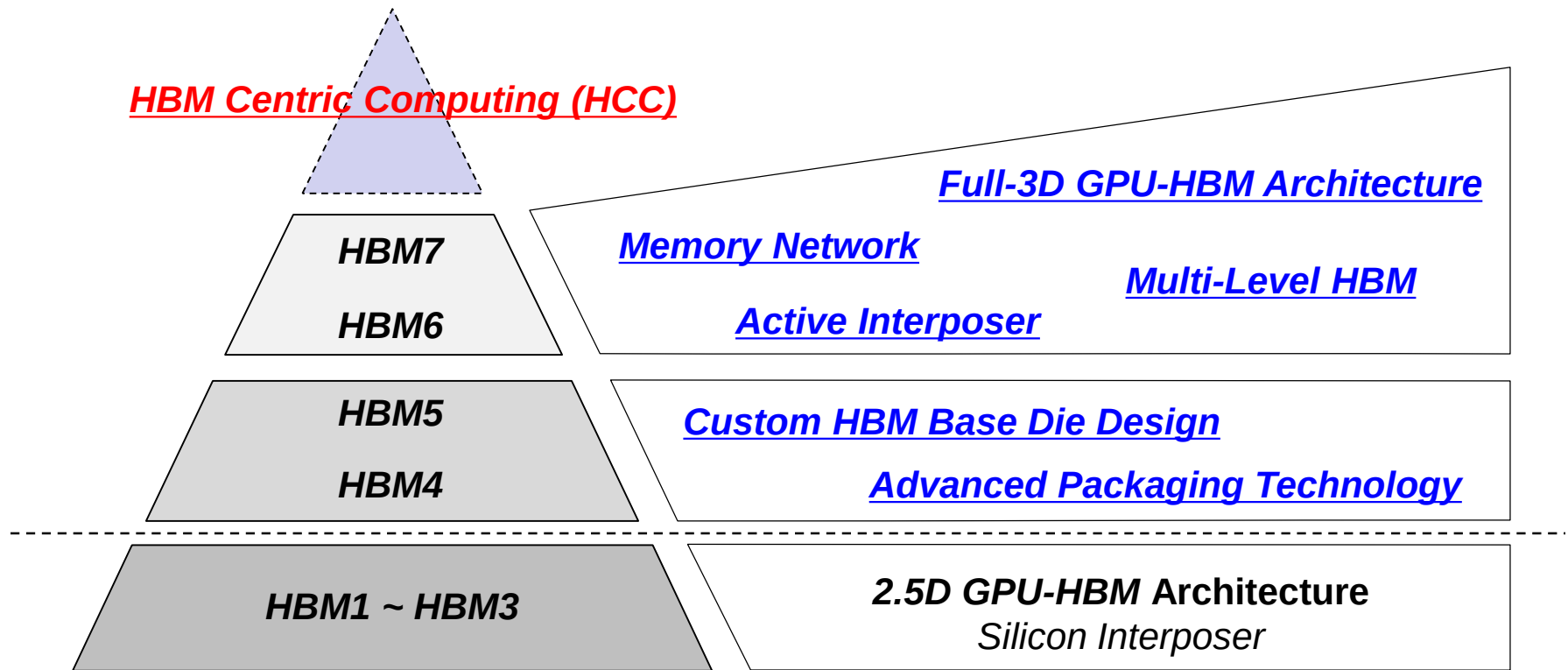
HBM8
Architecture



< 3D Memory Expansion Architecture with Embedded Cooling for HBM8 with Double-Sided Interposer using HBM >



- GPU is implemented at top layer of memory stack for heat dissipation. (Thermal Issue ↓)
- Additional silicon interconnect pillar die is embedded between HBMs to support power to GPU.



< Next-Generation HBM **Architecture** Roadmap >

Part3: Key Features in HBM4

Key Features of HBM4

1. Electrical Specification

- Data Rate : ~ 8 Gbps
- Number of I/Os : 2,048 (4096)
- Total Bandwidth : 2.0 TB/s
- Number of die stack : 12/16-Hi
- Capacity/die : 24 Gb

2. Packaging/Cooling Method

- Microbump (MR-MUF)
- Direct-to-Chip (D2C) Liquid Cooling

3. HBM Architecture

- Custom HBM Base Die
- NMC processor + LPDDR in Base Die

4. AI Design Agent

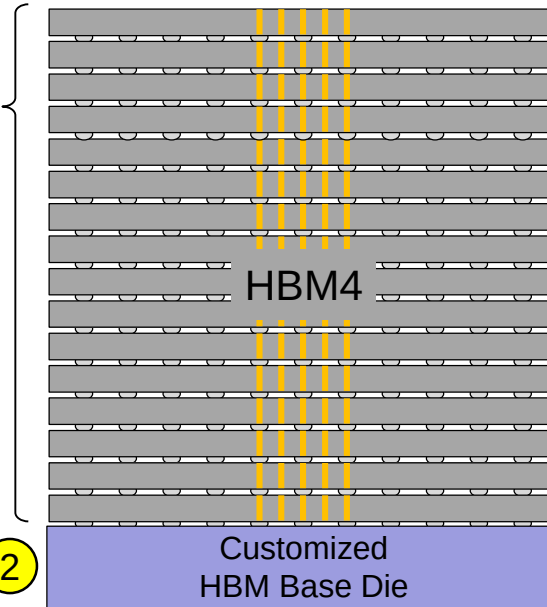
- AI assistant
- Microbump & TSV array, and Decap placement Optimization based on Reinforcement Learning

TSV(실리콘 관통전극)



① HBM 적층 수, 저장 용량 증가
8 ~ 12 단 → 12 ~ 16 단
16 ~ 24 GB → 36 ~ 48 GB

①



HBM4

GPU

② Customized
HBM Base Die

②

② 고객 맞춤형 (Customized)
Base Die 설계
일부 GPU 계산 기능이 이동

③

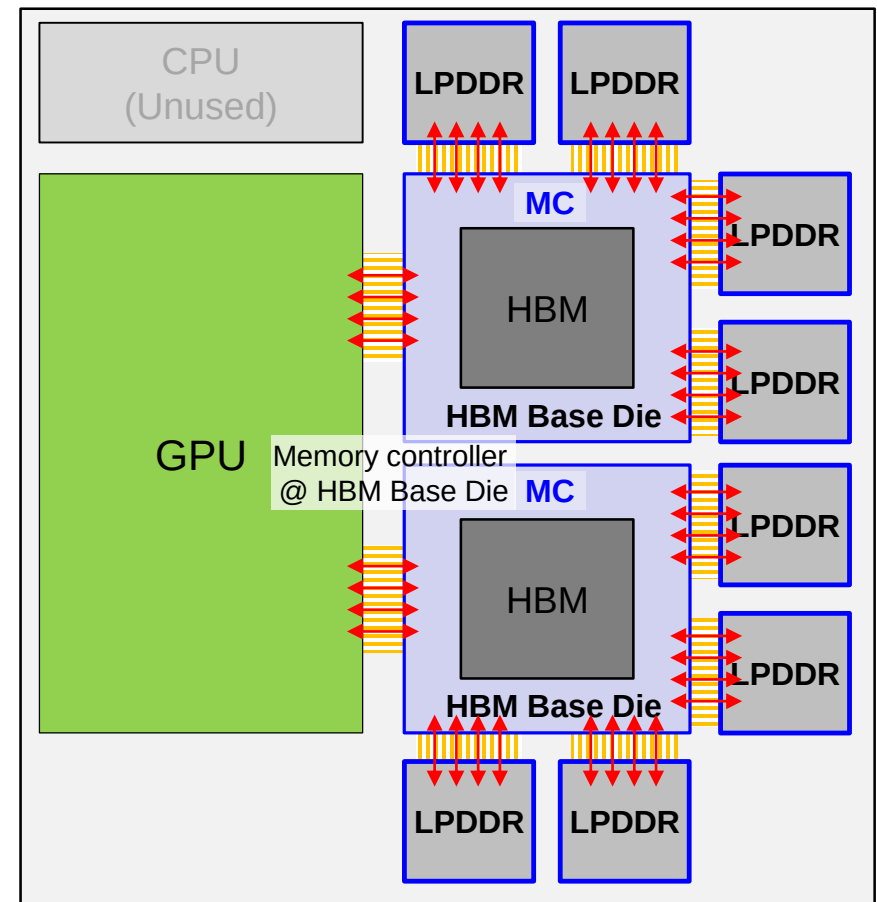
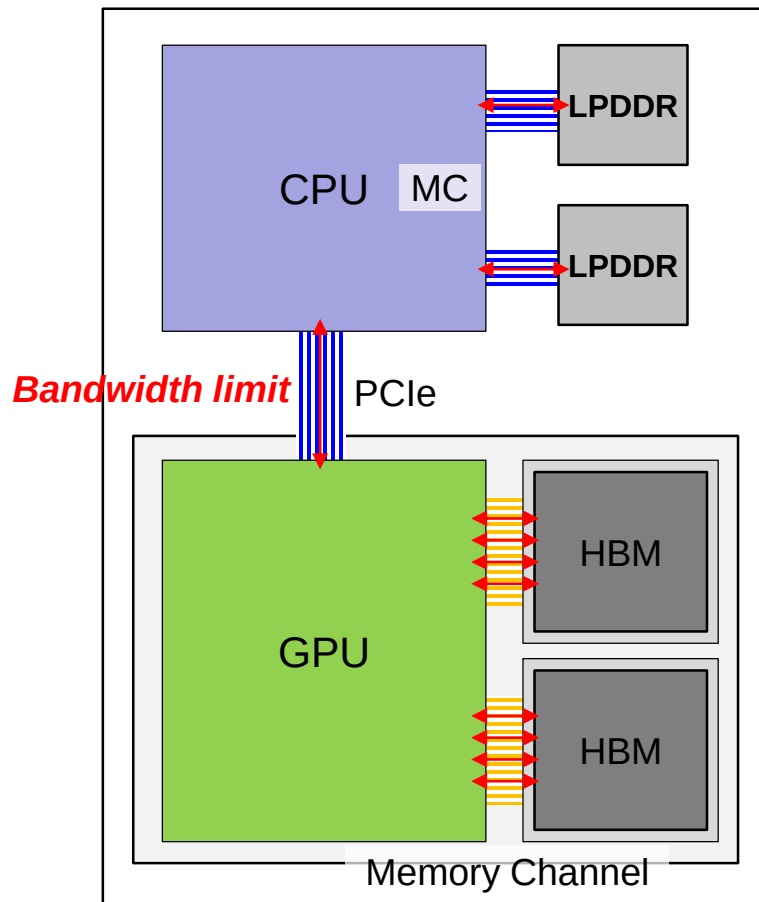
③

HBM3/3E 대비 2배 늘어난
HBM-GPU 연결선 (I/O)
2,048 개

실리콘 인터포저 (기판)

Custom HBM Base Die Design : LPDDR Memory Channel for High Capacity & Memory Bandwidth

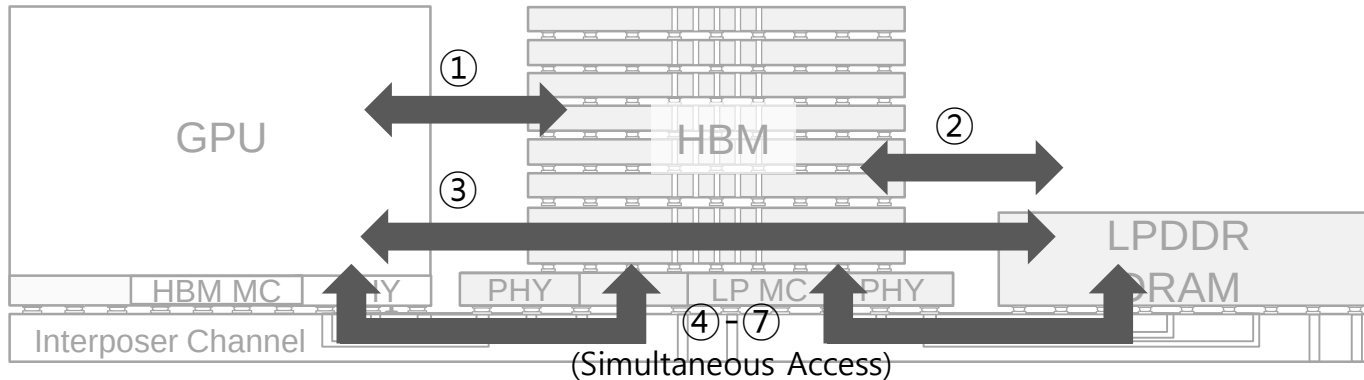
↔ : Low-Bandwidth ↔↔↔ : High-Bandwidth



< Conventional CPU-GPU Architecture with HBM3 >

< HBM4 Architecture with LPDDR >

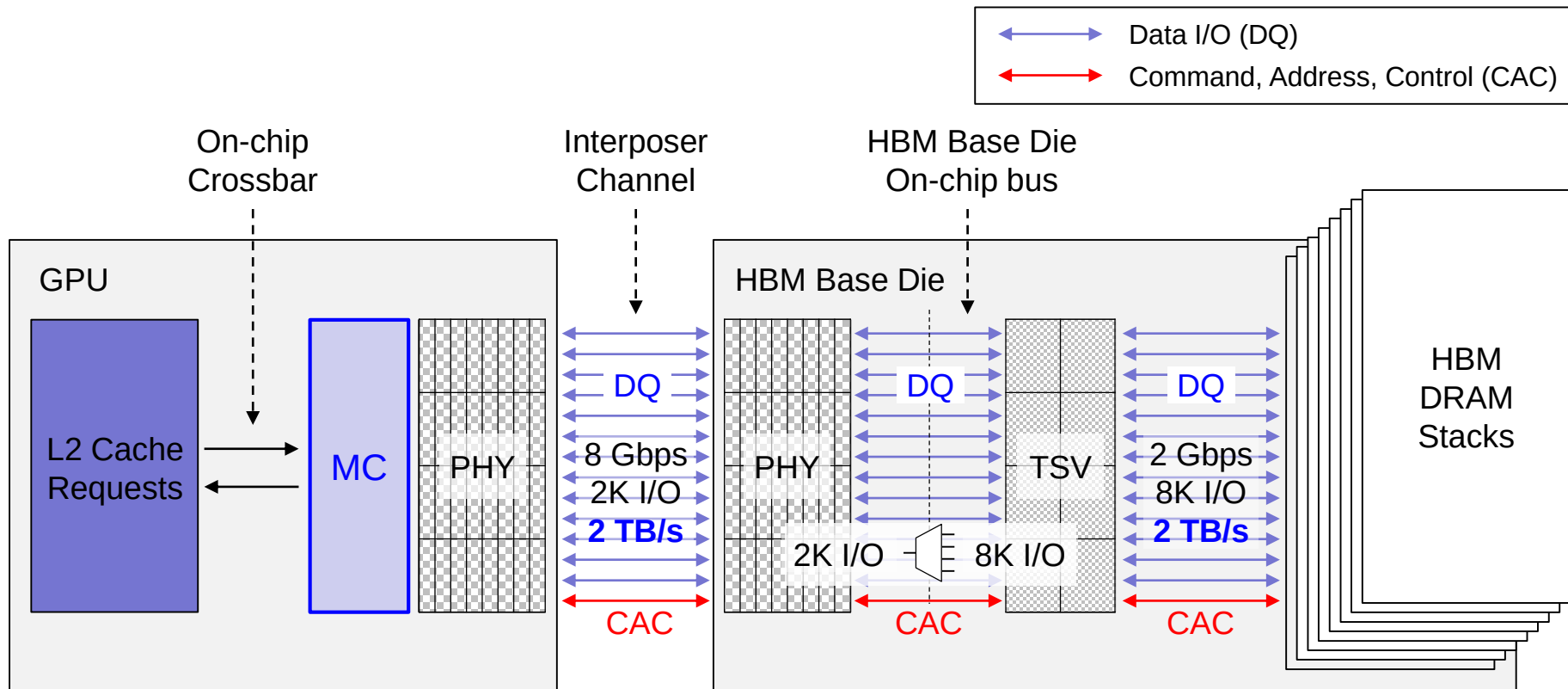
- The custom base die of HBM4 enables direct access to HBM and LPDDR, providing improved memory capacity without the CPU.



< Side-view of HBM-LPDDR Structure and its data path >

- Case 1: GPU ↔ HBM (Read/Write)
 - Case 2: HBM ↔ LPDDR (Read/Write)
 - Case 3: GPU ↔ LPDDR (Read/Write)
 - Case 4: GPU → HBM (Write) & HBM ← LPDDR (Read)
 - Case 5: GPU → HBM (Write) & HBM → LPDDR (Write)
 - Case 6: GPU ← HBM (Read) & HBM ← LPDDR (Read)
 - Case 7: GPU ← HBM (Read) & HBM → LPDDR (Write)
- Single-Command Execution
- Dual-Command Execution
(GPU ↔ HBM ↔ LPDDR)

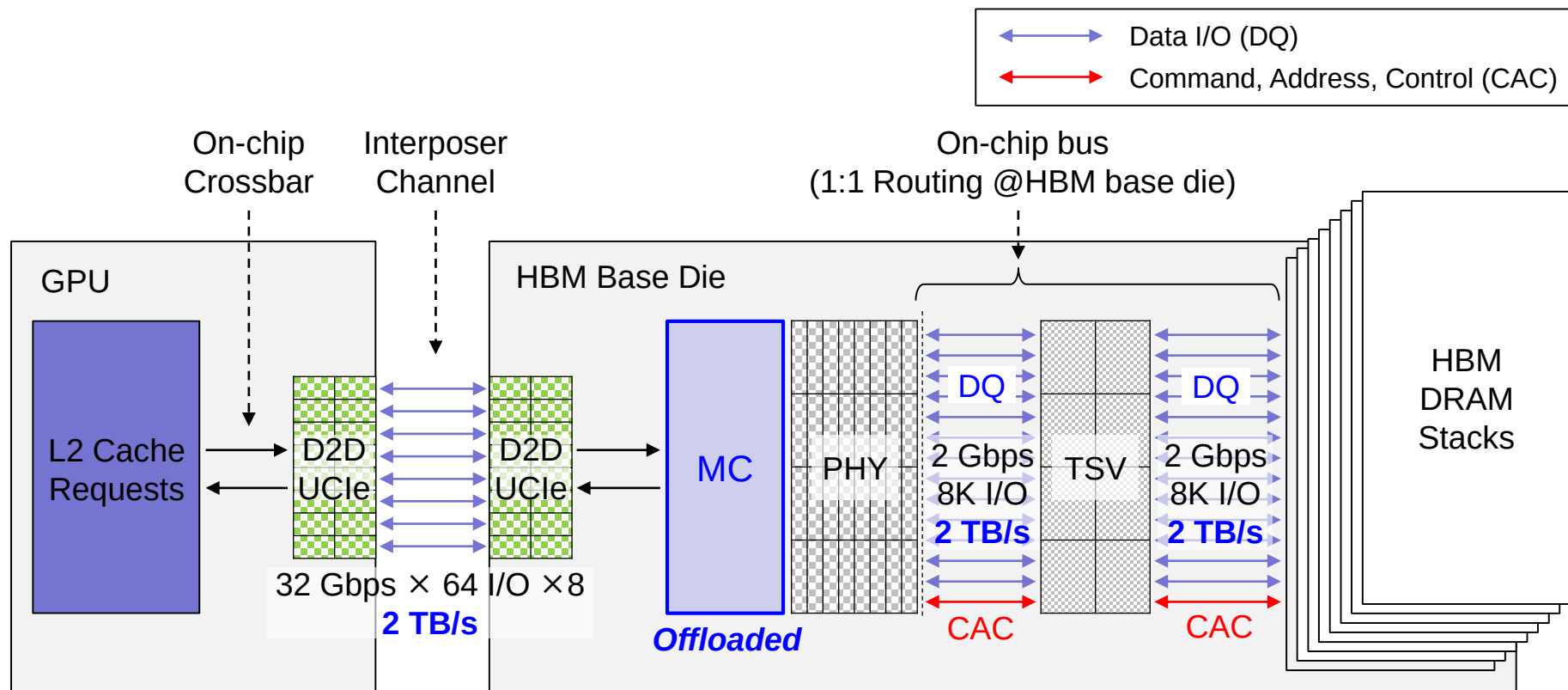
Conventional HBM Architecture : Memory Controller Located at GPU (HBM4)



< Hierarchic Bandwidth Teardown : Memory Controller at GPU >

- The memory controller of the conventional GPU-HBM architecture is located at the GPU.
- A total of 2,048 HBM4 interposer channels provides the required memory bandwidth of 2 TB/s.
 - ✓ Interposer Channel : 8 Gbps × 2,048 I/O
- Through the HBM base die, each channel at HBM PHY is mapped to its corresponding TSV.

Next-Generation HBM Architecture : Memory Controller Located at HBM Base Die

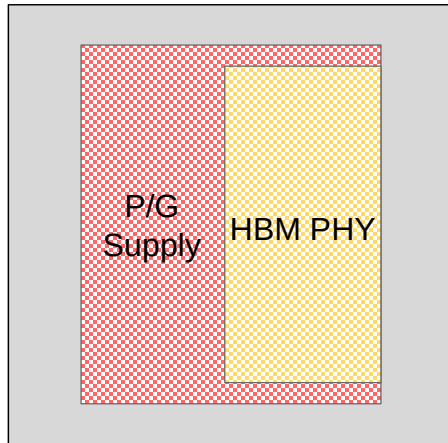


< Hierarchic Bandwidth Teardown : Memory Controller offloaded to HBM Base Die >

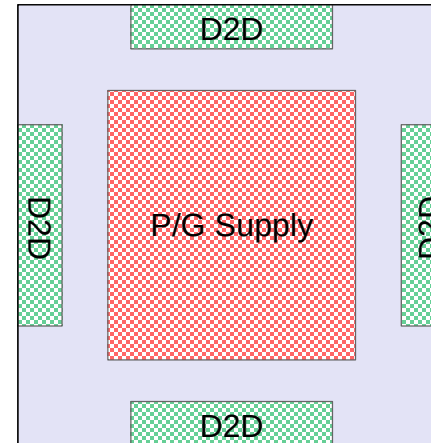
- As the memory controller is offloaded to the HBM base die, the previous long-range interposer channels are replaced by short-reach, serial based die-to-die (D2D) interconnect (UCle-A).
- Eight D2D interconnect (UCle-A x64) Tx-Rx units provide a total memory bandwidth of 2 TB/s.
 - ✓ UCle-A x64 Channel : 32 Gbps × 64 I/O × 8 (units)
- The MC maps each HBM PHY channel to each TSV interconnect, routed through the on-chip bus.

Advantages / Challenges of Offloading HBM Memory Controller to Base Die

Bump Map of Conventional HBM Base Die



Bump Map Concept of Custom HBM Base Die w/ MC



< Conceptual View of Conventional HBM & Custom HBM Base Die with Offloaded MC >

■ Advantages :

- ✓ Capable of increasing # of GPU Cores due to increased area from offloading MC & PHY
- ✓ High utilization of HBM die edge area (D2D Link)
- ✓ No need for additional channels for Command / Address / Clock data transfer.
- ✓ Improved fine-pitch channel design margin & decreased microbump area at PHY from lower number of I/Os (~ 50% ↓)

■ Challenges :

- ✓ Adoption of SerDes-based interface & new interface layers

→ Consideration of protocol conversion overhead is required. (AXI → UCle → AXI → DFI)

Comparison of GPU-HBM Architecture According to MC Position

	GPU-HBM (Conventional)	GPU-Customized HBM
Side View		
Memory Ctrl	Located at GPU	Located at HBM Base Die
Data Rate	8 Gbps	32 Gbps
# I/O	DQ : 64 bit × 32 ch = 2,048 + Cmd, Addr, Ctrl	DQ : 64 bit × 8 units × 2 (bidirectional) = 1,024 + Valid, Track, CLK
Signaling Scheme	Bi-Directional / Direct Parallel	Uni-directional / D2D
Bandwidth	2048 GB/s	2048 GB/s
Channel Length	5 ~ 6 mm	~ 2 mm

< Comparison of GPU-HBM Architecture with Different MC Position >

Interconnection Efficiency between GPU and HBM (1)

- 1) Interconnection number from 2048 to 2048: Same
- 2) 8 Gbps (Type I) to 32 Gbps (Type 2) (Need High speed I/O circuits)
- 3) Length limit from 5-6 mm (Type I) to 2mm (Type II)
- 4) Type II Bidirectional :interconnection efficiency 50%
- 5) Differential line pair: interconnection efficiency 50%
- 6) Type I Summary: $8\text{GBps} \times 2048 \times 1.0 \times 1.0 = 16.384 \text{ Tbps} = 2.048 \text{ TB/s}$
per GPU-HBM Interface
- 7) Type II Summary: $32\text{GBps} \times 2048 \times 0.5 \times 0.5 = 16.384 \text{ Tbps} = 2.048 \text{ TB/s}$
per GPU-HBM Interface
- 8) Same interconnection efficiency

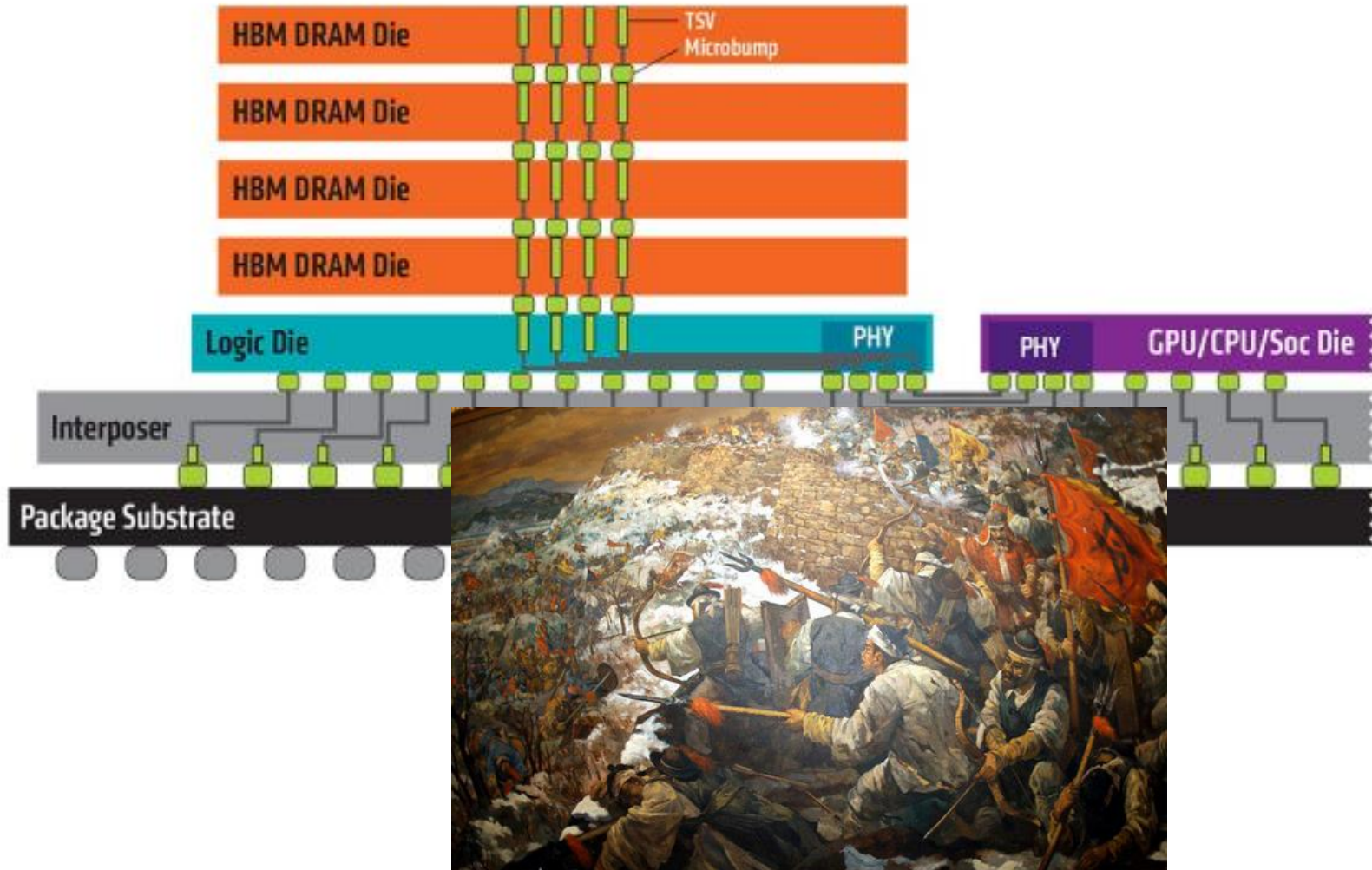
Interconnection Efficiency between GPU and HBM (2)

- 1) Interconnection number from 2048 to 2048: Same
- 2) 8 Gbps (Type I) to 64 Gbps (Type 2) (Need High speed I/O circuits)
- 3) Length limit from 5-6 mm (Type I) to 2mm (Type II)
- 4) Type II Bidirectional :interconnection efficiency 50%
- 5) Differential line pair: interconnection efficiency 50%
- 6) Type I Summary: $8\text{GBps} \times 2048 \times 1.0 \times 1.0 = 16.384\text{ Tbps} = 2.048\text{ TB/s}$
per GPU-HBM Interface
- 7) Type II Summary: $64\text{GBps} \times 2048 \times 0.5 \times 0.5 = 32.768\text{ Tbps} = 4.096\text{ TB/s}$
per GPU-HBM Interface
- 8) X2 interconnection efficiency

Interconnection Efficiency between GPU and HBM (3)

- 1) Interconnection number from 2048 to 2048: Same
- 2) 8 Gbps (Type I) to 128 Gbps (Type 2) (Need High speed I/O circuits)
- 3) Length limit from 5-6 mm (Type I) to 2mm (Type II)
- 4) Type II Bidirectional :interconnection efficiency 50%
- 5) Differential line pair: interconnection efficiency 50%
- 6) Type I Summary: $8\text{GBps} \times 2048 \times 1.0 \times 1.0 = 16.384\text{ Tbps} = 2.048\text{ TB/s}$
per GPU-HBM Interface
- 7) Type II Summary: $128\text{GBps} \times 2048 \times 0.5 \times 0.5 = 65.536\text{ Tbps} = 8.192\text{ TB/s}$
per GPU-HBM Interface
- 8) X4 interconnection efficiency

Initiative War between GPU and HBM



Part4: Key Features in HBM5

Key Features of HBM5

1. Electrical Specification

- Data Rate : 8 Gbps
- Number of I/Os : 4,096
- Total Bandwidth : 4.0 TB/s
- Number of die stack : 16-Hi
- Capacity/die : 40 Gb

2. Packaging/Cooling Method

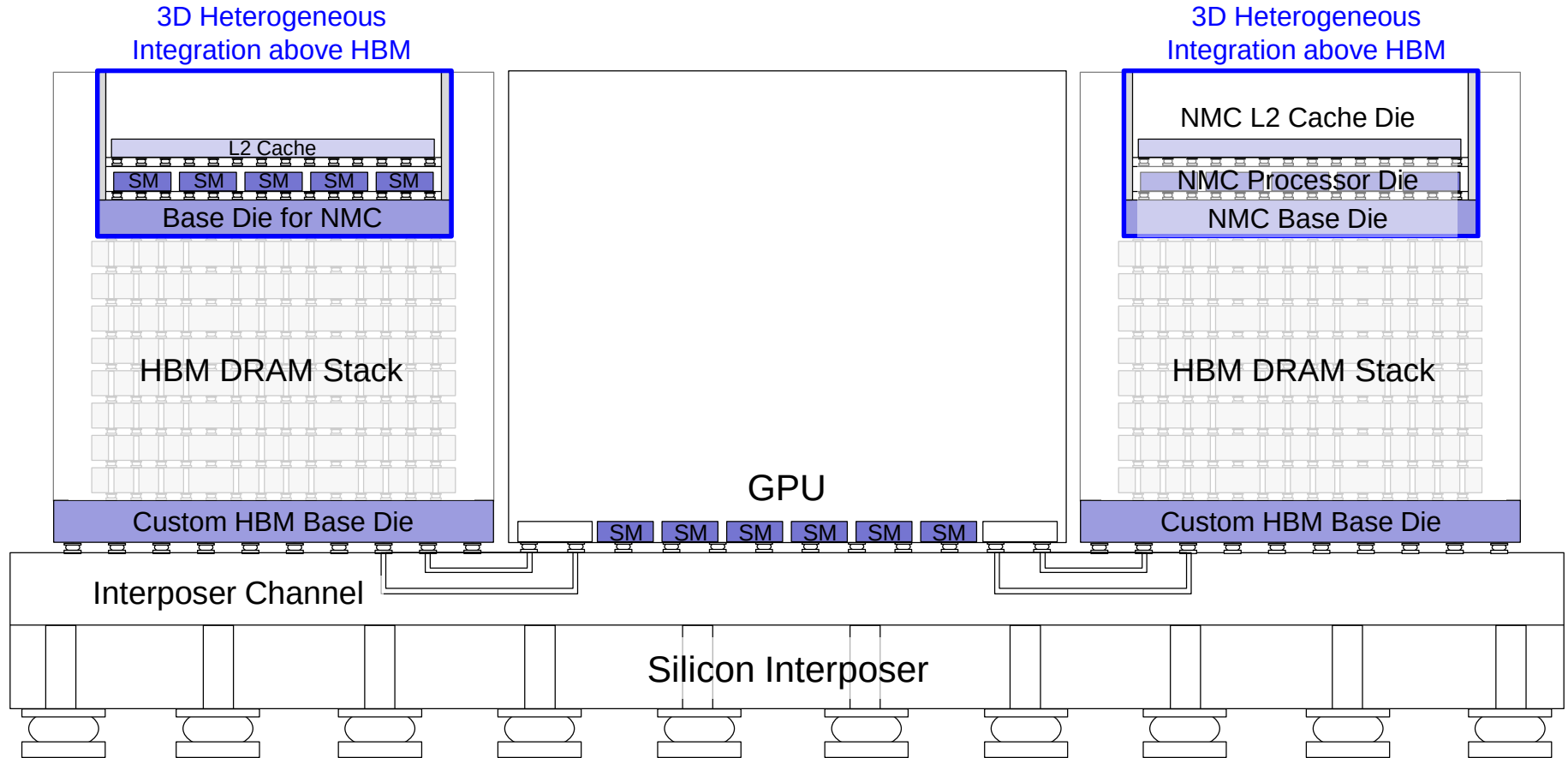
- Microbump (MR-MUF)
- Immersion Cooling, Thermal Via (TTV), Thermal Bonding
- Temperature sensors in base die

3. HBM Architecture

- Dedicated decoupling capacitor chip die stack
- Custom HBM Base Die w/ 3D NMC-HBM & Stacked Cache
- NMC + Cache in HBM PKG
- LPDDR + CXL in Base Die
- Separated TSV, TGV, TPV, TTV designs

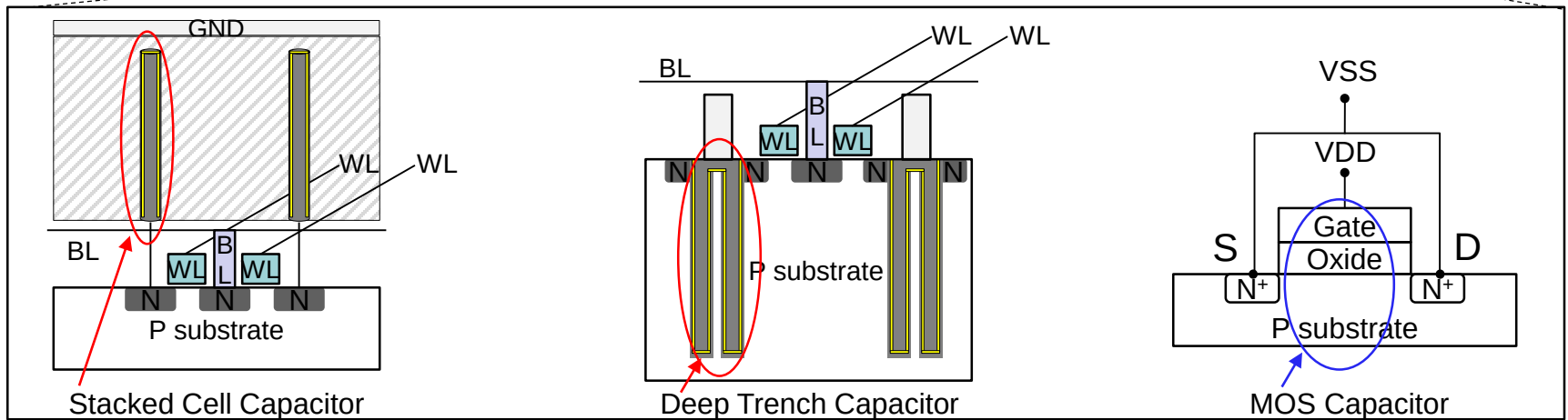
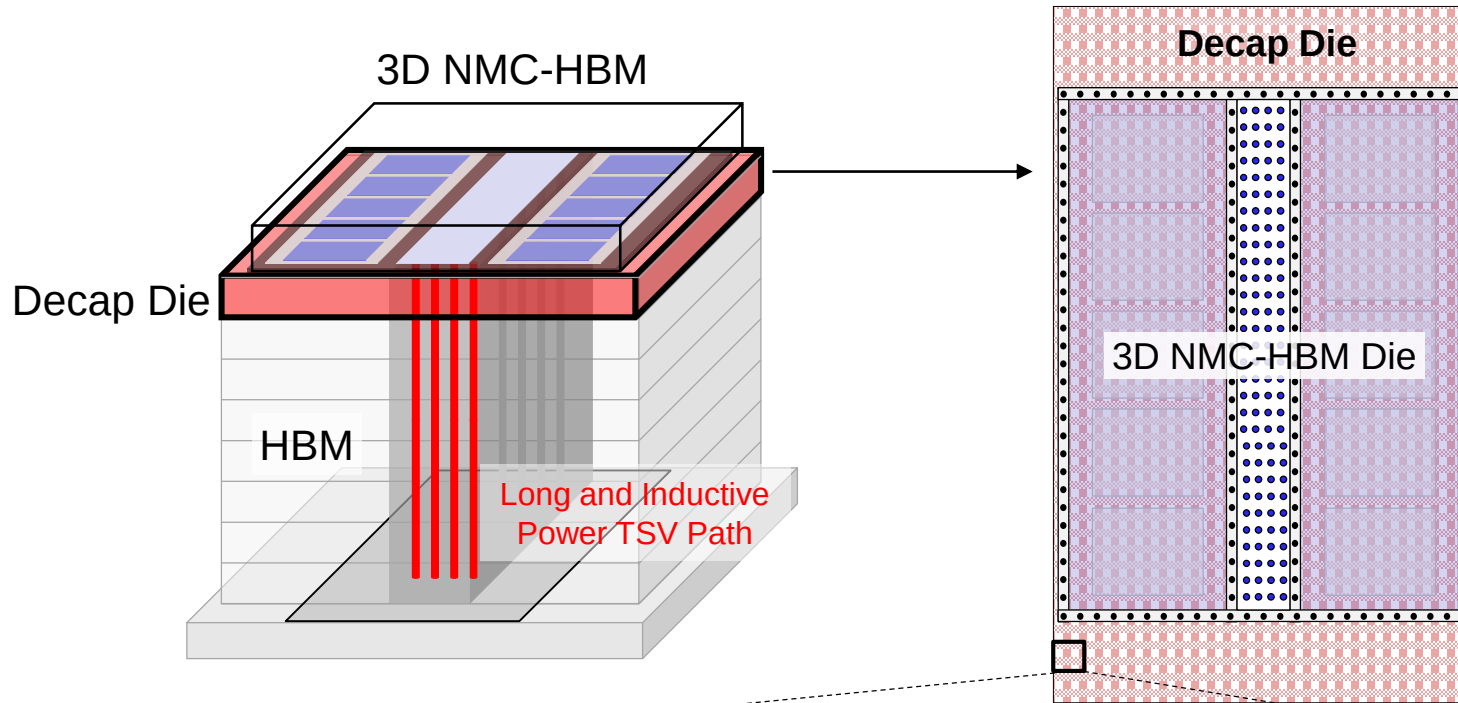
4. AI Design Agent

- AI Agent
- I/O Interface Optimization considering PSIJ based on Reinforcement Learning

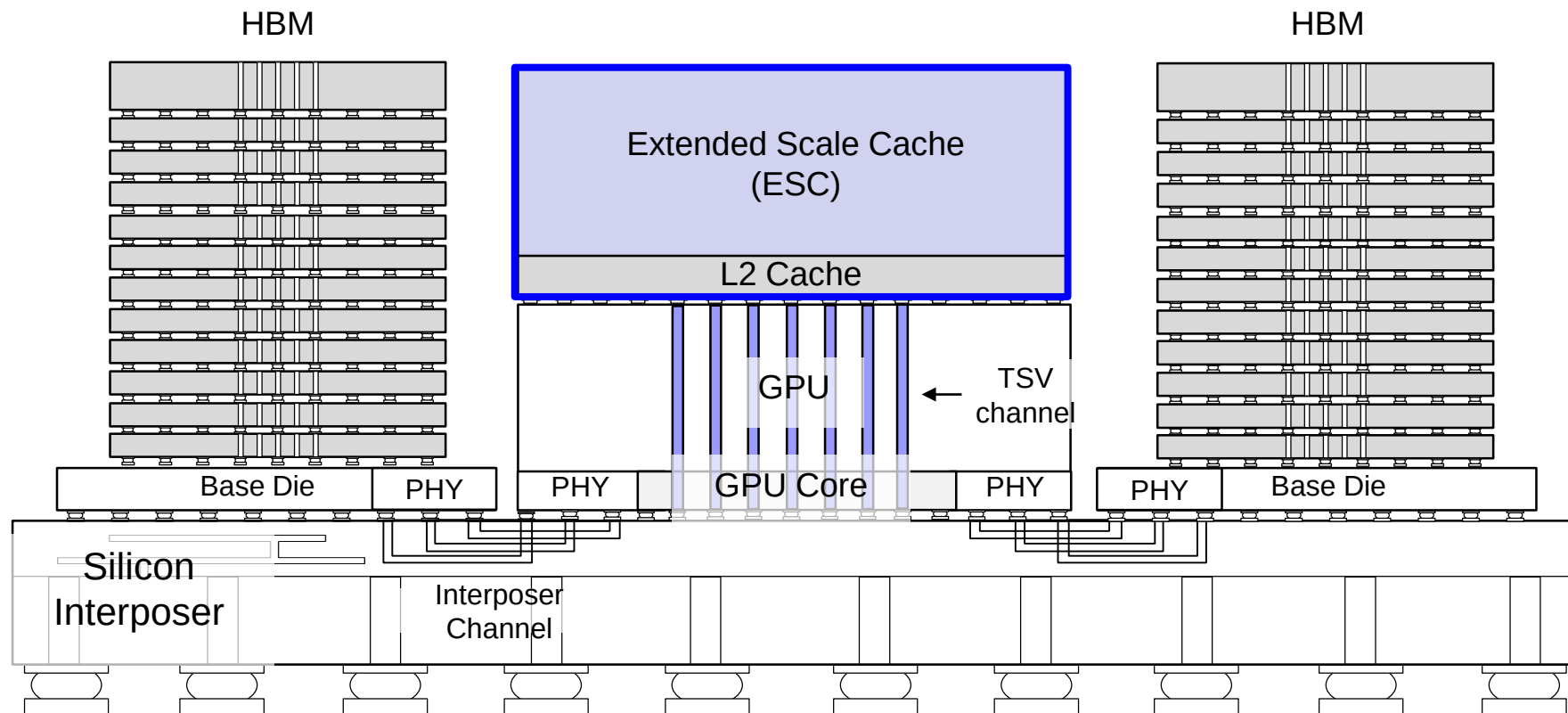


< 3D Integrated NMC-HBM Architecture above HBM DRAM >

- By integrating a NMC processor die and cache die above HBM, the proposed 3D NMC-HBM achieves high performance and energy efficient computing through dedicated TSV interconnection and power supply network.

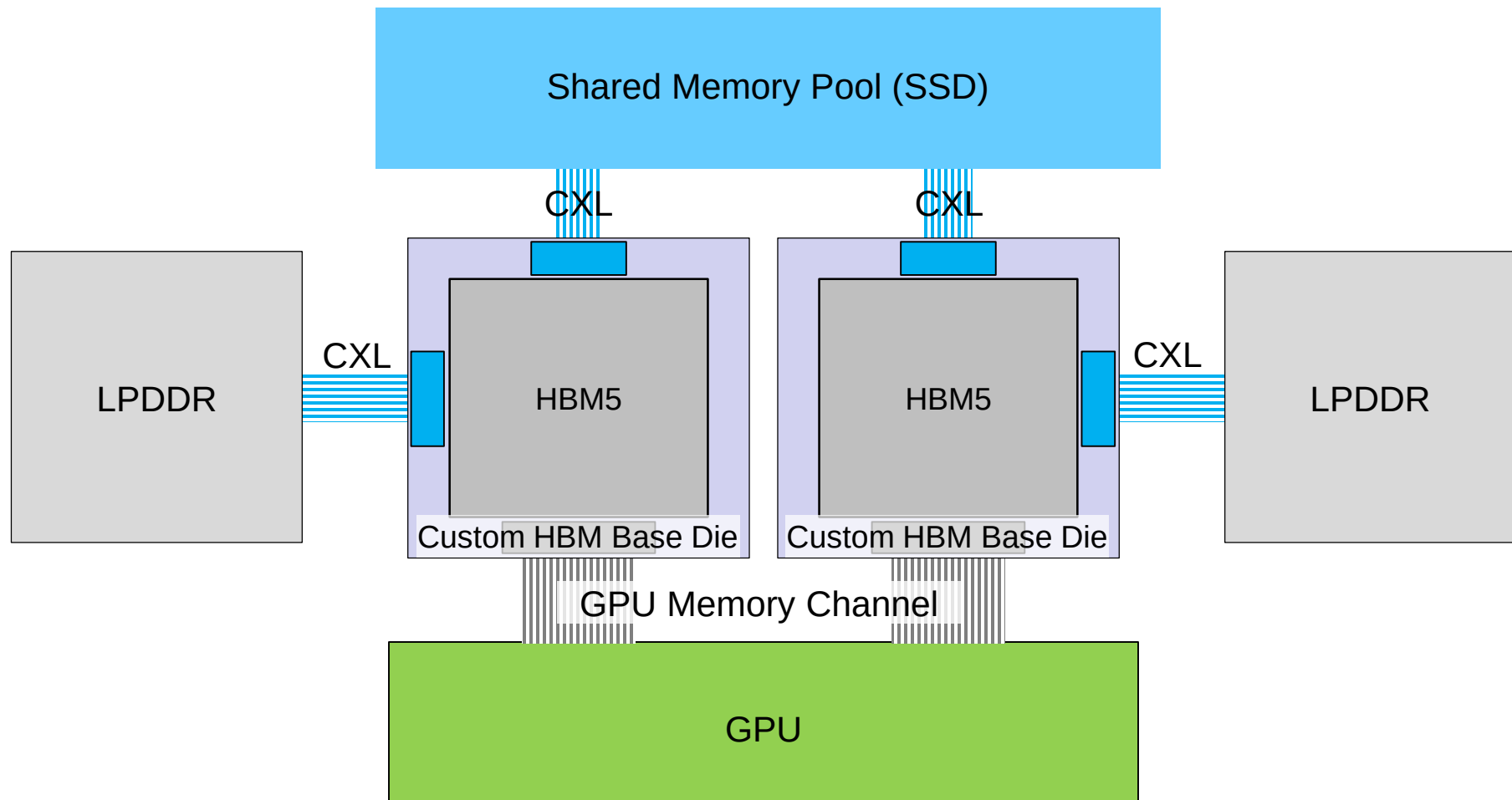


< Designated Decoupling Capacitor Die in 3D Stacked NMC-HBM5 Architecture >



< Extended Scale Cache (ESC) Stacked-GPU HBM Architecture >

- The proposed architecture utilizes an extended L2 cache, which is the last-level cache of the GPU, stacked above the GPU integrated using through silicon vias (TSVs).



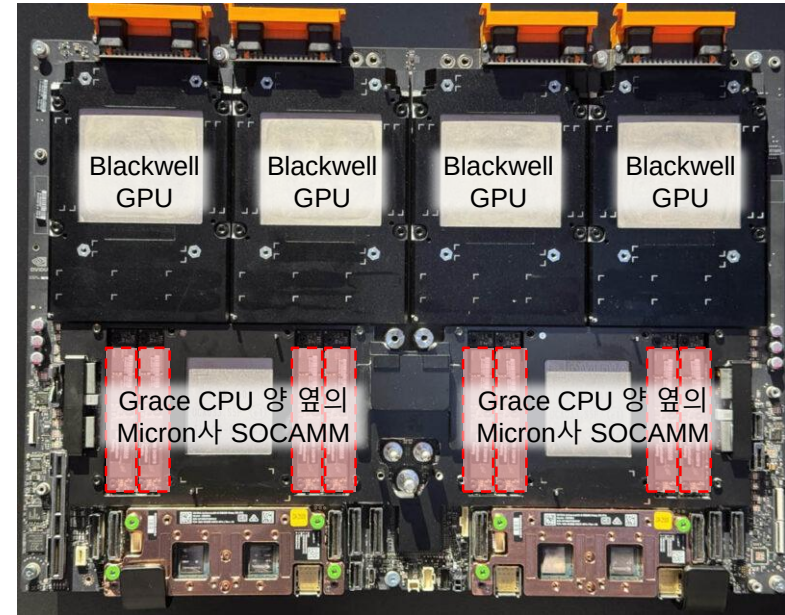
< HBM5 Architecture with CXL >

- The custom base die of HBM5 enables direct access to shared memory pool through CXL, providing a single unified memory with improved bandwidth and capacity.

Introduction to Small Outline Compression Attached Memory Modules (SOCAMM)



< (상) Micron 128GB SOCAMM 모듈,
(하) SK hynix 시연용 SOCAMM 모듈 >

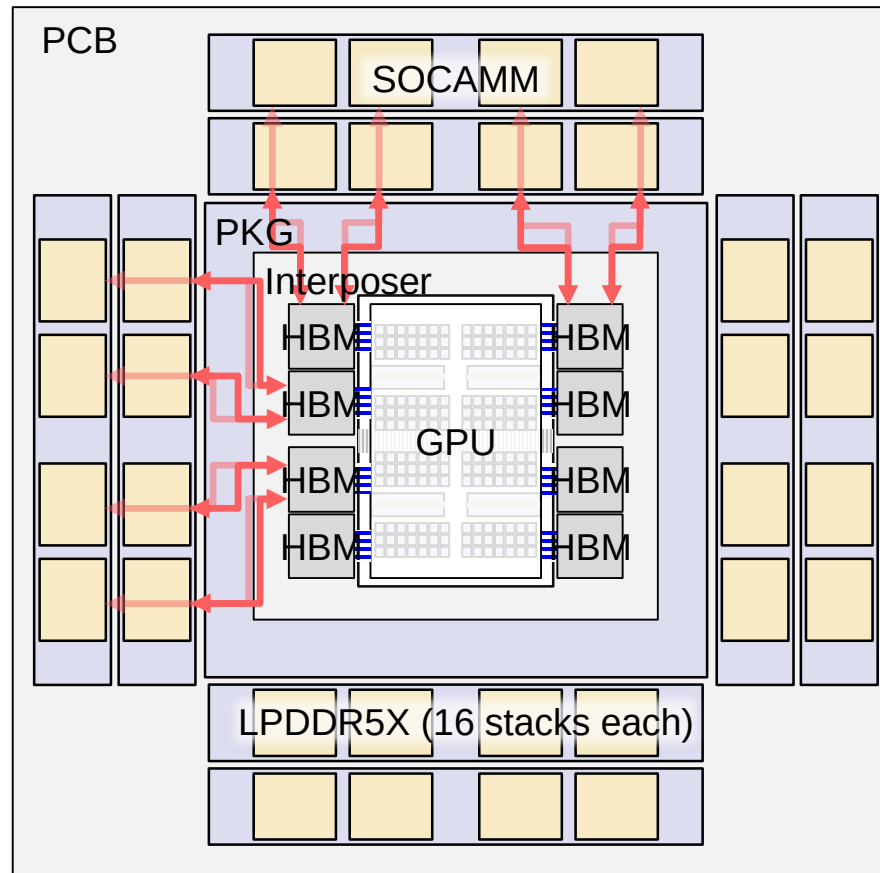


< SOCAMM 탑재 Grace Blackwell 기반 NVIDIA GB300 보드 >

- SOCAMM:
 - ✓ Small Outline Compression Attached Memory Modules
 - ✓ CAMM2 표준 기반의 차세대 메모리 모듈로, 슬림한 폼팩터와 슬롯형 장착 구조를 제공
 - ✓ LPDDR5X 모듈형 메모리 폼팩터로, 모바일용 저전력 DRAM(LPDDR)의 이점을 서버에 도입
- NVIDIA 주도로 개발되었으며, Grace Blackwell 플랫폼에 최초 적용
 - ✓ Micron: GB300 Grace Blackwell Ultra Superchip에 SOCAMM 탑재
 - ✓ SK Hynix, Samsung: 개발 중 및 프로토타입 테스트 단계

SOCAMM이 포함된 GPU 구조 제안 (2) HBM + SOCAMM (Off-interposer)

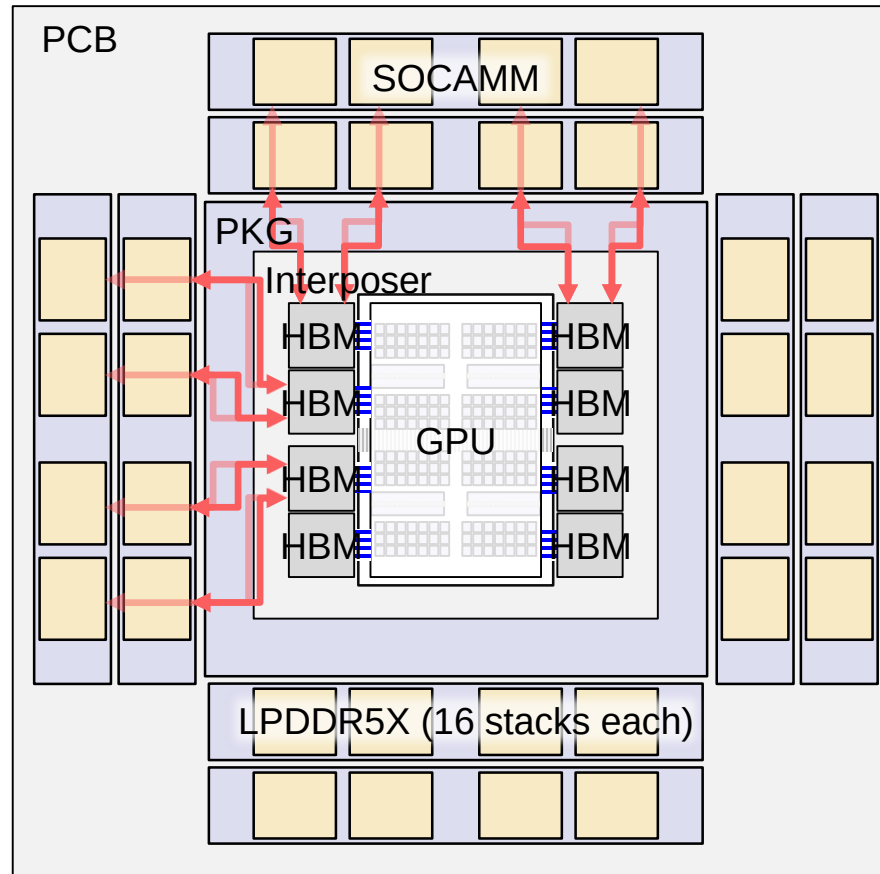
Interposer routing
PCB routing



< Proposal of GPU-HBM-SOCAMM System: HBM-Centric Architecture with Off-Interposer SOCAMM Expansion >

- HBM은 인터포저 위에, SOCAMM은 인터포저 외부(PCB)에 배치된 구조로, 고대역폭과 대용량을 동시에 확보하는 아키텍처
- GPU-HBM 간은 고속 인터포저 연결, SOCAMM은 PCB 라우팅을 통해 추가 확장 가능

 Interposer routing
 PCB routing



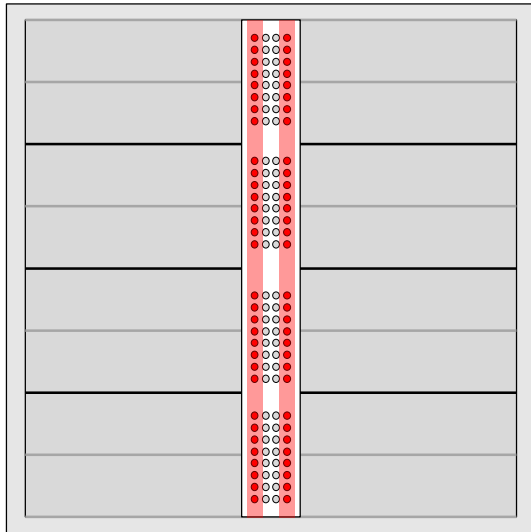
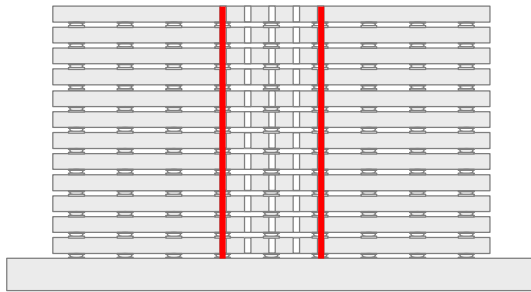
< Proposal of GPU-HBM-SOCAMM System: HBM-Centric Architecture with Off-Interposer SOCAMM Expansion >

- HBM은 인터포저 위에, SOCAMM은 인터포저 외부(PCB)에 배치된 구조로, 고대역폭과 대용량을 동시에 확보하는 아키텍처
- GPU-HBM 간은 고속 인터포저 연결, SOCAMM은 PCB 라우팅을 통해 추가 확장 가능

Next-Generation HBM Roadmap : Distributed Power + Thermal HBM TSV Array Placement

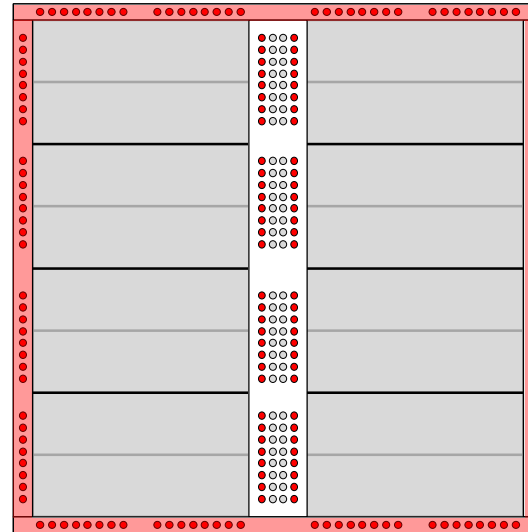
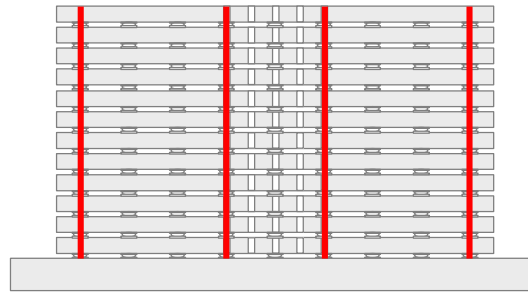
**HBM5
Architecture**

HBM3



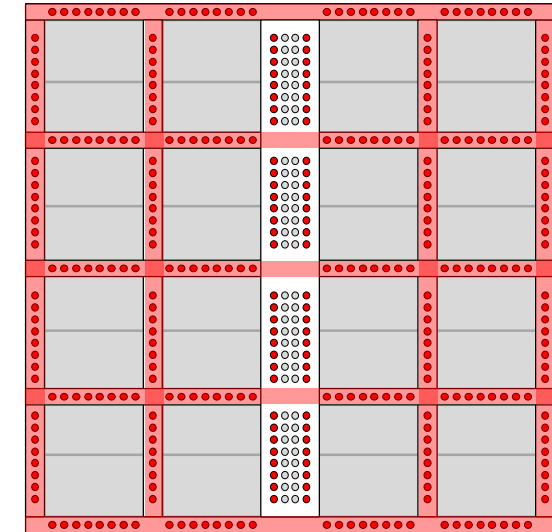
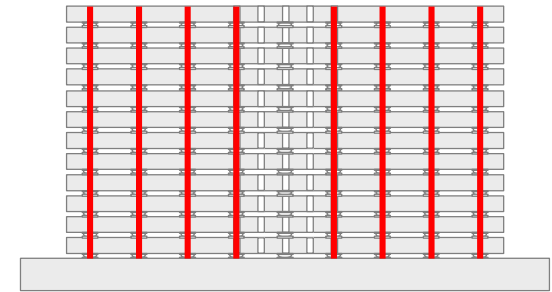
Center Array
P/G + Thermal TSV

HBM4

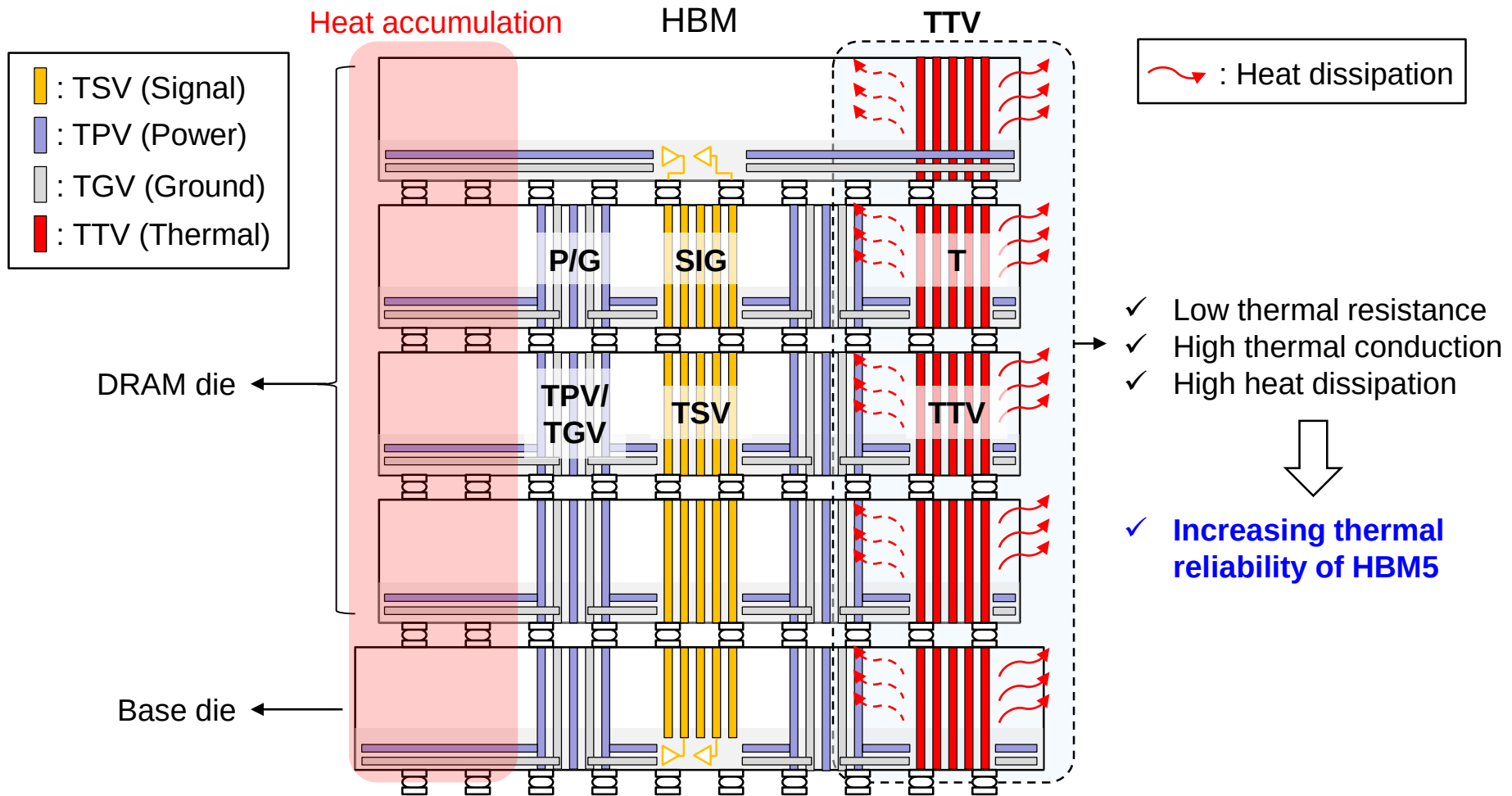


Center Array + Ring
P/G + Thermal TSV

HBM5 ~ HBM8



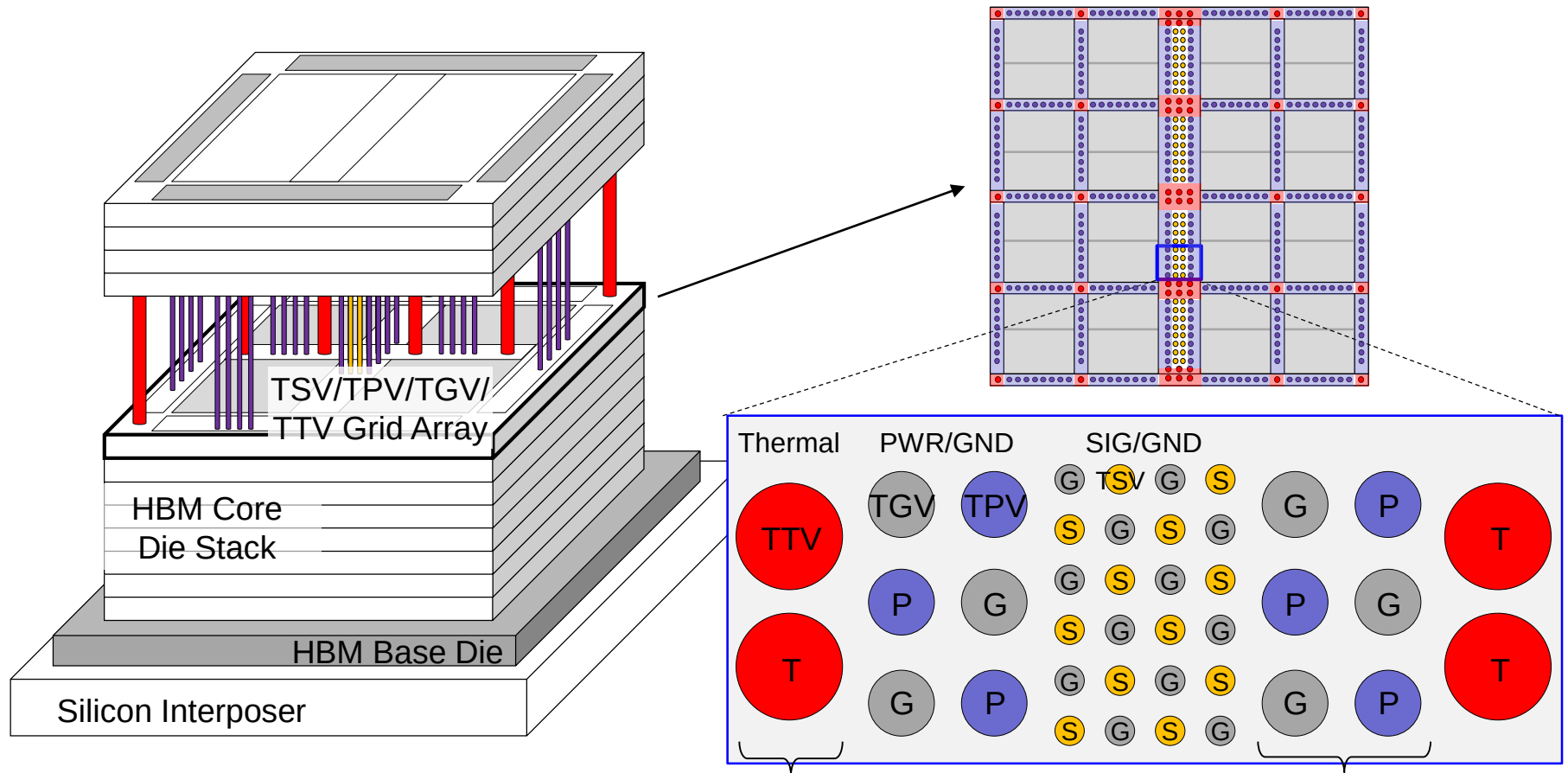
Distributed Grid Array
P/G + Thermal TSV



< Through Thermal Via (TTV) for Increasing Heat Dissipation in HBM5 >

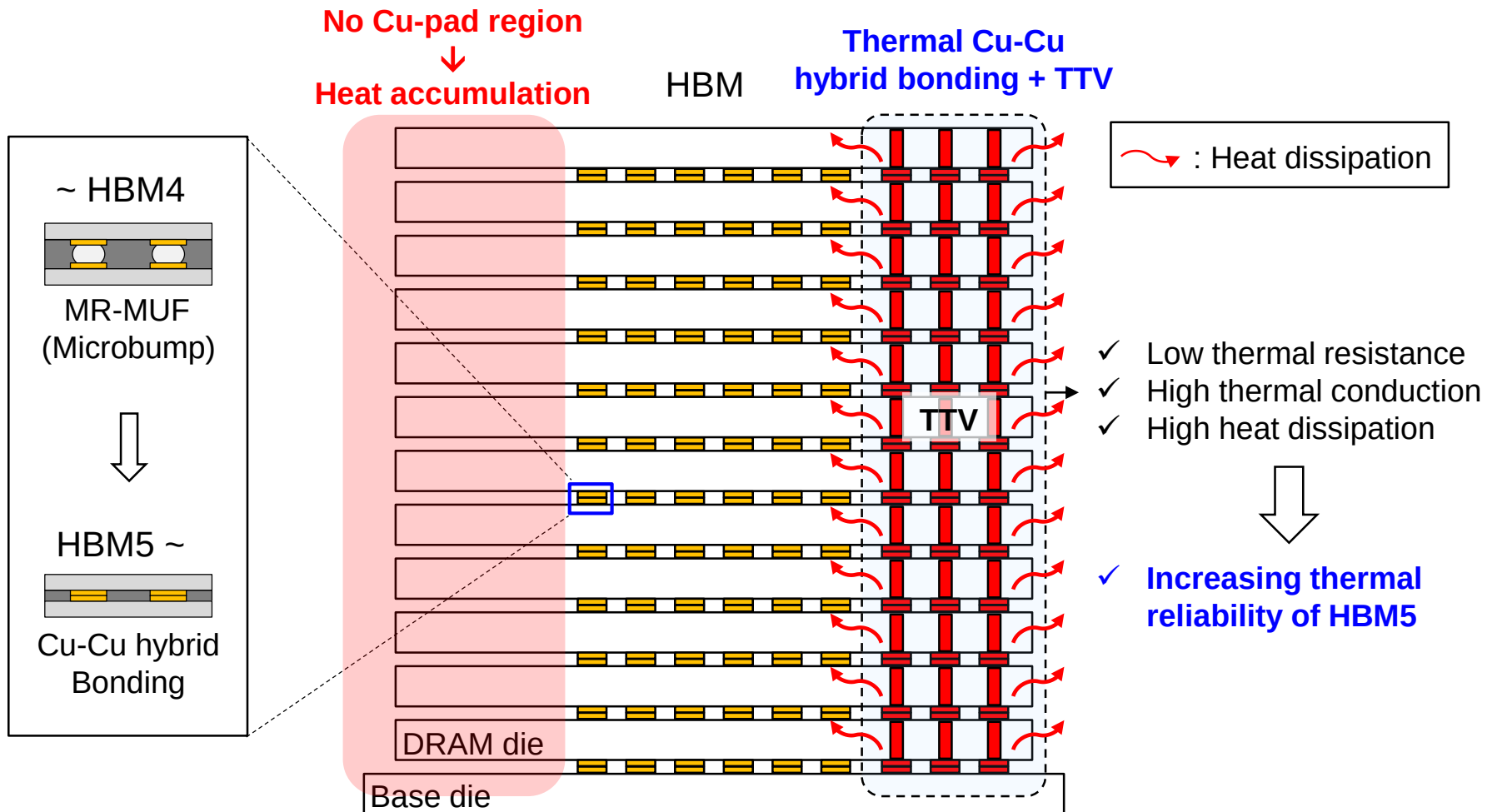
- Through Thermal Vias (TTVs) are implemented in the unused areas of the HBM silicon substrate, excluding the regions occupied by signal, power, and ground vias.

Distributed S/P/G/T Via Grid Array at HBM5



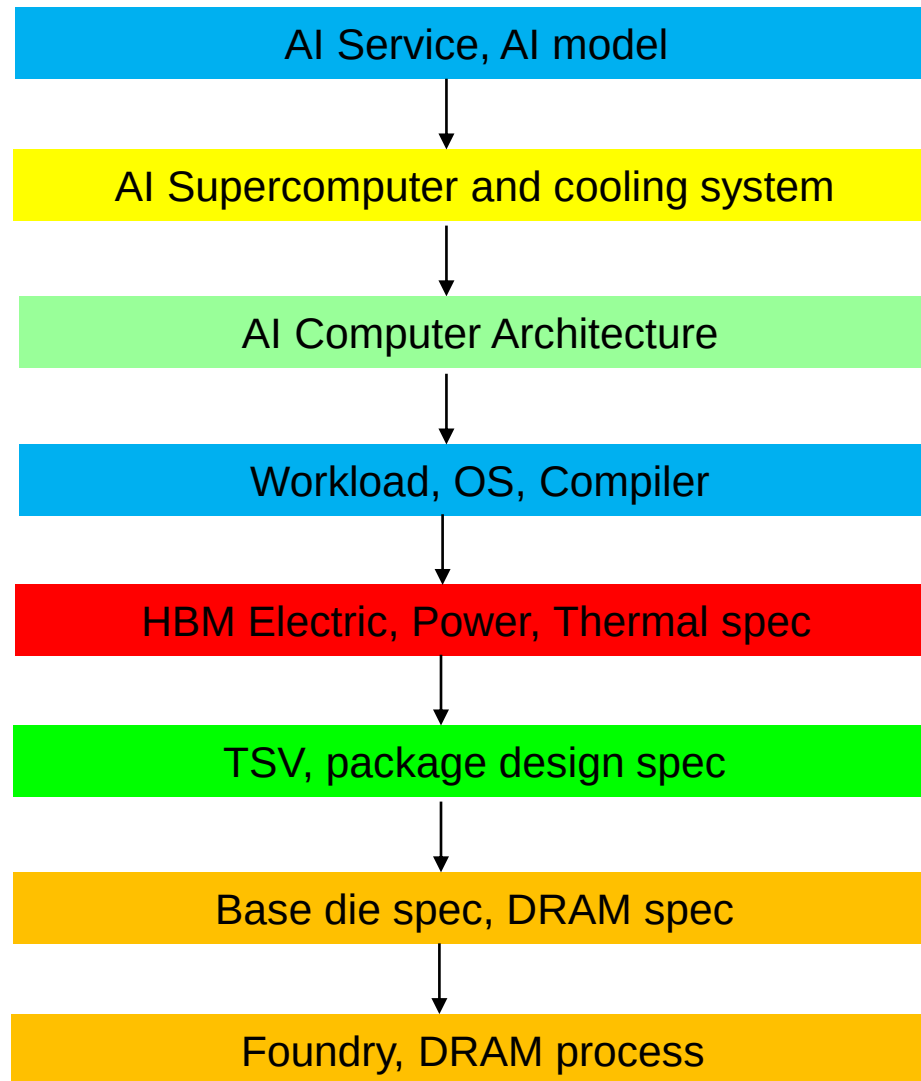
- ✓ Low thermal resistance
- ✓ High thermal conduction
- ✓ Low inductance
- ✓ Low electrical resistance

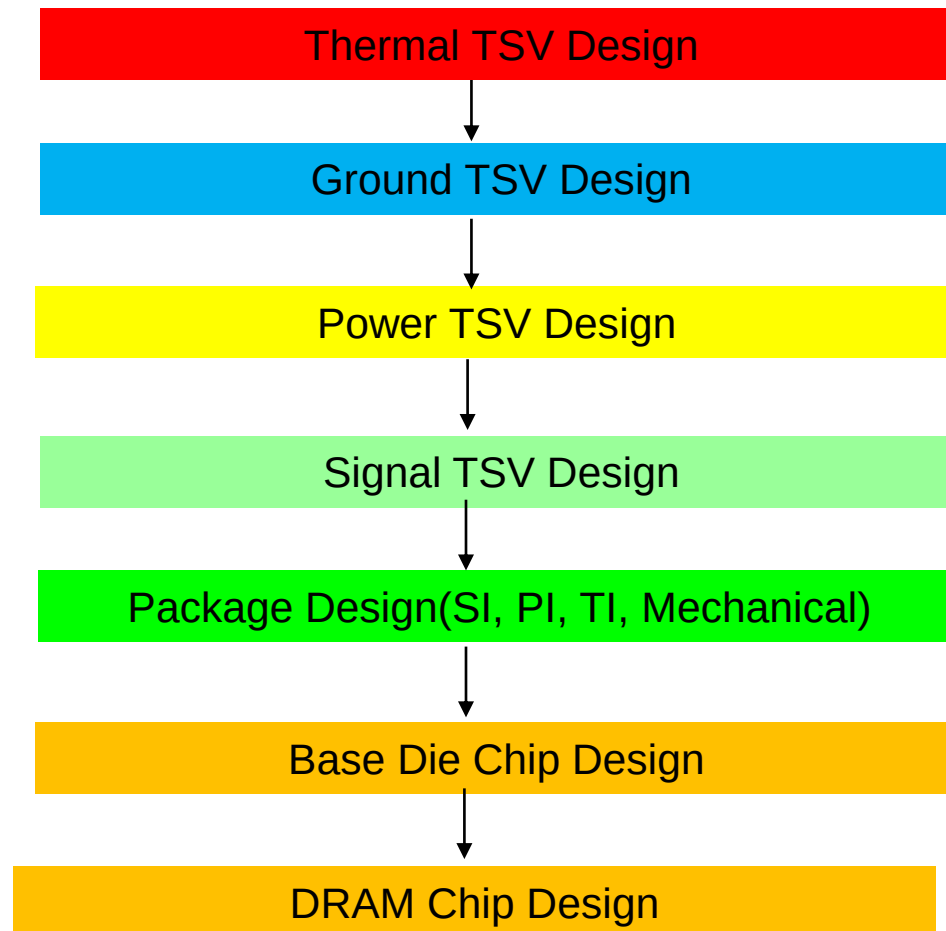
< Asymmetric Distributed TSV/TPV/PGV/TTV Grid Array Design for HBM5 >

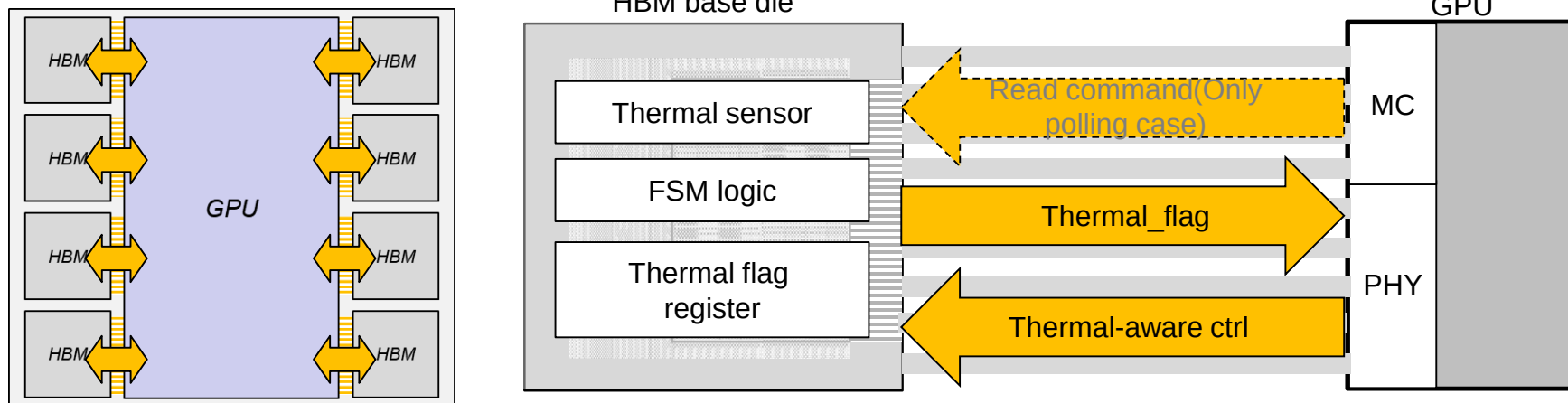


< Thermal Cu-Cu Hybrid Bonding with TTVs for Thermal Reliability in HBM5 >

- Thermal Cu-Cu hybrid bondings are implemented on no Cu-pad (no bump) region of HBM except S/P/G Cu-pad.
- Thermal Cu-Cu hybrid bondings are interconnected to TTVs to maximize heat dissipation of HBM.





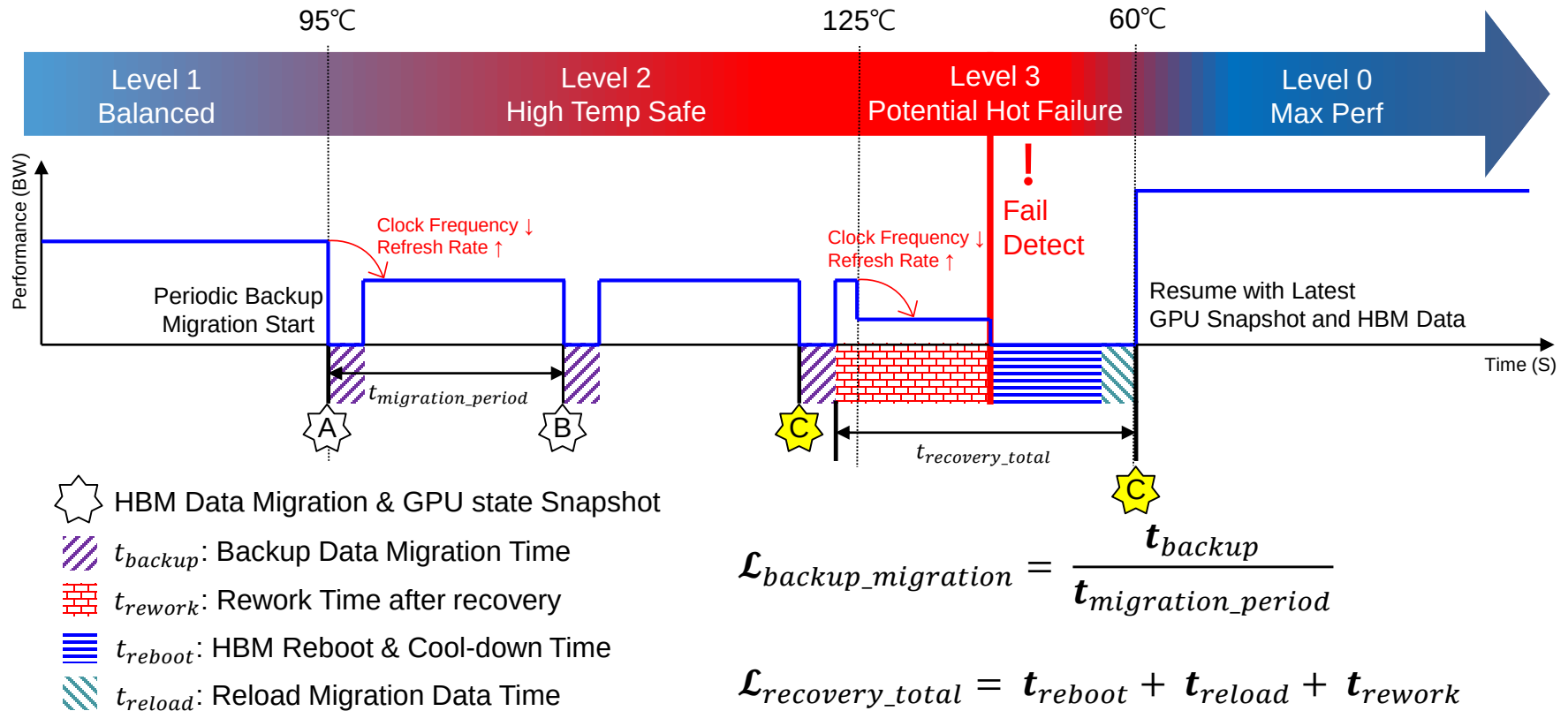


<HBM base die thermal sensing and FSM-based control flag interface>

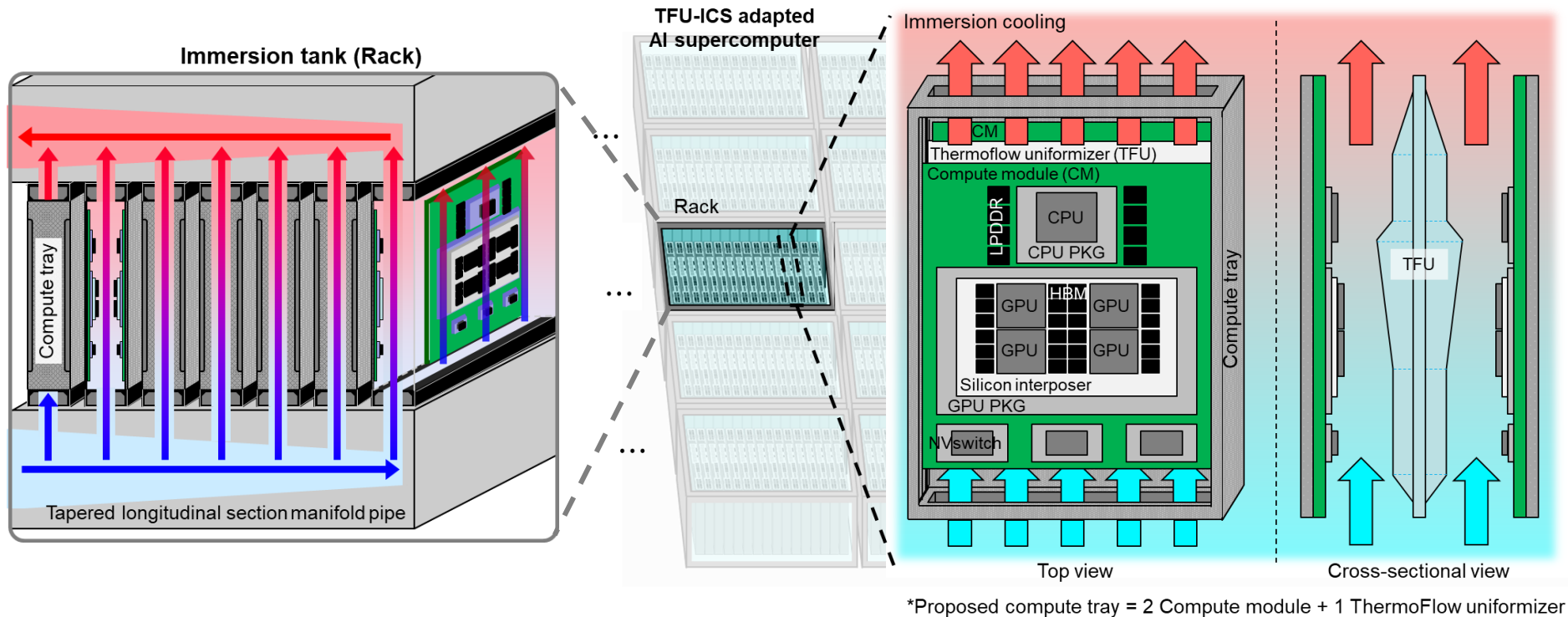
- HBM base die에는 Thermal sensor와 FSM이 통합되어 있으며, 온도를 실시간으로 모니터링해 thermal level (Level 0/1/2)을 판단하고 해당 상태에 따라 내부 thermal flag register에 값을 기록함
 - ✓ Base die 내 thermal-aware block 예상 면적: 약 0.02–0.05 mm² 이하 < 0.5% 추정
- GPU 내 Memory Controller는 다음 두 가지 방식 중 하나로 thermal flag를 인지하여 정책을 적용할 수 있음
 - ✓ Polling 방식: Controller가 HBM command bus를 통해 주기적으로 register를 읽는 구조
 - ✓ Push 방식: Base die FSM에서 상태 변화 시 thermal_flag를 data path에 실어 controller로 즉시 전달 (interrupt-like signaling)

→ 이 구조는 JEDEC burst protocol을 유지하면서도, thermal 대응을 위한 light-weight signaling 경로로 사용 가능함.

HBM Shutdown and Recovery Strategy - Common



- Thermal Level 2 or 3: Periodic Backup Migration Enable (to HBF/LPDDR)
- Aggressive DFS로 제어되지 않는 Thermal이 Fail을 유발하는 경우 (ECC Detect)
 - Step 1) HBM Reboot & Cool-down
 - Step 2) Reload Latest Backup Data to HBM (from HBF/LPDDR)
 - Step 3) Reload Latest GPU State Snapshot → **Resume**



< Concept of the proposed immersion cooling structure for GPU-HBM module >

- The proposed Thermal-Flow-Uniformized (TFU)-based immersion cooling structure provides a uniform cooling solution to the GPU-HBM module by submerging it in an immersion tank.
- The proposed TFU is designed to ensure uniform flow in order to maintain temperature uniformity across the high-dense GPU-HBM module.

Part5: Key Features in HBM6

Key Features of HBM6

1. Electrical Specification

- Data Rate : 16 Gbps
- Number of I/Os : 4,096
- Total Bandwidth : 8.0 TB/s
- Number of die stack : 16/20-Hi
- Capacity/die : 48 Gb

2. Packaging/Cooling Method

- Bump-less Cu-Cu Direct Bonding
- Immersion Cooling

3. HBM Architecture

- Custom Multi-tower HBMs
- Active / Hybrid (Silicon+Glass) Interposer
- Network Switch + Bridge Die
- BS-PDN

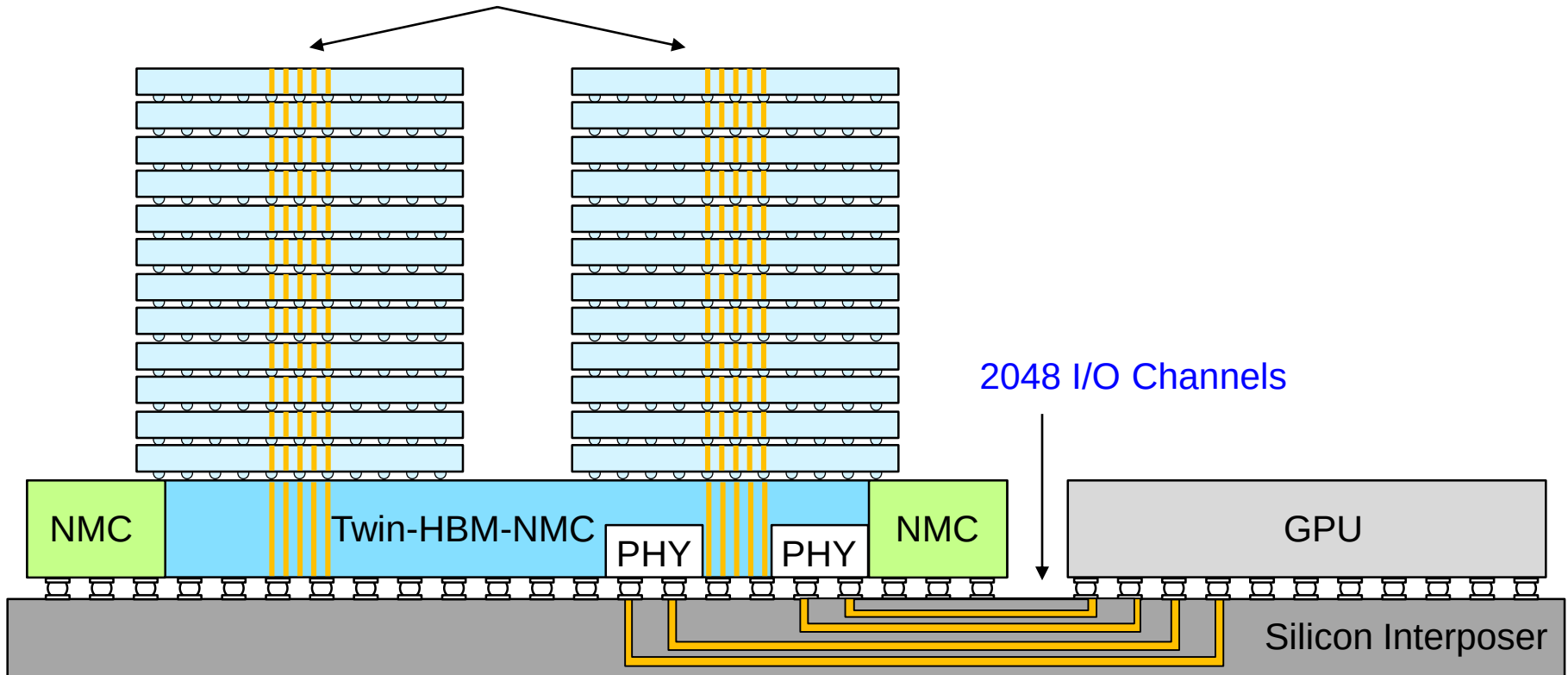
4. AI Design Agent

- Hybrid Equalizer + Generative AI based SI/PI Metric Estimation
- AGI (Artificial General Intelligence)

Proposal of Twin Tower High Bandwidth Memory with Near-Memory Computing (Twin-HBM-NMC) Architecture

HBM6
Architecture

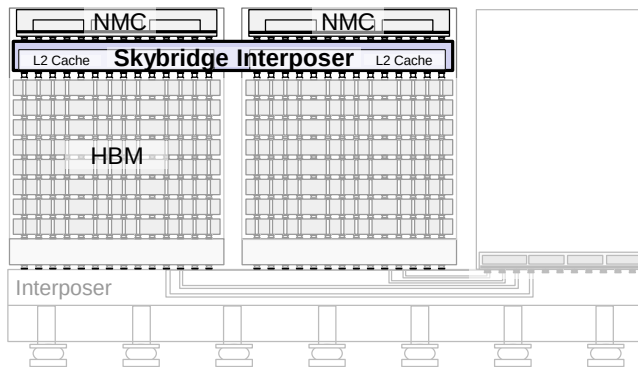
Two DRAM Stacks



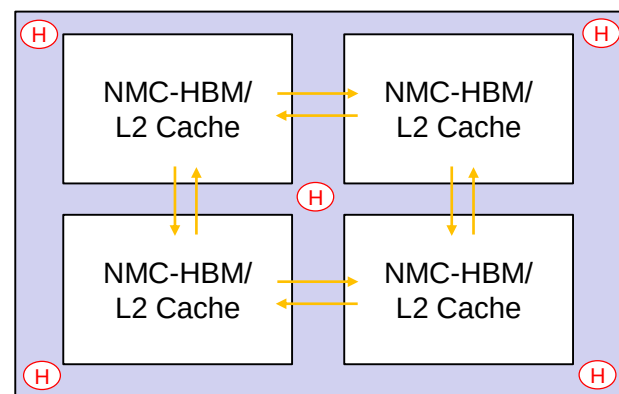
< Overview of Twin Tower HBM with NMC Architecture >

- In twin tower high bandwidth memory with near-memory-computing (Twin-HBM-NMC) architecture, two DRAM stacks are located on top of the large logic die.
- The logic die include NMC units and is connected to the GPU via 2048 interposer channels.

Multi HBM-HBM Skybridge Interposer for Efficient Near Memory Computing Performance



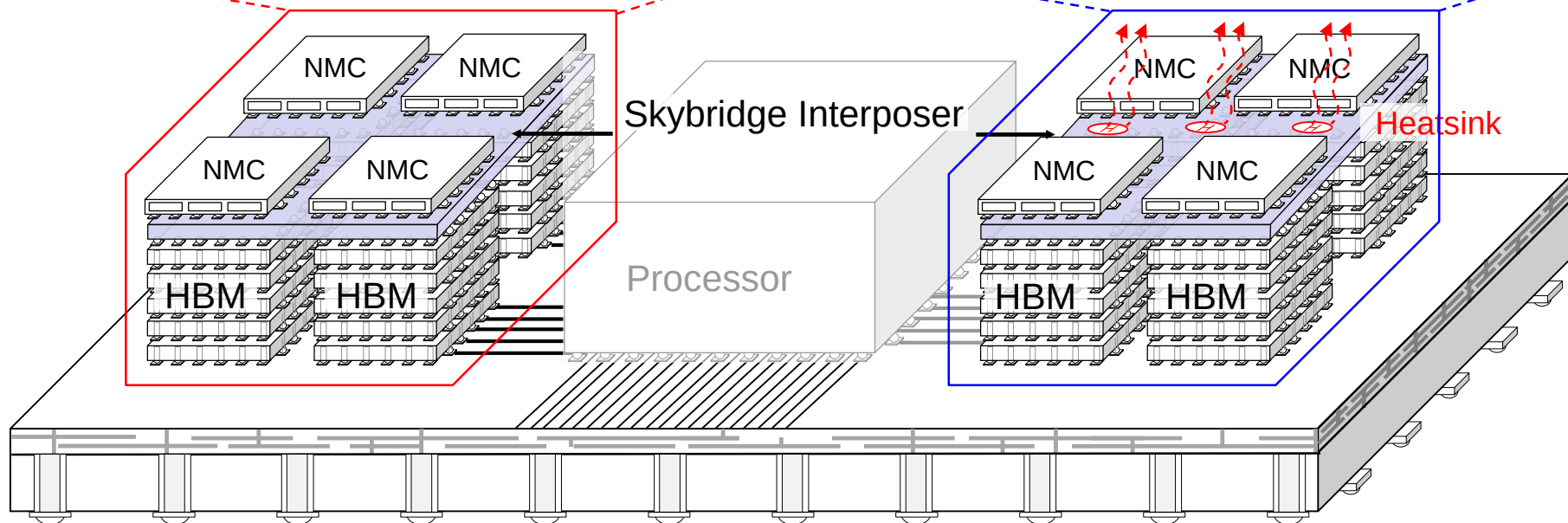
< Side view of Skybridge Interposer Architecture >

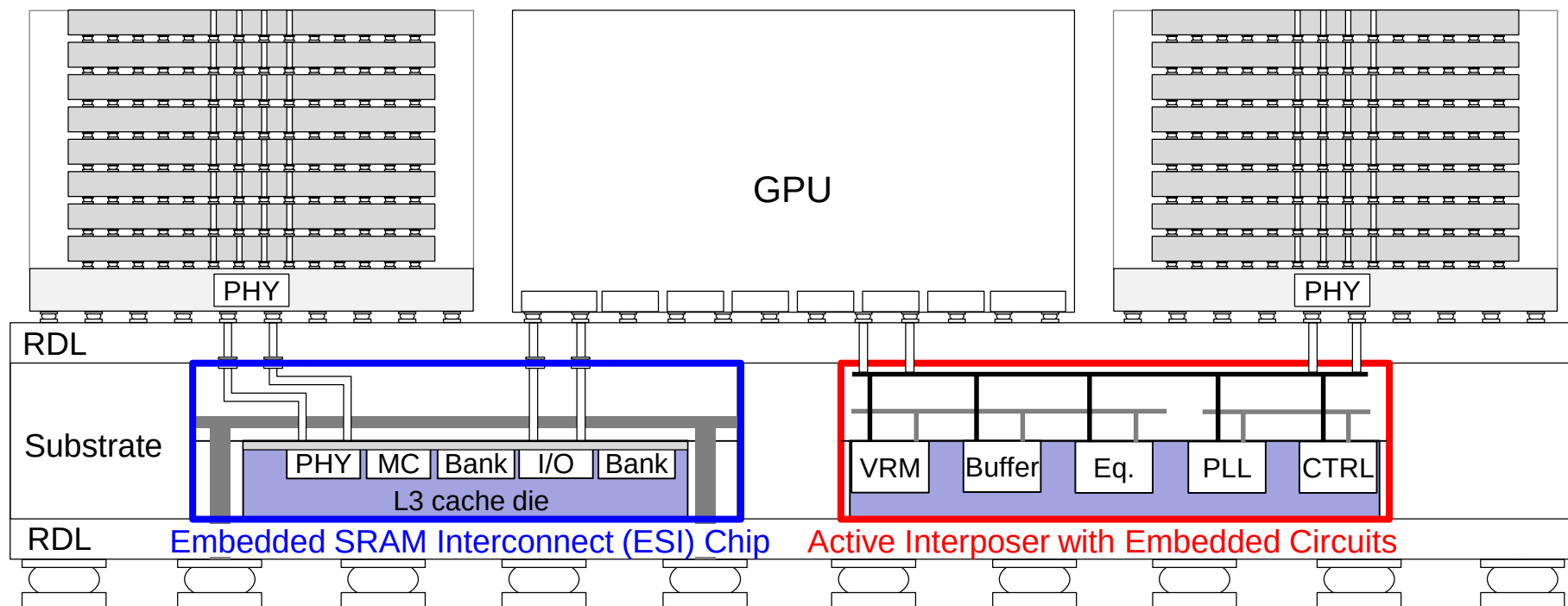


< Top view of Skybridge Interposer >

Through
Skybridge
Interposer...

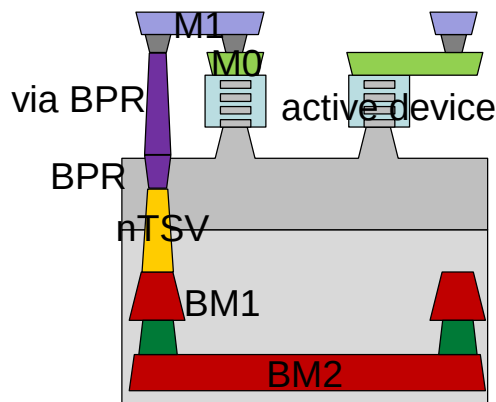
- ✓ NMC-NMC
- ✓ HBM-HBM
- ✓ NMC-HBM
- ✓ L2 cache
- ✓ Heatsink



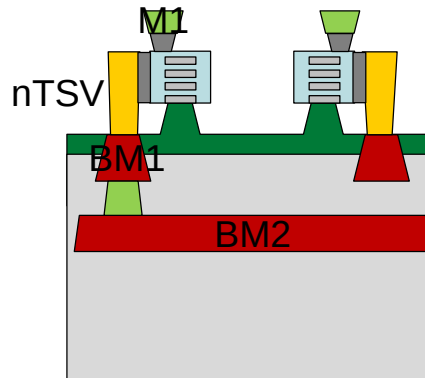


< Embedded GPU-HBM Architecture with Active Interposer >

- The local silicon interconnect (LSI) die and L3 cache die is hybrid bonded to form embedded SRAM interconnect (ESI) chip, which is placed inside the interposer using COWOS-L technology.
- Embedded circuit components including VRM, equalizer, PLL, and control circuits are embedded in active interposer to enhance electrical performance.

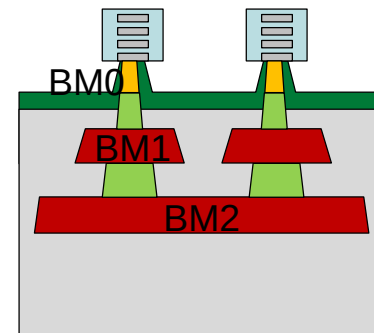


Buried Power Rail



Power Via

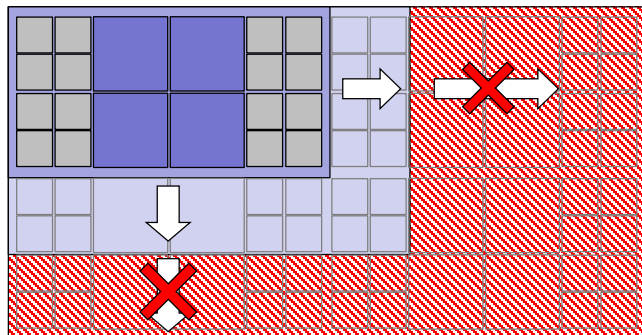
< 기본적인 BSPDN 구조 >



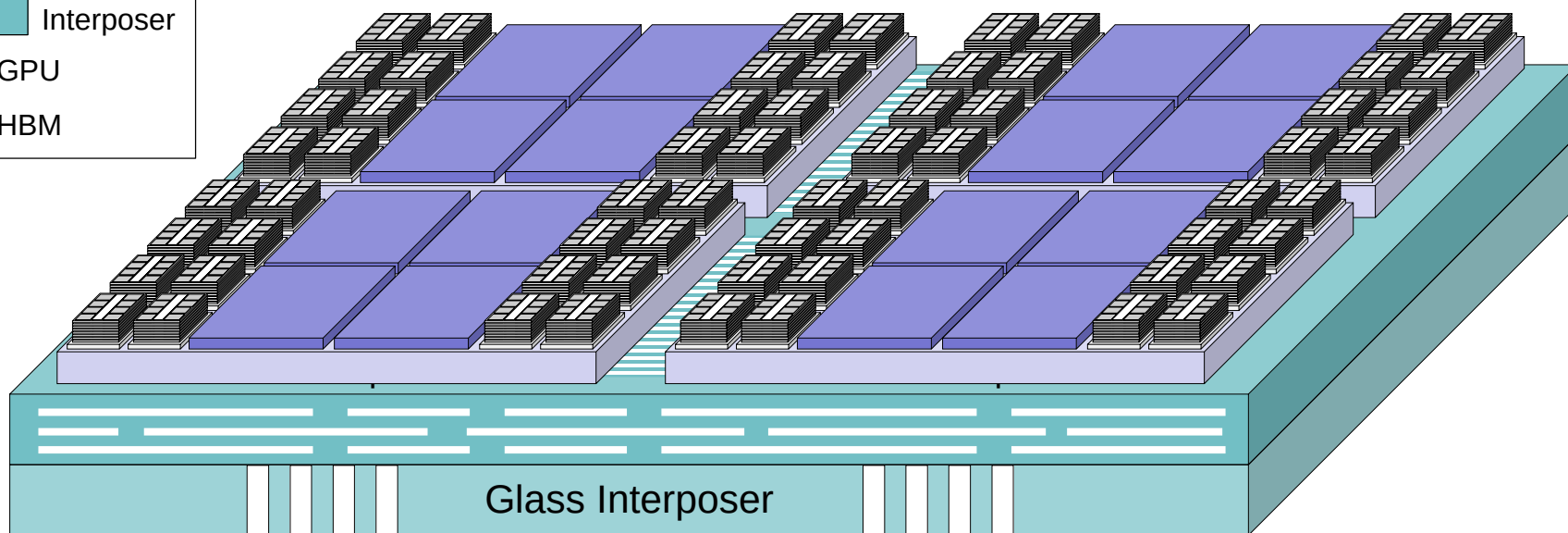
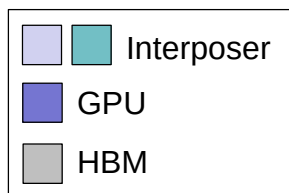
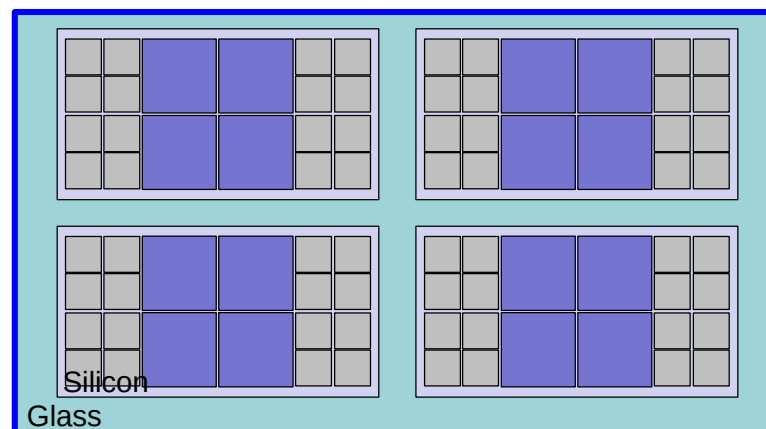
Super Power Rail

- 산업에서 개발된 Active 소자와 backside metal을 interconnection 하는 방식의 차이로 총 3가지 구조 Buried Power Rail(BPR), Power Via, Super Power Rail(SPR)로 나뉘어 짐
 - BPR : backside metal – nTSV - BPR – VBPR – frontside metal(M0) – active devices
 - Power Via : backside metal(BM1) – nTSV – active devices
 - SPR : backside metal(BM1) – BM0 – active devices

Limitation in increasing
silicon interposer die size

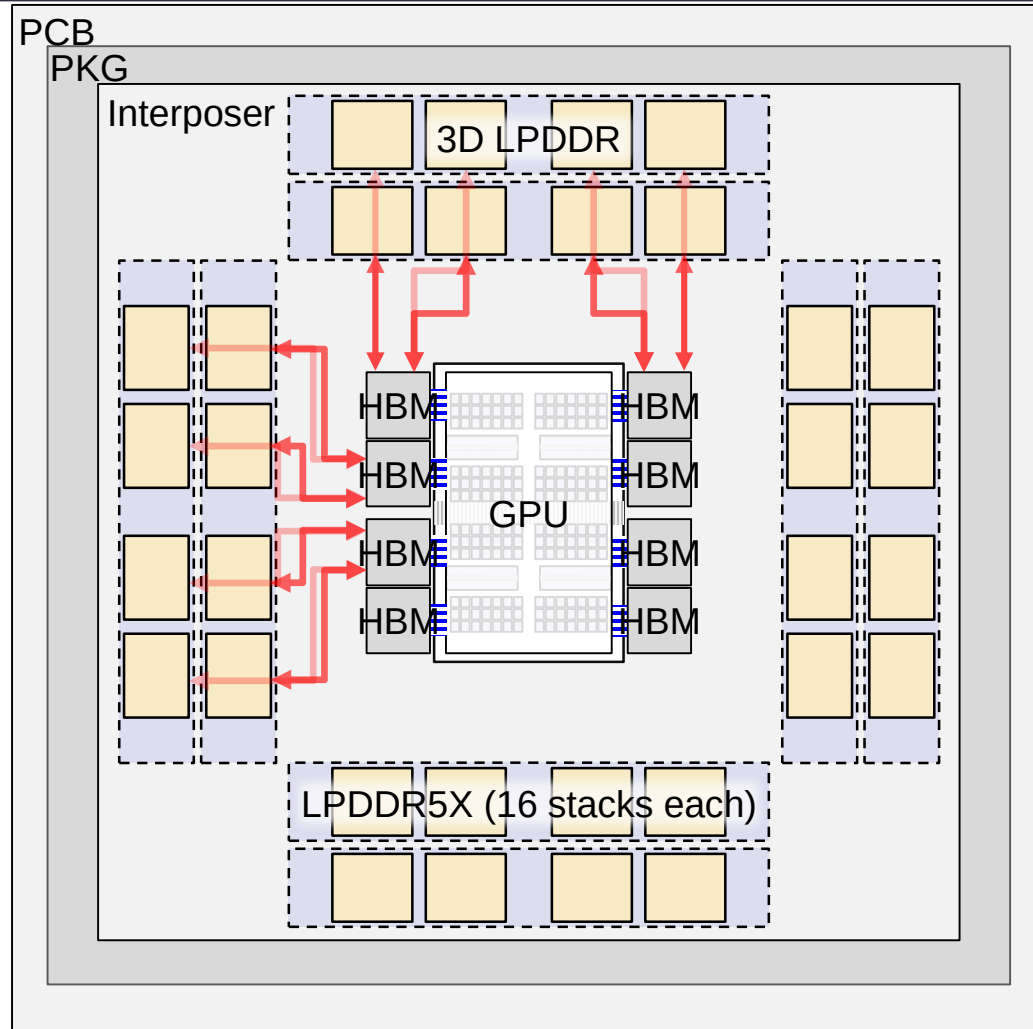


Large Scale Hybrid (Silicon+Glass) interposer



< Silicon-Glass Hybrid Interposer for Ultra Large-Scale Next Generation HBM >

 Interposer routing (HBM-GPU)
 Interposer routing (HBM-SOCAMM)



< Proposal of GPU-HBM-LPDDR5X System: Interposer-Integrated 3D LPDDR Expansion with SOCAMM >

- HBM과 GPU는 기존처럼 인터포저 상에서 직접 연결되며, LPDDR5X는 HBM을 통해 GPU와 연결되어 확장된 고용량·저전력 메모리 시스템 구성 가능

Part6: Key Features in HBM7

Key Features of HBM7

1. Electrical Specification

- Data Rate : 24 Gbps
- Number of I/Os : 8,192
- Total Bandwidth : 24.0 TB/s
- Number of die stack : 20/24-Hi
- Capacity/die : 64 Gb

2. Packaging/Cooling Method

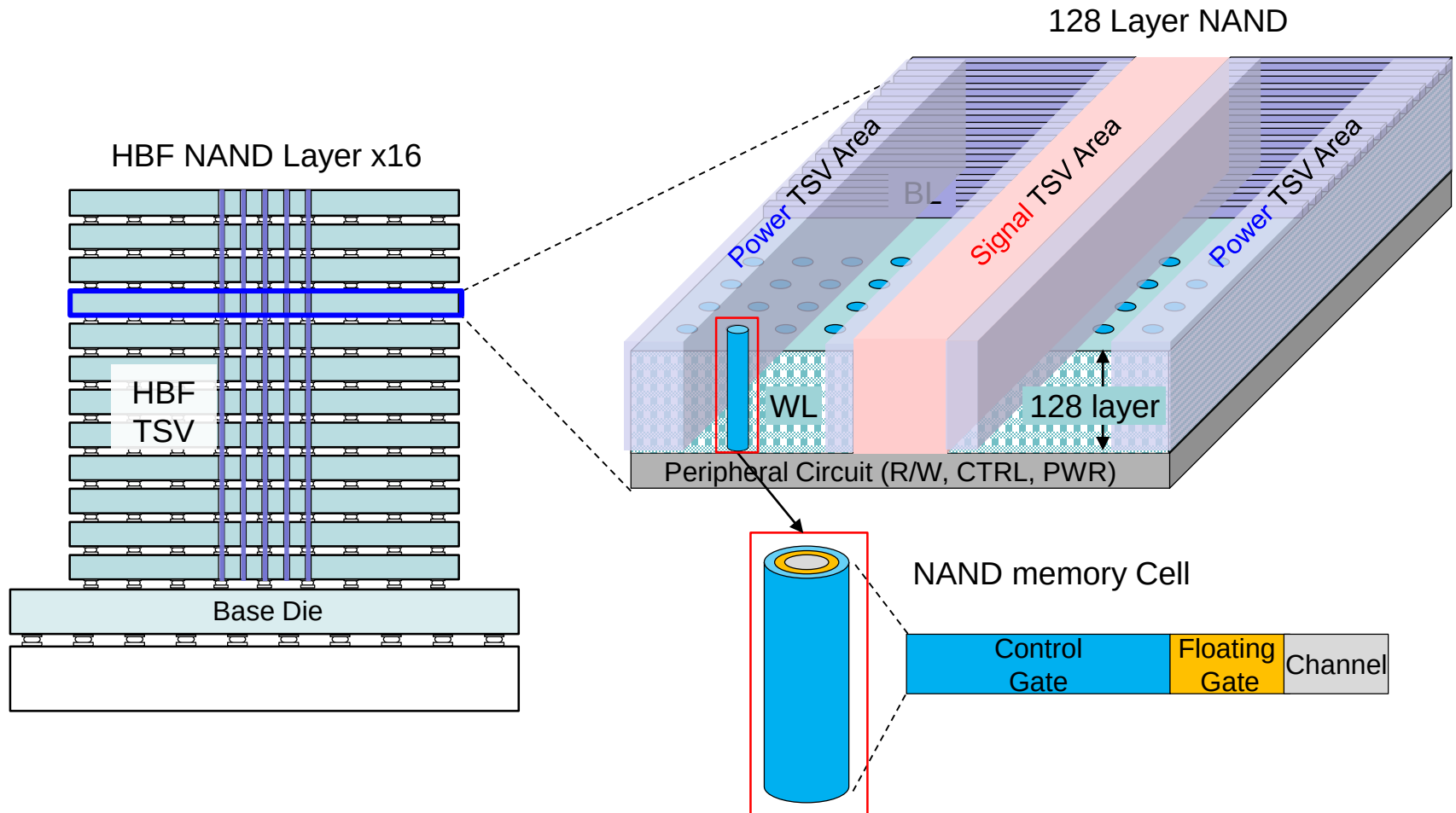
- Bump-less Cu-Cu Direct Bonding
- Embedded Cooling

3. HBM Architecture

- Hybrid HBM Architecture
- HBM-HBF
- HBM-LPDDR
- Buffer dies in HBM stack

4. AI Design Agent

- LLM based Human Interactive AI Design Agent

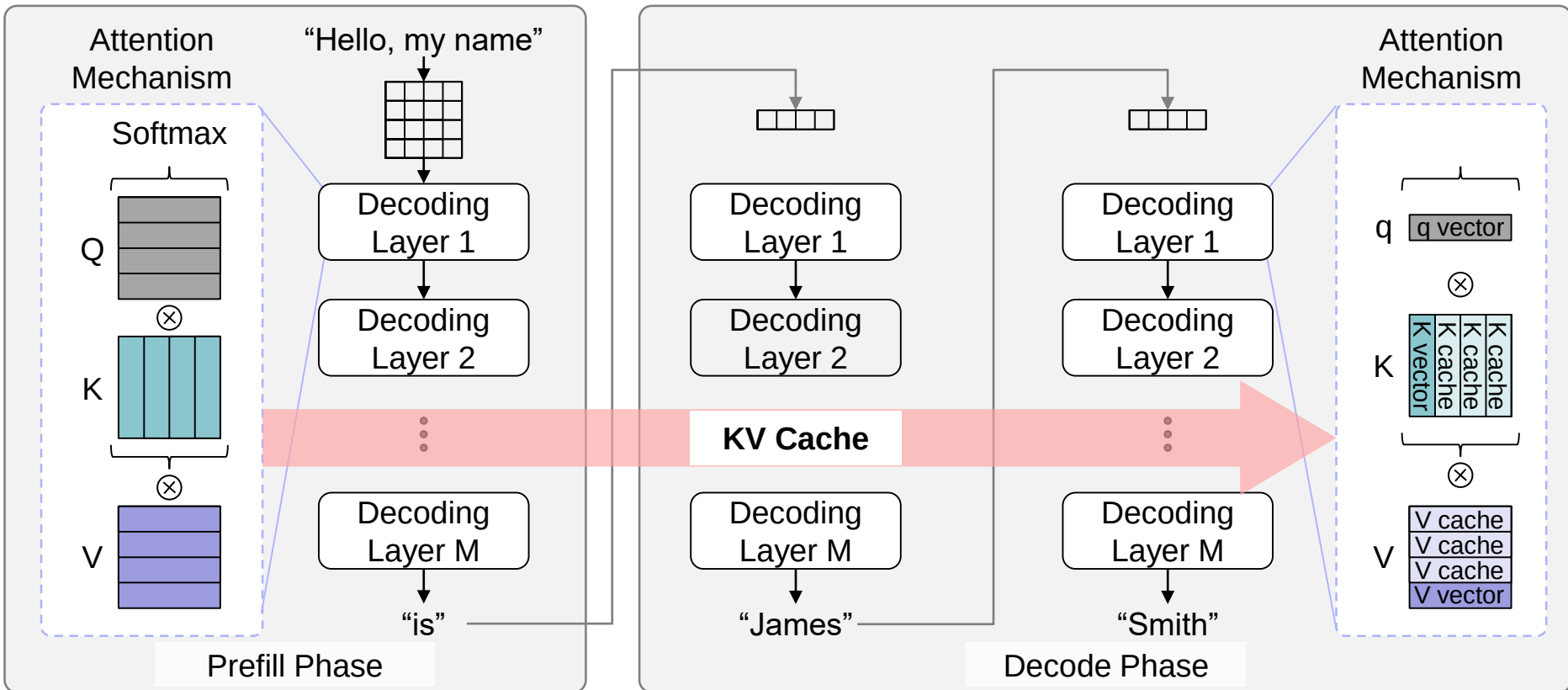


< High Bandwidth Flash (HBF) Architecture for Memory Intensive LLM Inference >

➤ 17.6 TB



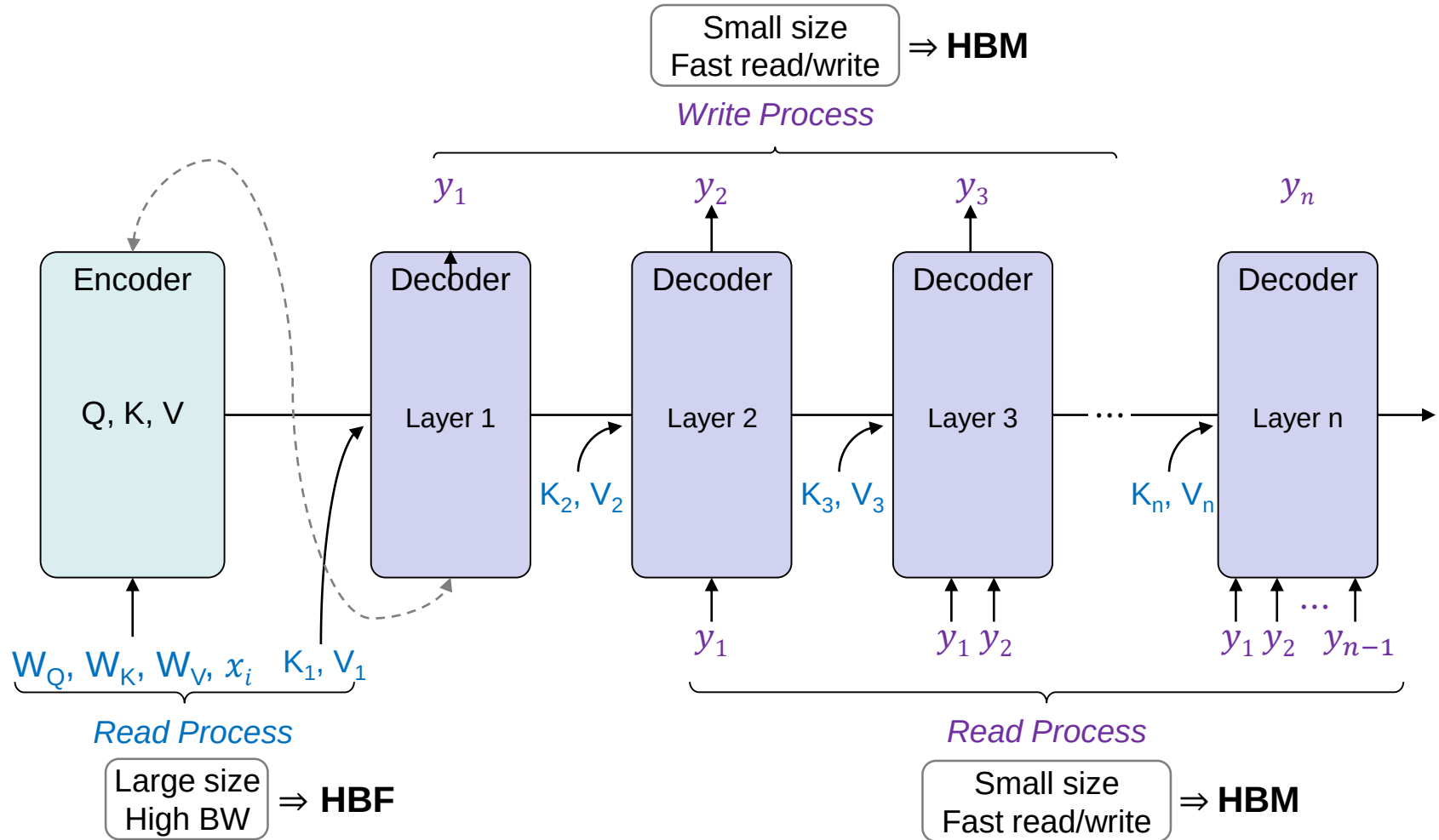
Transformer-based LLM Inference Pipeline and Acceleration via Reuse of KV Cache



< Reuse of KV Cache in Autoregressive LLM Inference >

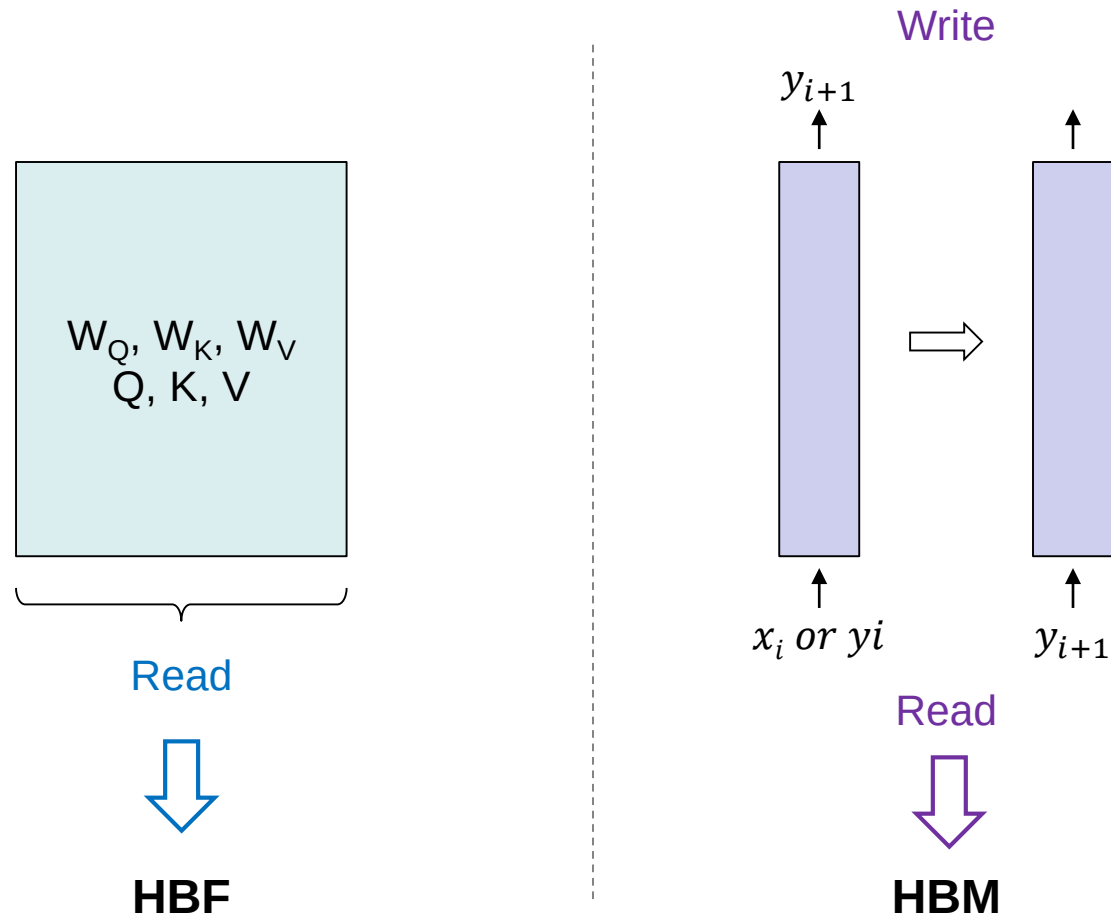
- Inference in LLMs follows an autoregressive process, typically divided into two phases:
- Prefill phase – processes the input prompt and stores key-value (KV) matrices.
- Decode phase – reuses stored KV cache to generate new tokens sequentially.
- The KV cache avoids redundant computation and significantly accelerates the inference process.

HBM-HBF Memory Allocation in Inference Pipeline



< Memory allocation strategy for encoding and decoding phases >

Memory Hierarchy Optimization for Autoregressive Inference



< Optimized memory bandwidth utilization through hierarchical placement >

Required Memory Capacity for LLM and sLLM Inference : Benchmark Model Weight & KV Cache

✓ Precision : FP16, size of FP16 = 2Byte

✓ Total size of KV cache in bytes = (batch_size) * (sequence_length) * 2 * (num_layers) * (dimension) * sizeof(FP16)

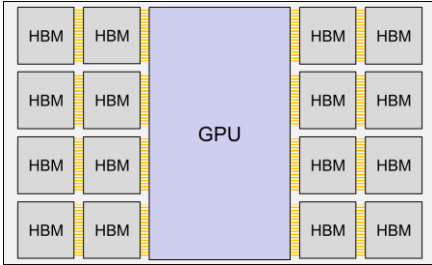
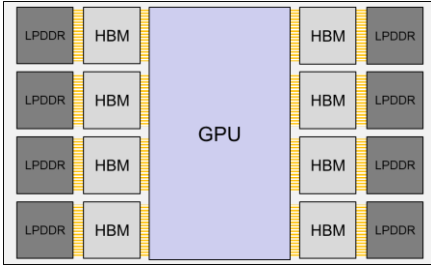
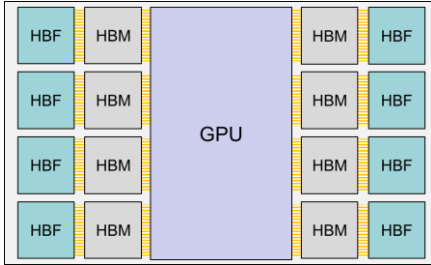
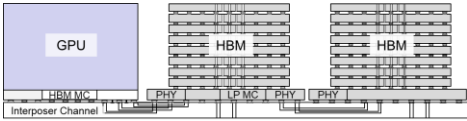
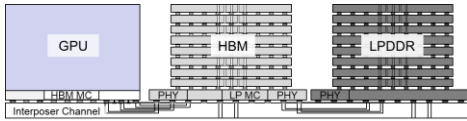
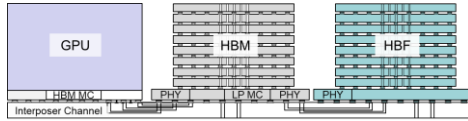
	LLM	sLLM
Benchmark Model	Llama3-400B	Llama3-8B
Model Dimension	16,384	4,096
Number of Layer	126	32
Required memory size for storing model weights (FP16)	800 GB	16 GB
Required memory size for storing KV Cache (Batch Size : 1, Sequence Length : 4,096)	33.8 GB	2.2 GB

< Required Memory Capacity for Storing LLM / sLLM Weight & KV Cache >

- Two Llama3 LLM models with different parameter sizes are chosen for LLM / sLLM memory capacity & latency evaluation during inference.

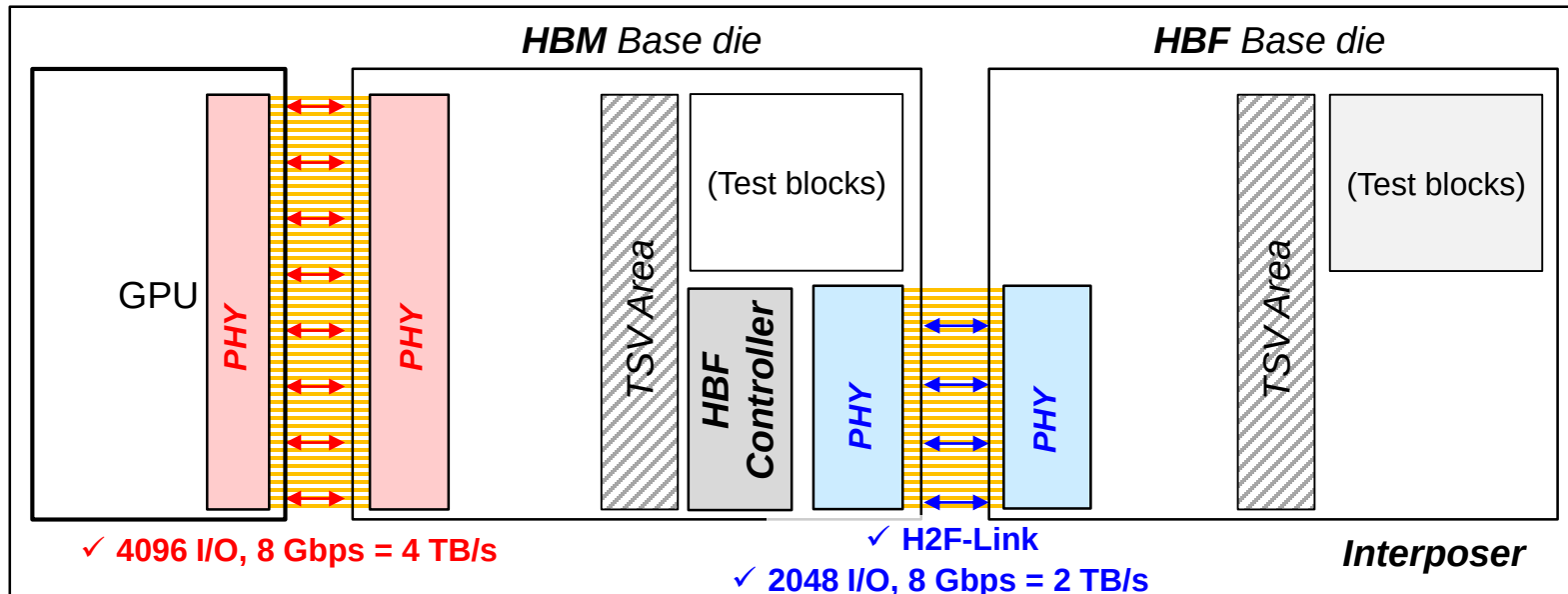
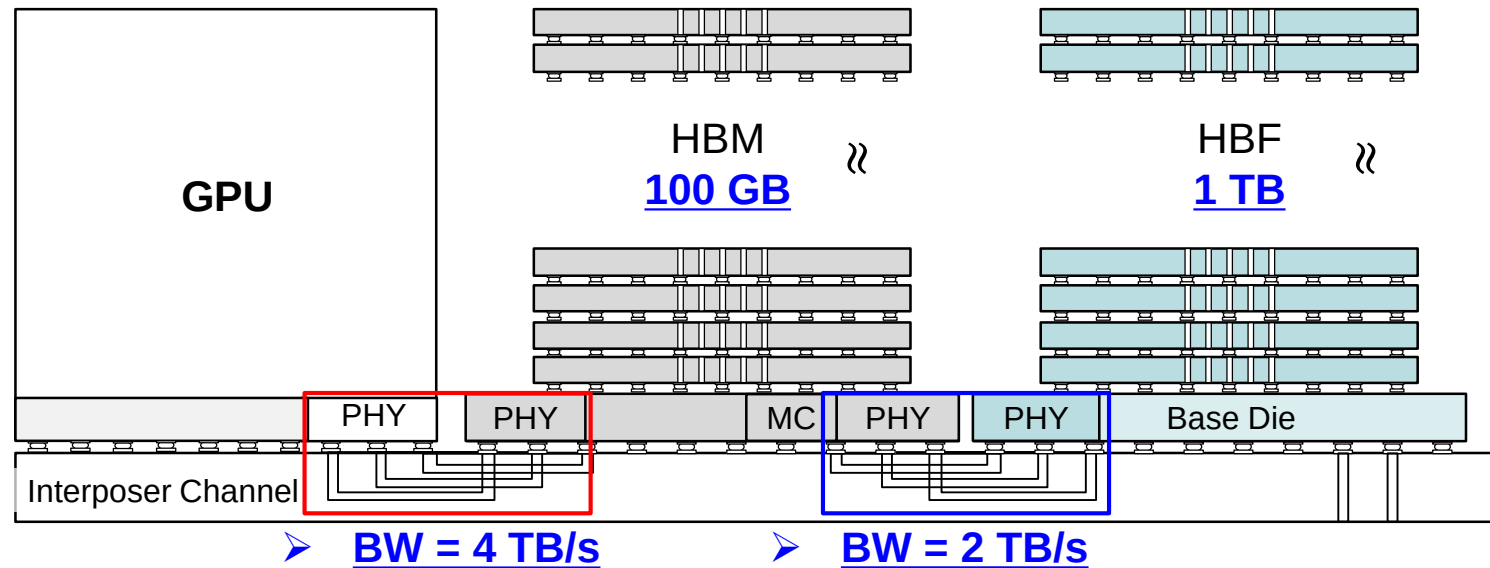
Ref) <https://developer.nvidia.com/blog/mastering-llm-techniques-inference-optimization/>

HBM-based Memory Capacity Expansion for LLM / sLLM Inference

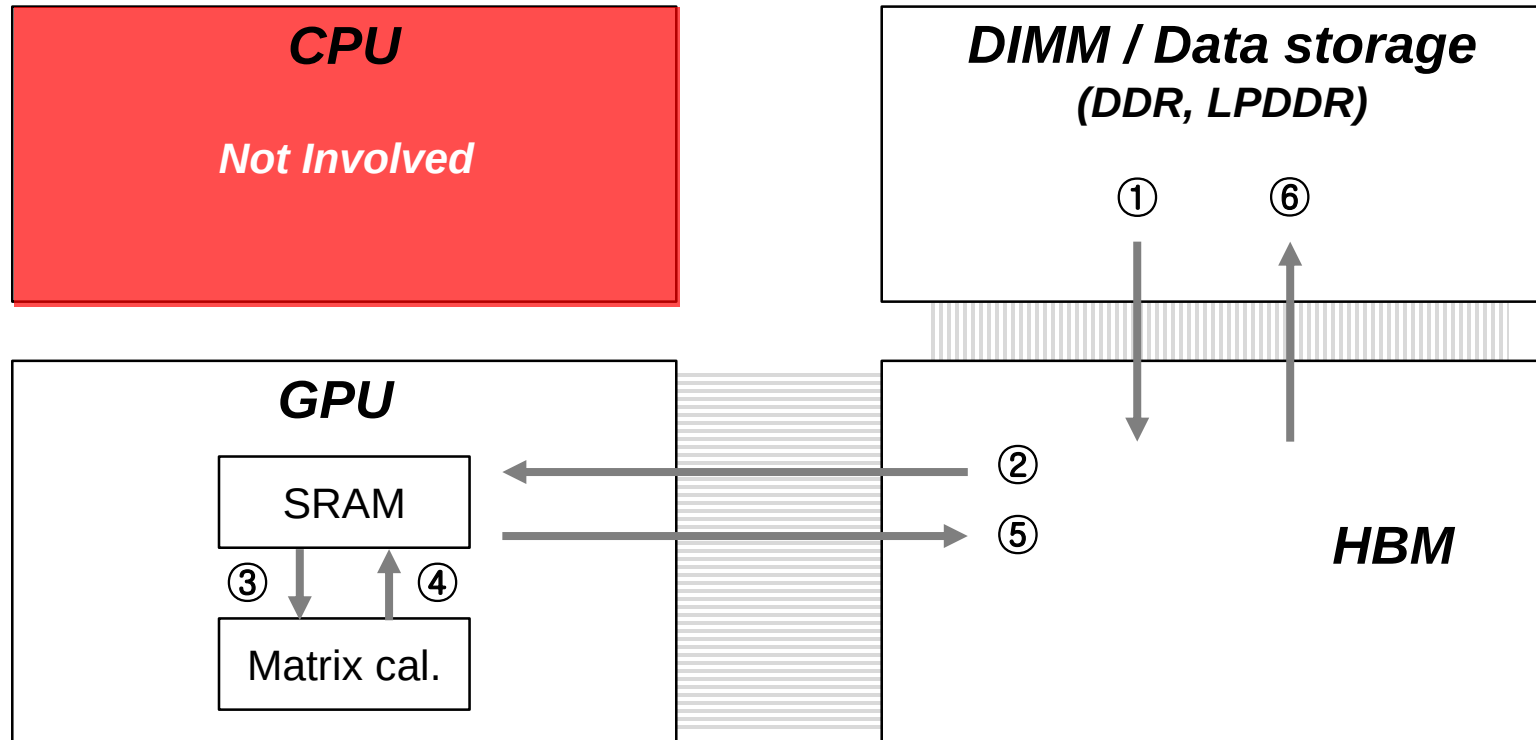
	HBM (Multi-tower)	HBM-LPDDR	HBM-HBF
Top View			
Side View			
Memory Capacity	HBM 400 GB	HBM 200 GB + LPDDR 400 GB	HBM 200 GB + HBF 2 TB
Memory Bandwidth	4 TB/s	4 TB/s	12 TB/s

< HBM-based Architectures for Memory Capacity Expansion >

- GPU SRAM 및 기존 HBM 의 부족한 용량을 극복하고자 Multi-stack HBM, HBM-LPDDR, HBM-HBF 를 비롯한 HBM 기반 시스템을 이용하여 High Bandwidth + Memory Capacity Expansion이 가능함.
- LLM Inference 시 재사용률이 높은 KV Cache 및 Parameter 를 최대한 GPU Module 내 저장.

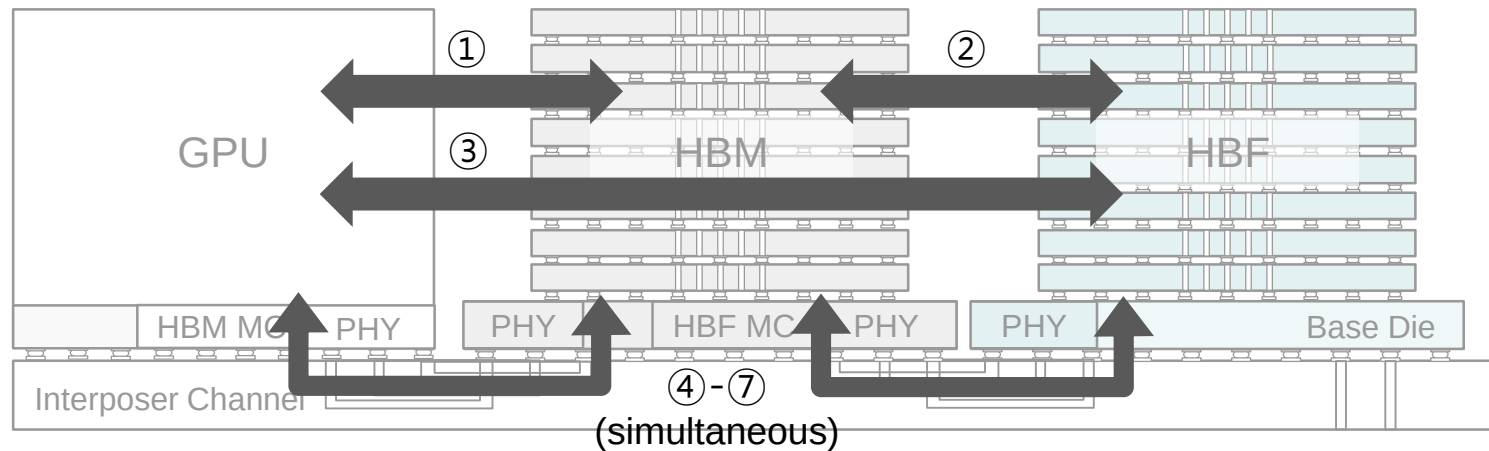


New Data Flow in HCC with GPU Co-Existence



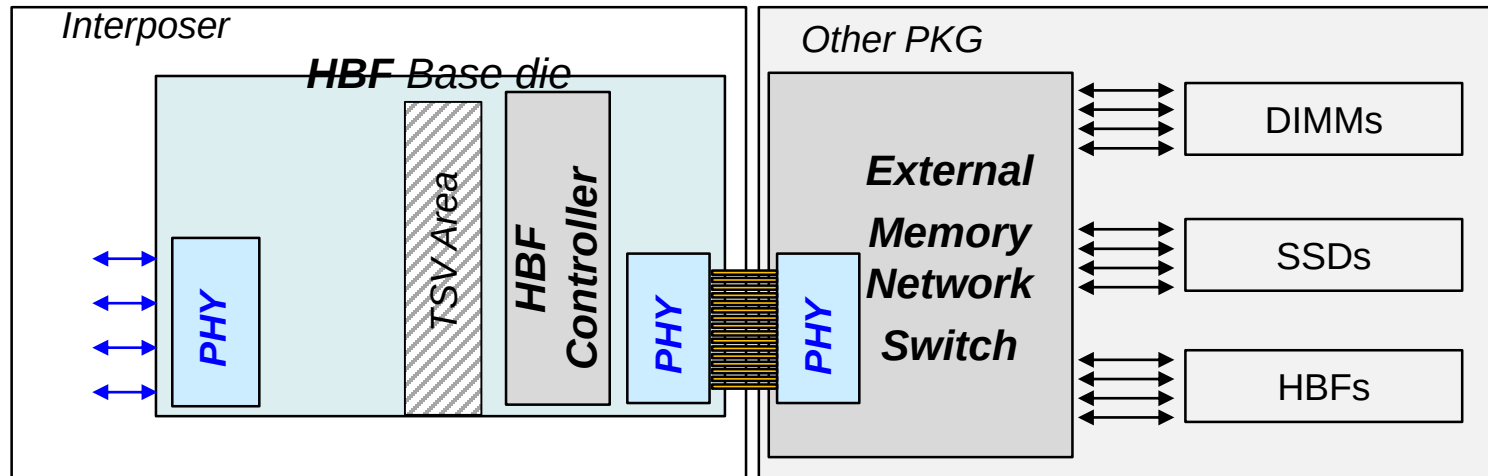
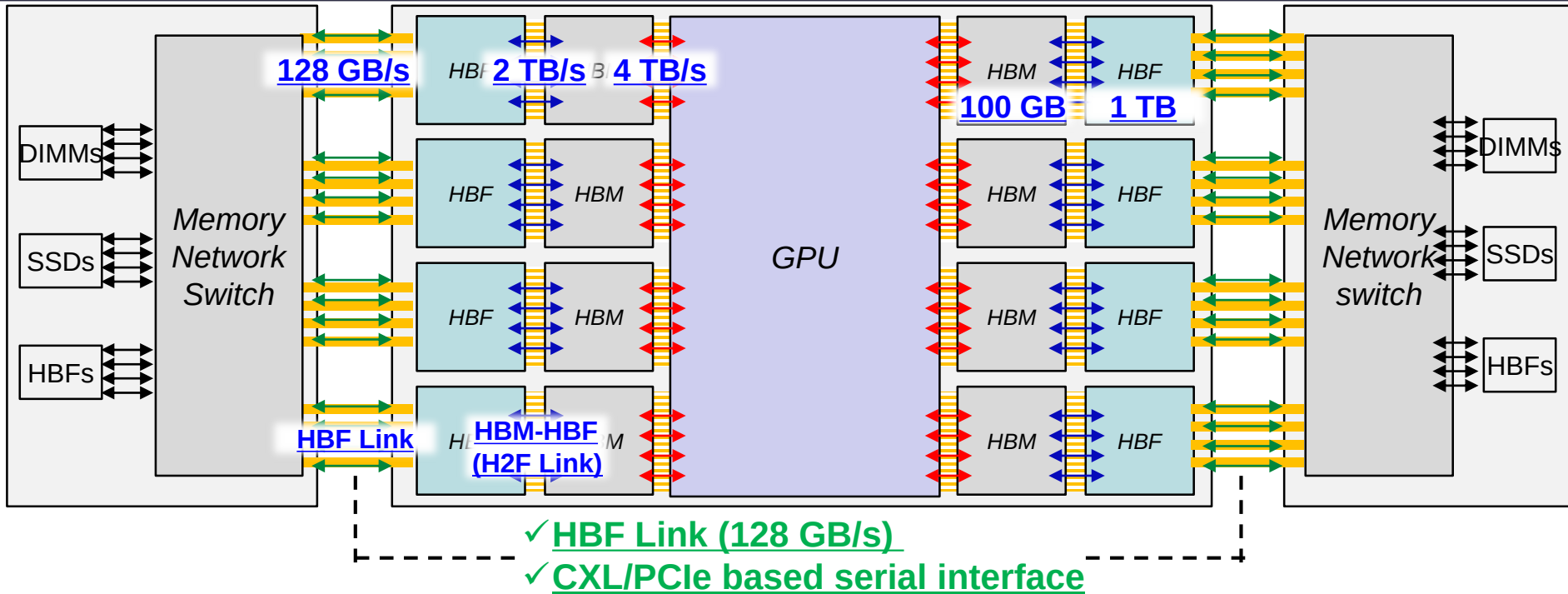
- ① Data (Input x , Input h , Weight w_{ij}) from DIMM → copy to → HBM / Path: DIMM → HBM
- ② Data (Input x , Input h , Weight w_{ij}) from HBM to cache in GPU / Path: HBM → GPU
- ③ Matrix calculation in GPU
- ④ Save matrix multiplication results to SRAM
- ⑤ Save it to HBM / Path: GPU → HBM
- ⑥ Copy it to DIMM / Path: HBM → DIMM

→ Since the *CPU is not involved* in the memory transfer path, *delays can be reduced*, and it has the advantage of *fewer interconnection steps and shorter lengths*.



< Side-view of GPU-HBM-HBF Structure and Its Data Path Cases >

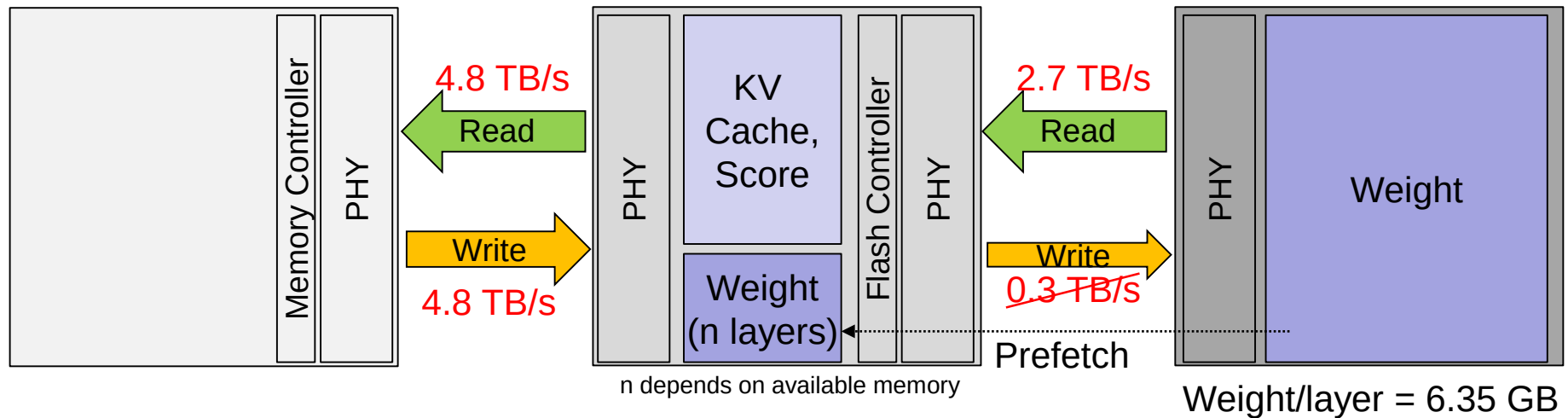
- Case 1: GPU ↔ HBM (Read/Write)
 - Case 2: HBM ↔ HBF (Read/Write)
 - Case 3: GPU ↔ HBF (Read/Write)
- Single-Command Execution
-
- Case 4: GPU → HBM (Write) & HBM ← HBF (Read)
 - Case 5: GPU → HBM (Write) & HBM → HBF (Write)
 - Case 6: GPU ← HBM (Read) & HBM ← HBF (Read)
 - Case 7: GPU ← HBM (Read) & HBM → HBF (Write)
- Dual-Command Execution
(GPU ↔ HBM ↔ HBF)



GPU

HBM: 192 GB (24 GB x 8)

HBF: 4 TB (512 GB x 8)



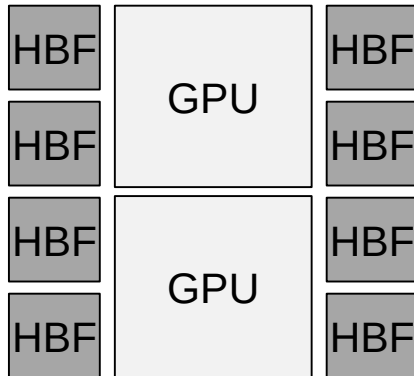
	Prefill	Decode	Llama3-400B
HBF→HBM	$4d^2$	$4d^2 * N$	8.8 TB
HBM→GPU	$4d^2 + (s^2 + 4sd) * B$	$(4d^2 + 2sd * B) * N$	9.9 TB
GPU→HBM	$(s^2 + 4sd) * B$	$2sd * B * N$	1.1 TB
R/W ratio	9:1 (HBM), inf. for HBF		

< Read and Write operations in GPU-HBM-HBF Architecture >

- HBF's main weakness is write operation; however, by storing weights in HBF and KV cache + other memory in HBM, no writes are done to HBF during inference.
- Hence, it gives the best performance, since most loads are from KV cache.
- However, reading weights require more latency to reach GPU.
- Also, additional area for interposer is required.

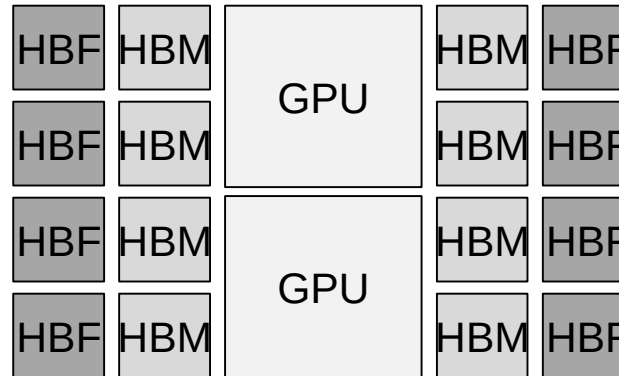
Comparison of HBM, HBF, and HBM-HBF Architecture

4TB NAND

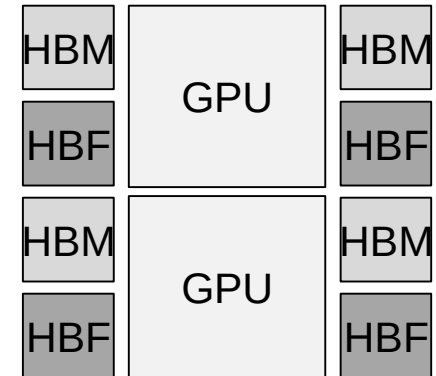


< GPU-HBF Architecture >

192GB DRAM + 4TB NAND

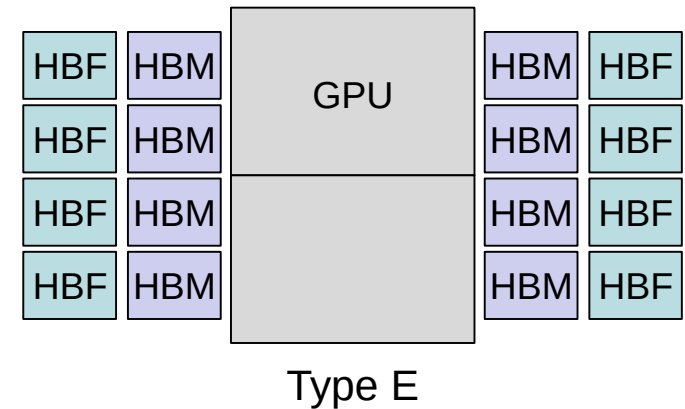
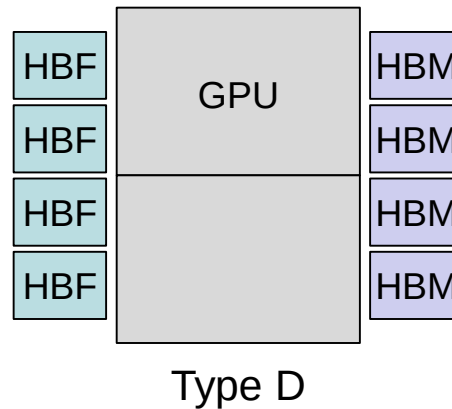
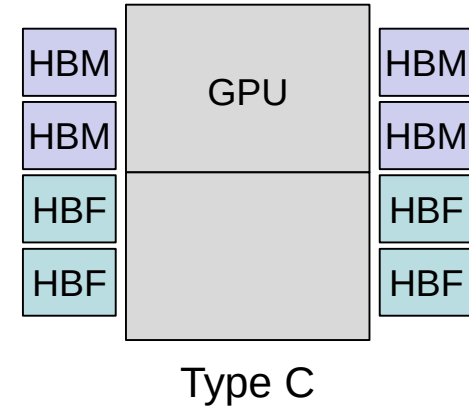
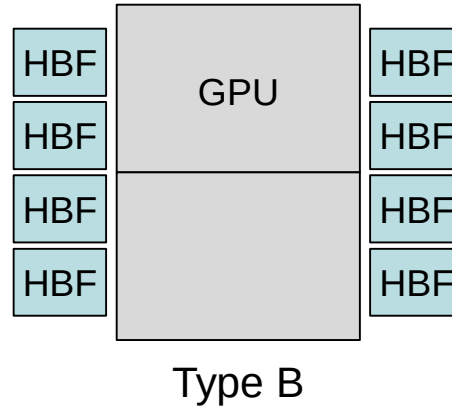
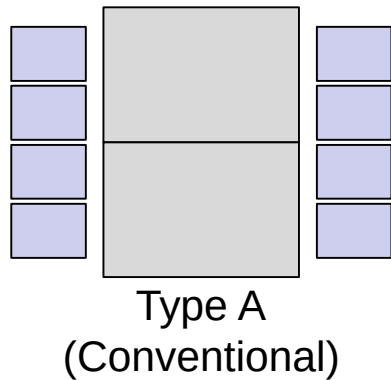


< GPU-HBM-HBF Architecture >

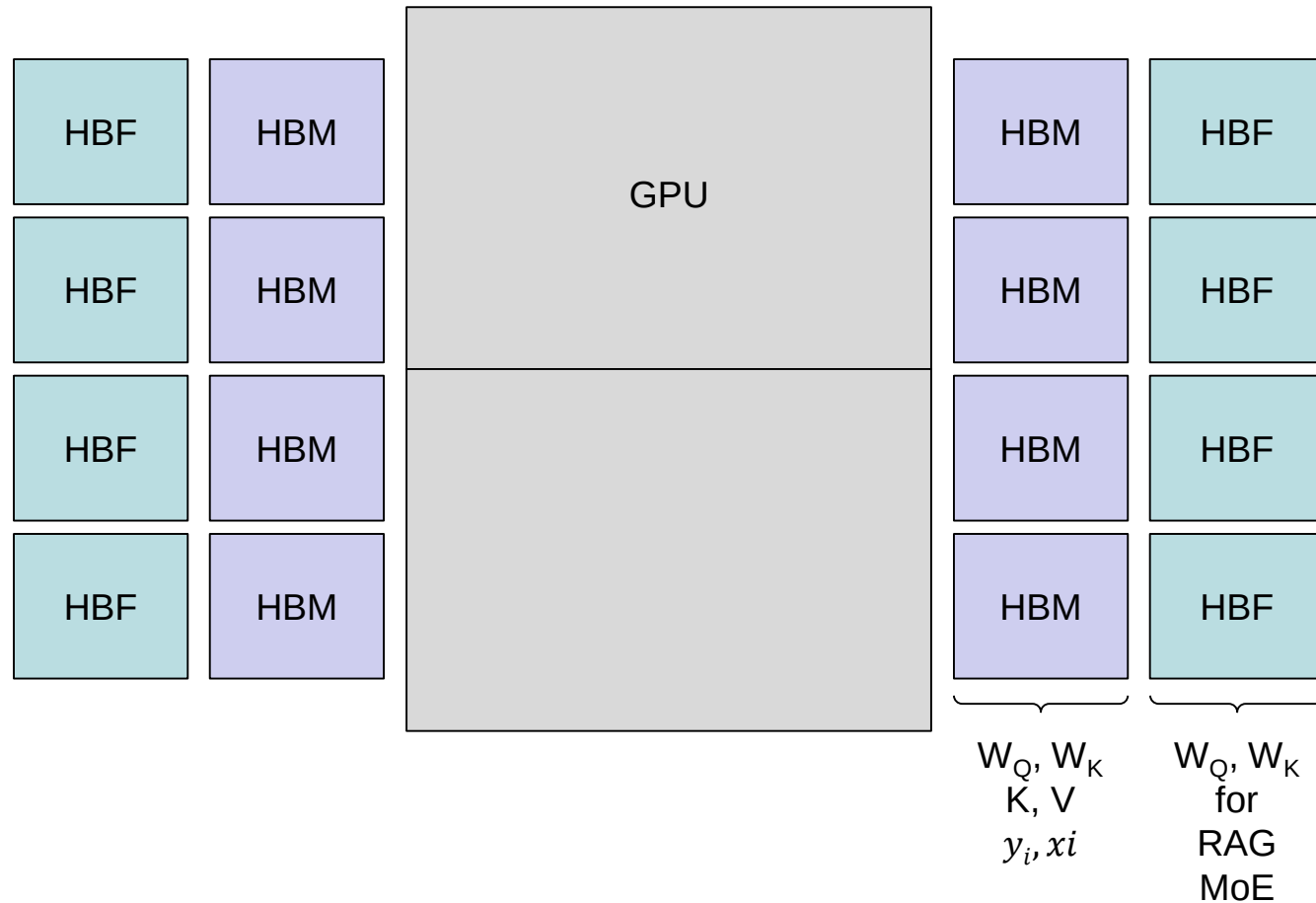


< Hybrid Architecture >

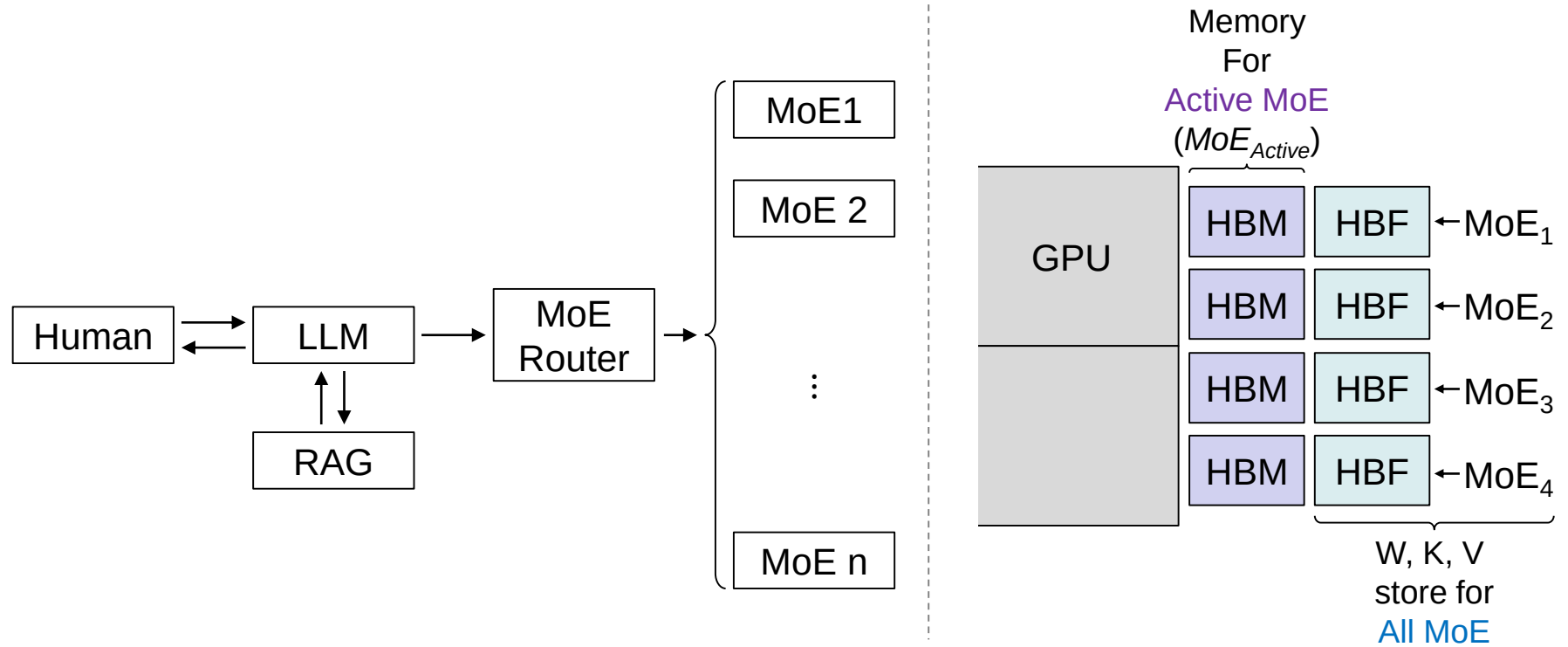
	HBM	HBF	HBM-HBF	Hybrid (1:1)
Capacity	192 GB	4 TB	4.2 TB	2.1 TB
Bandwidth	6.4 TB/s	2.7 TB/s	Mixed, Depends	Depends
Token/s	High	Medium	High	High
Cost	Medium	Low (\$ per bit is low)	High (Larger Interposer)	Medium-Low
Reliability	High	Low (HBF Write is bad)	Medium (No Write)	High



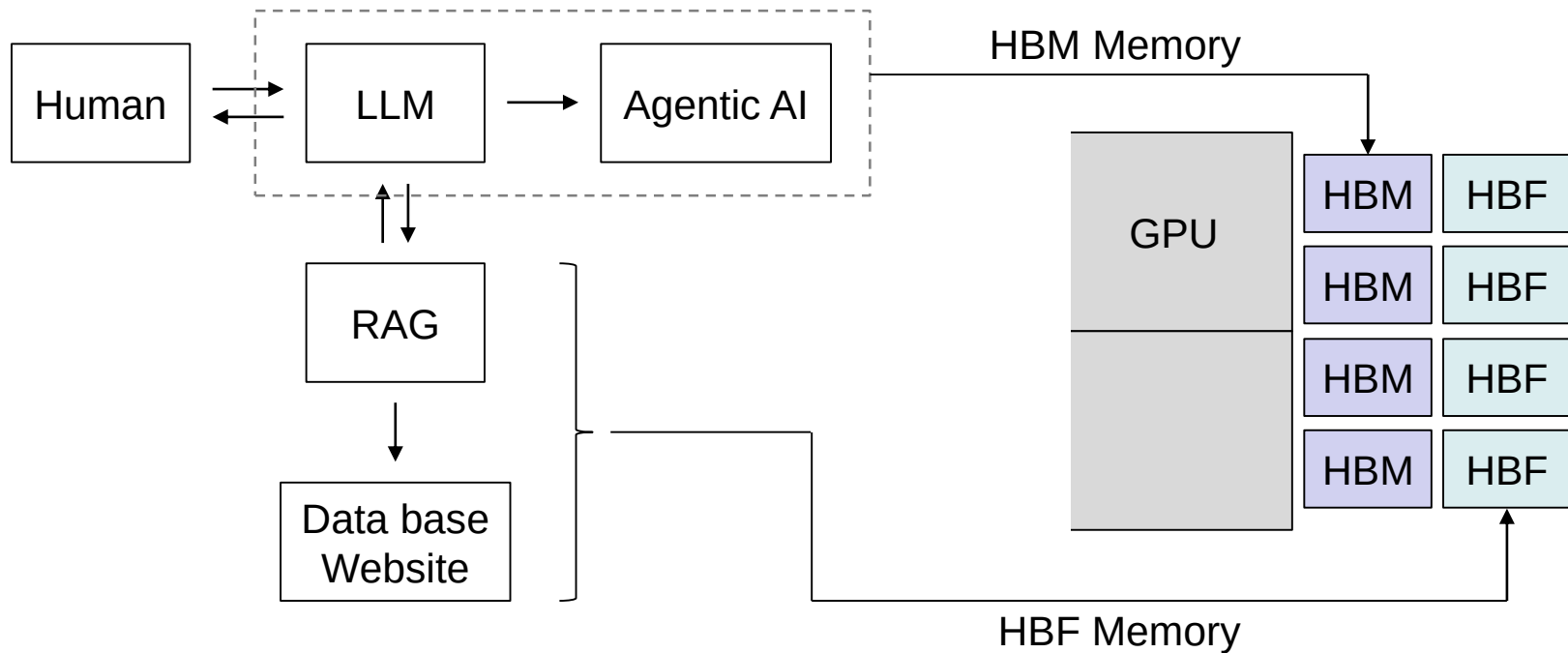
< 4 Types of HBM & HBM hybrid computing architecture >



< Workload-specific memory partitioning for RAG and MoE >



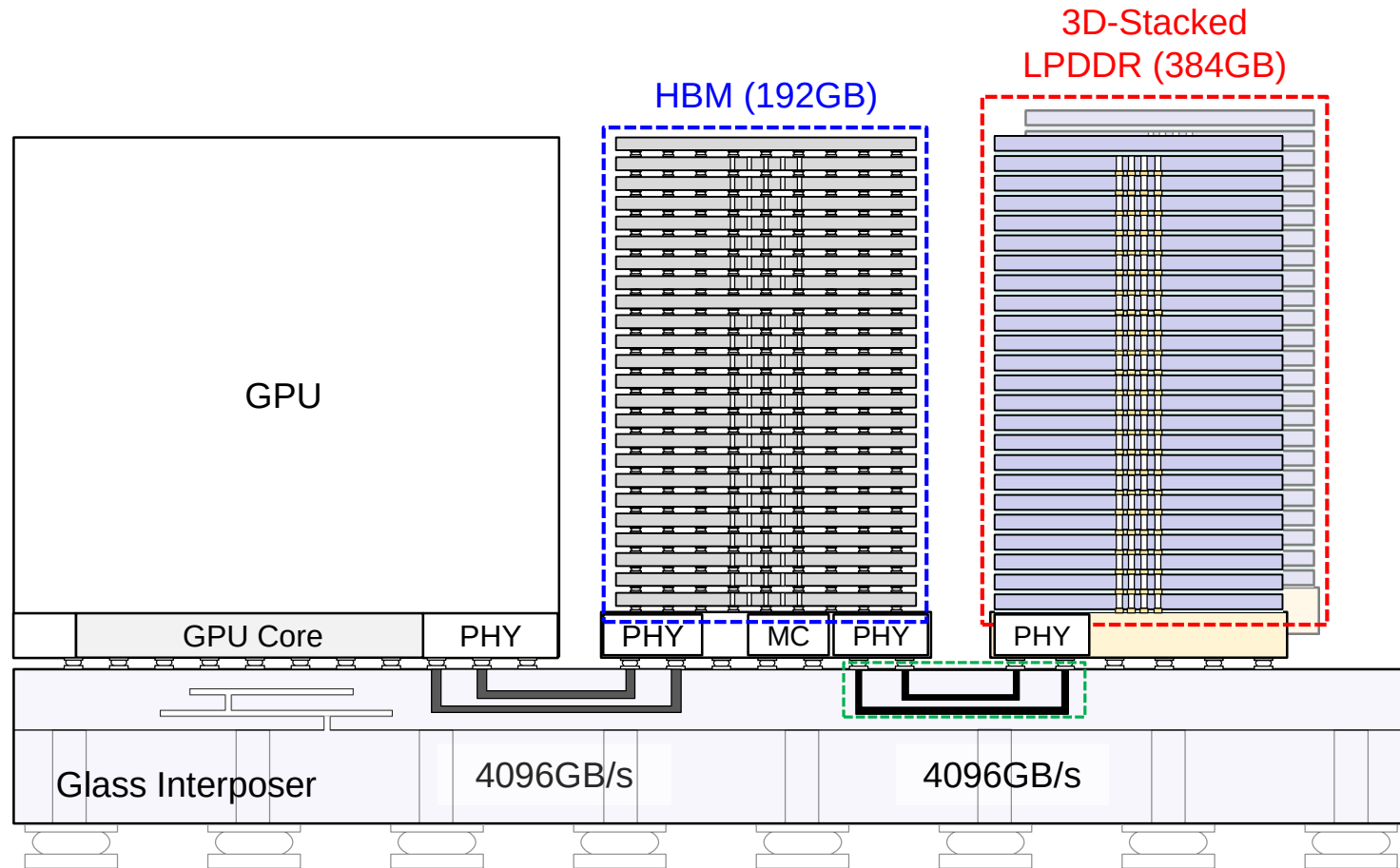
< Dynamic expert allocation with HBM-HBF hybrid memory architecture >



< External knowledge integration with HBM-HBF hybrid memory architecture >

HBM7 Architecture Integrated with High-Capacity 3D-Stacked LPDDR on Glass Interposer

HBM7
Architecture



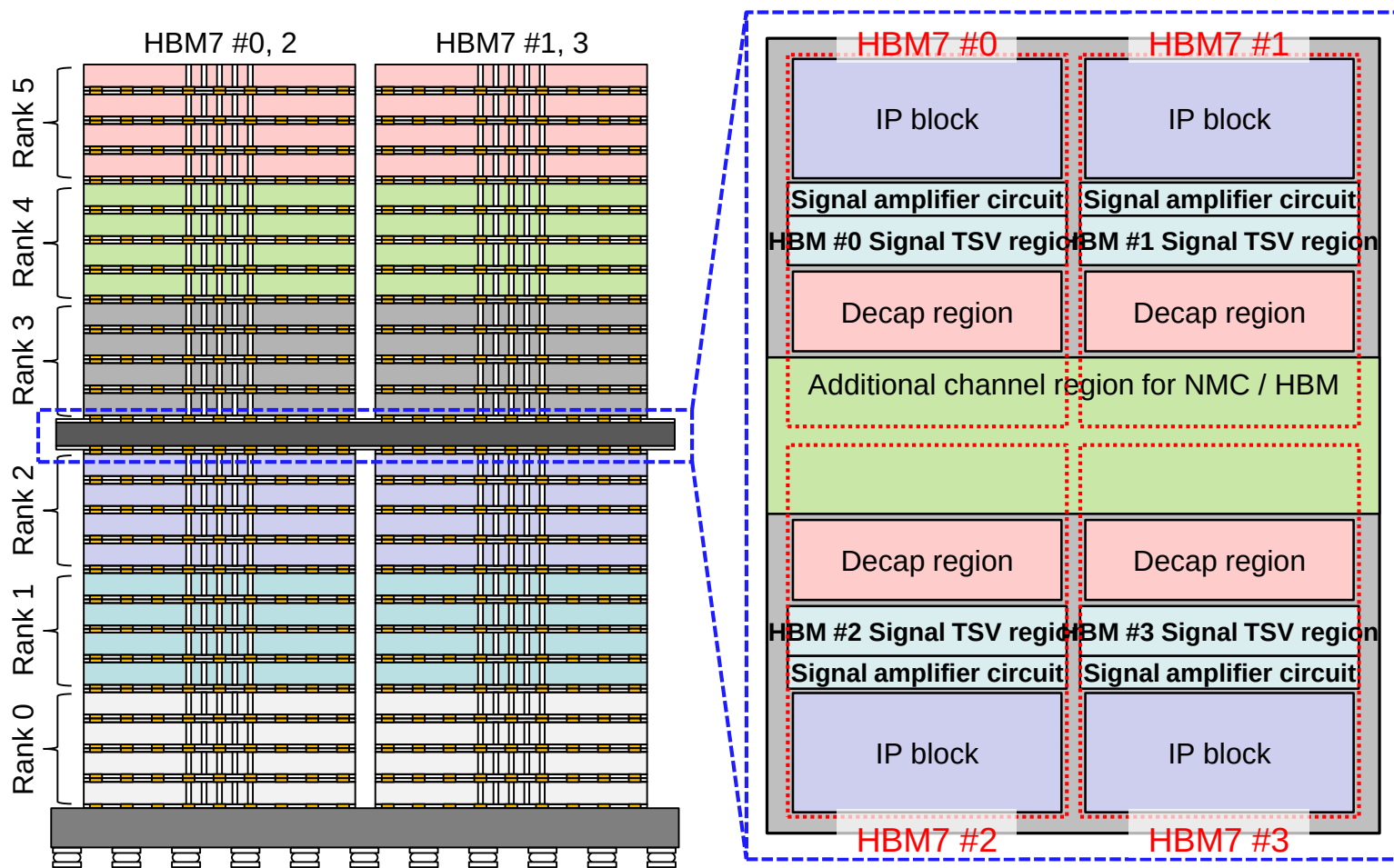
< HBM7 Architecture Integrated with High-Capacity 3D-Stacked LPDDR on Glass Interposer >

LPDDR 기반 LLW 성능 및 특징 비교 분석

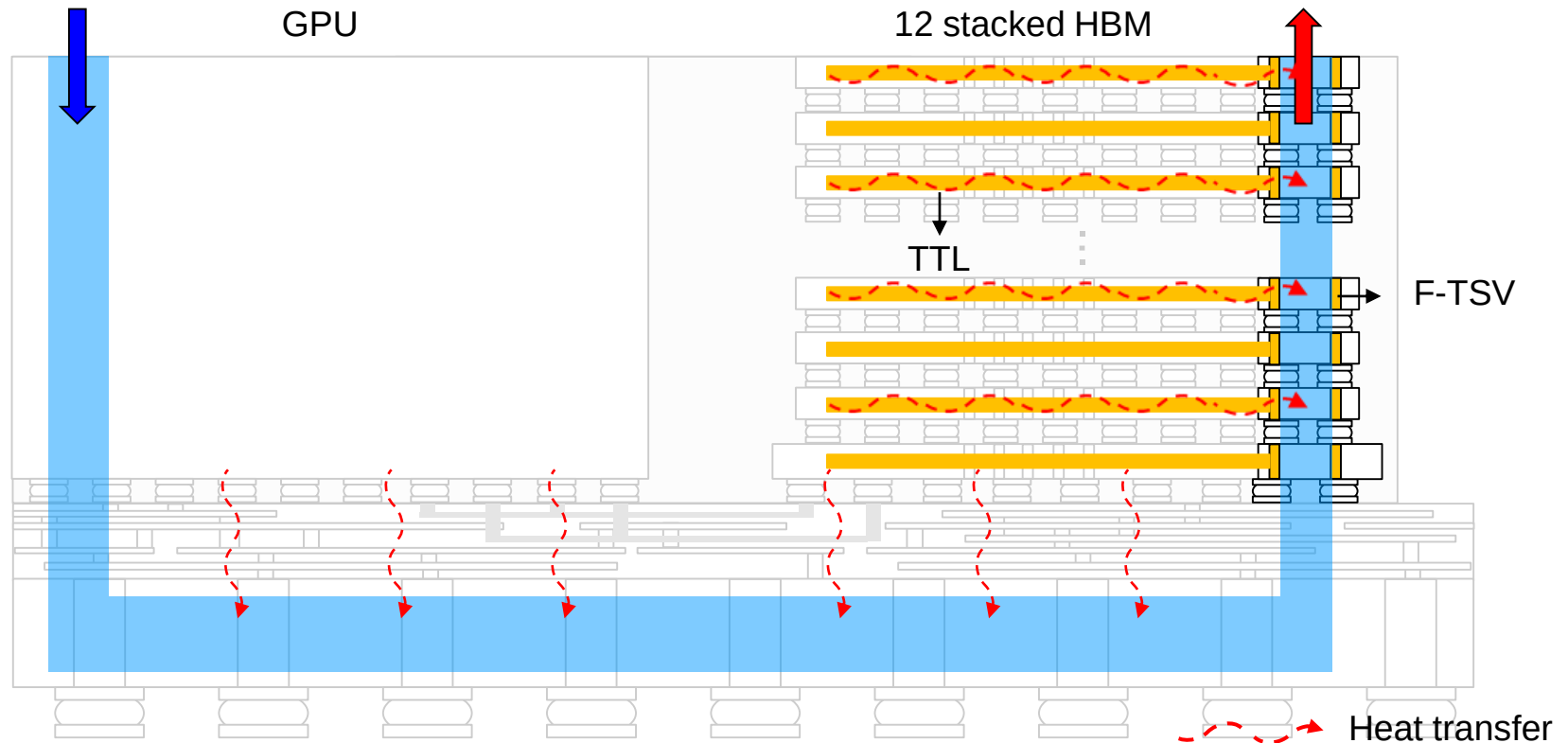
	LPDDR * LPDDR4 기준	DDR HBM * HBM2 기준	LLW (LPDDR HBM-like Memory)
Latency	약 60 ns (tRC)	약 47 ns (tRC)	약 32 ns (tRCW)
Bandwidth	25.6 GB/s	307 GB/s	128 GB/s
Energy per bit	4 pJ/bit (1.1V)	5 pJ/bit (1.2V)	1 pJ/bit (1V)
Protocol	JEDEC LPDDR Spec 기반	JEDEC HBM Spec 기반	전용 컨트롤러 또는 비 JEDEC 사양 기반 예상
Burst length	16 / 32	4	2~4 (추정)
Standby 전류	수 mA 이하	수십 mA	수 mA 이하
주요 응용처	모바일, 태블릿, 노트북 등	AI 가속기, HPC 서버	On-device AI, 가상현실, 자율주행 SoC
전력 최적화 수준	매우 우수	중간	HBM 대비 우수, LPDDR 대비 유사 또는 개선

< LPDDR, DDR-HBM, LLW 성능 및 특징 비교 분석 >

- LLW는 LPDDR의 저전력성과 HBM 수준의 대역폭과 낮은 지연 시간 특성을 구현한 차세대 wide I/O 기반 DRAM으로 정의됨.
- HBM 대비 에너지 효율이 우수하고, LPDDR 대비 높은 대역폭과 낮은 지연시간 등 성능을 제공하여, 고속성과 저전력 특성을 모두 요구하는 연산 환경에 최적화되어 있음.



< Side and top view of proposed multi functional bridge-die >



< Concept of the proposed embedded cooling structure for GPU-HBM module >

- The proposed Thermal Transmission Line (TTL) and Fluidic TSV (F-TSV) can cool the HBM module efficiently by circulating cooling fluid through the GPU to the interposer and HBM.
- The proposed TTL transfers the internal heat within HBM die to the fluid flowing in F-TSV

Part7: Key Features in HBM8

Key Features of HBM8

1. Electrical Specification

- Data Rate : 32 Gbps
- Number of I/Os : 16,384
- Total Bandwidth : 64.0 TB/s
- Number of die stack : 20/24-Hi
- Capacity/die : 80 Gb

2. Packaging/Cooling Method

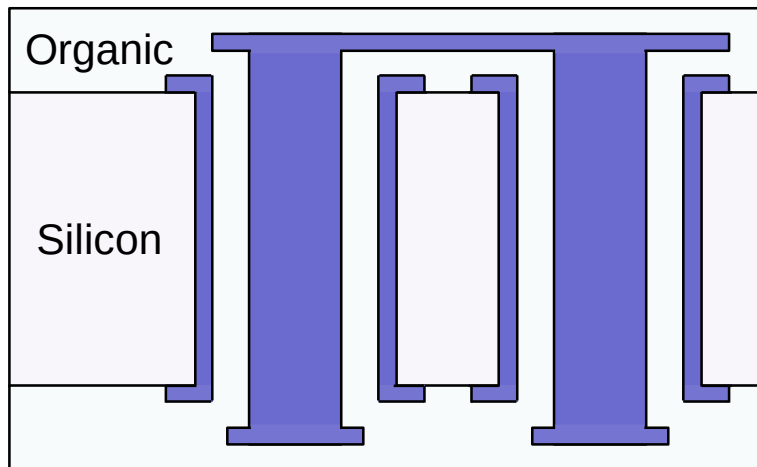
- Bump-less Cu-Cu Direct Bonding
- Embedded Cooling

3. HBM Architecture

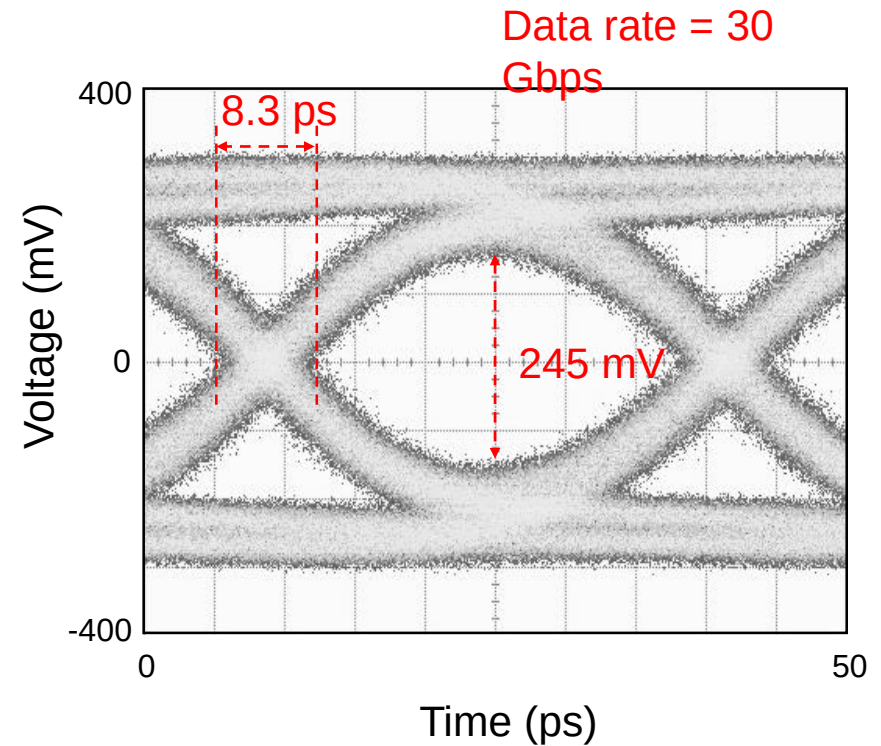
- Coaxial TSV
- Full-3D GPU-HBM
- HBM Centric Computing
- Full Memory Network
- Double Sided Interposer

4. AI Design Agent

- LLM based Human Interactive AI Design Agent



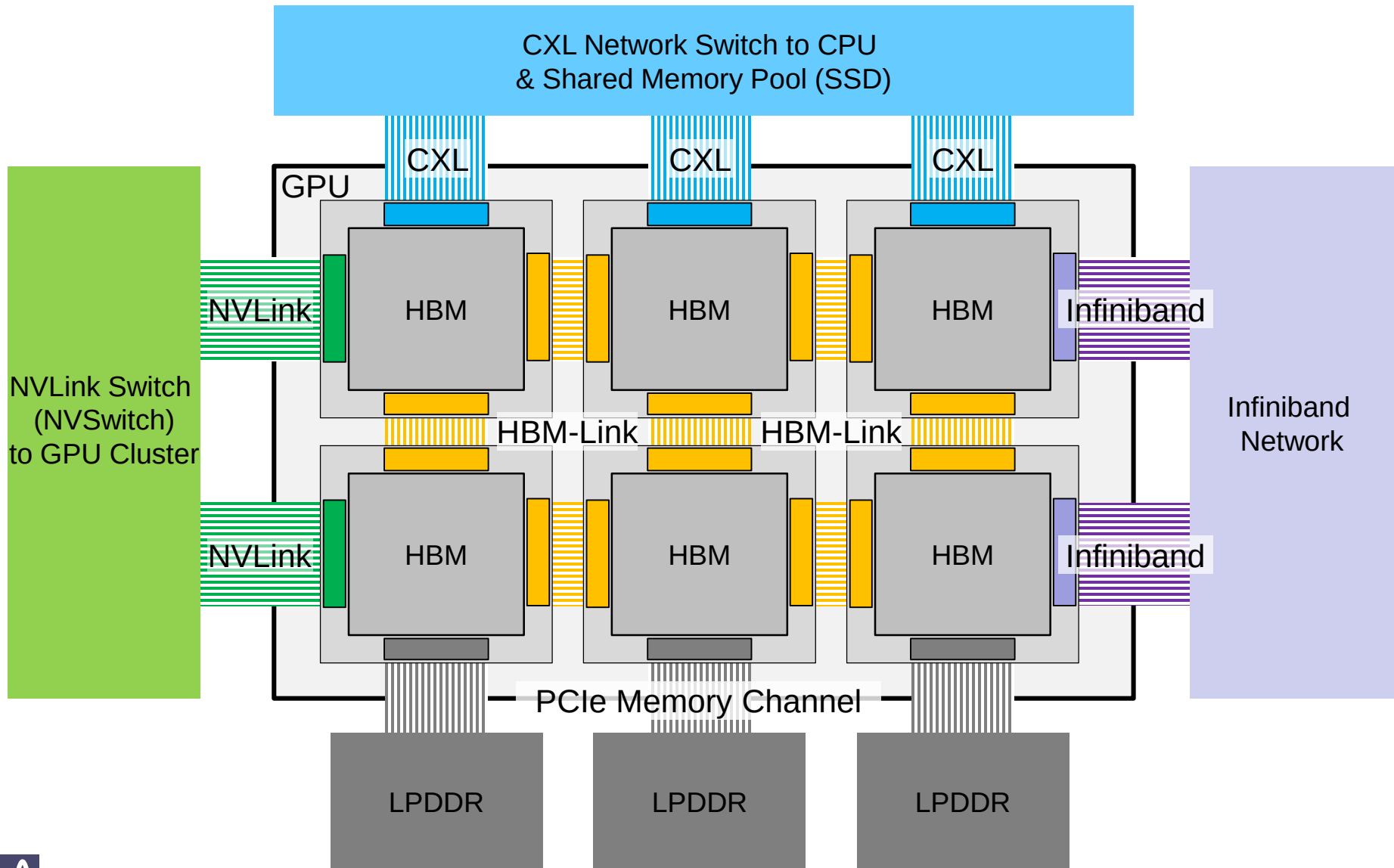
< Coaxial Organic Line Via (COLV) >



< Eye-diagram of Coaxial TSV channel >

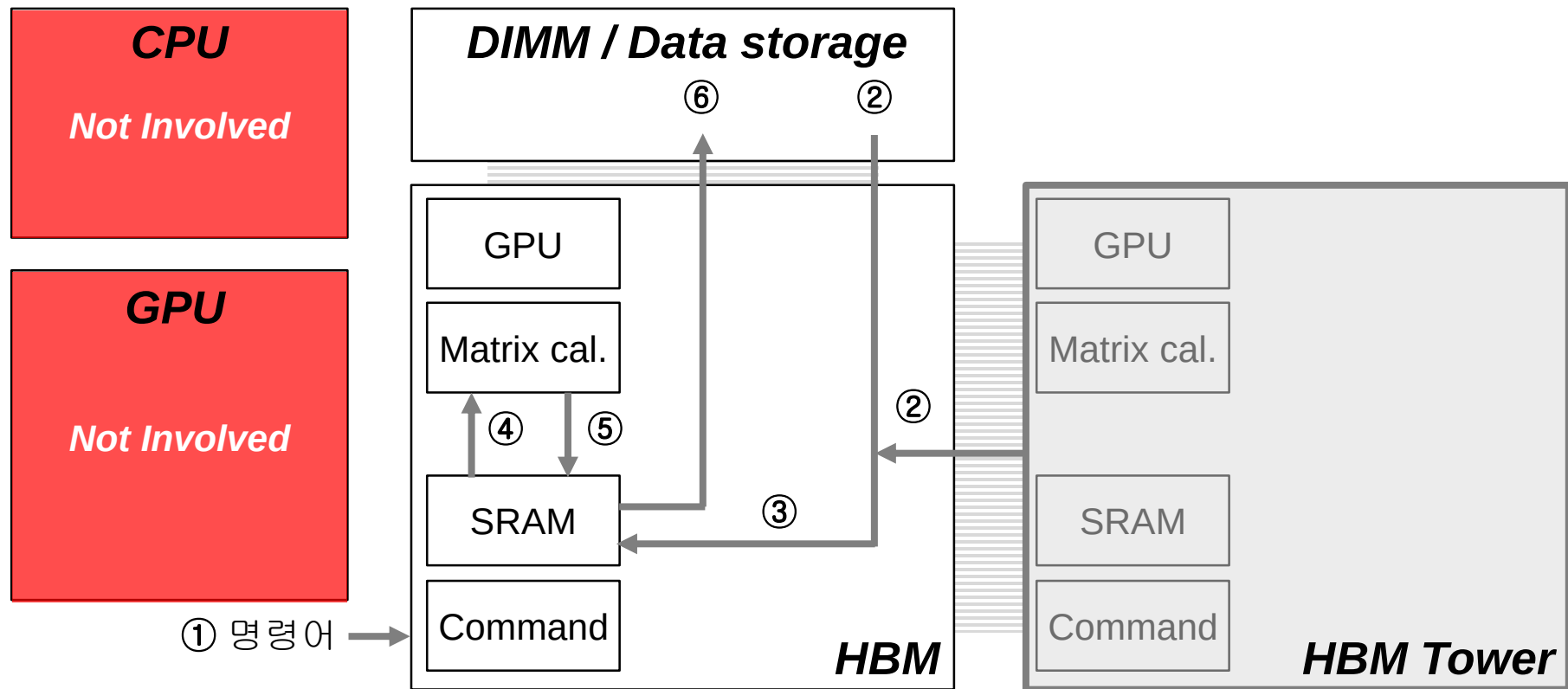
Next-Generation HBM Roadmap : HCC Architecture with System-Level Memory Network

HBM8
Architecture



< System-Level Memory Network for HCC Architecture >

New Data Flow in HCC with HBM Centric Computing

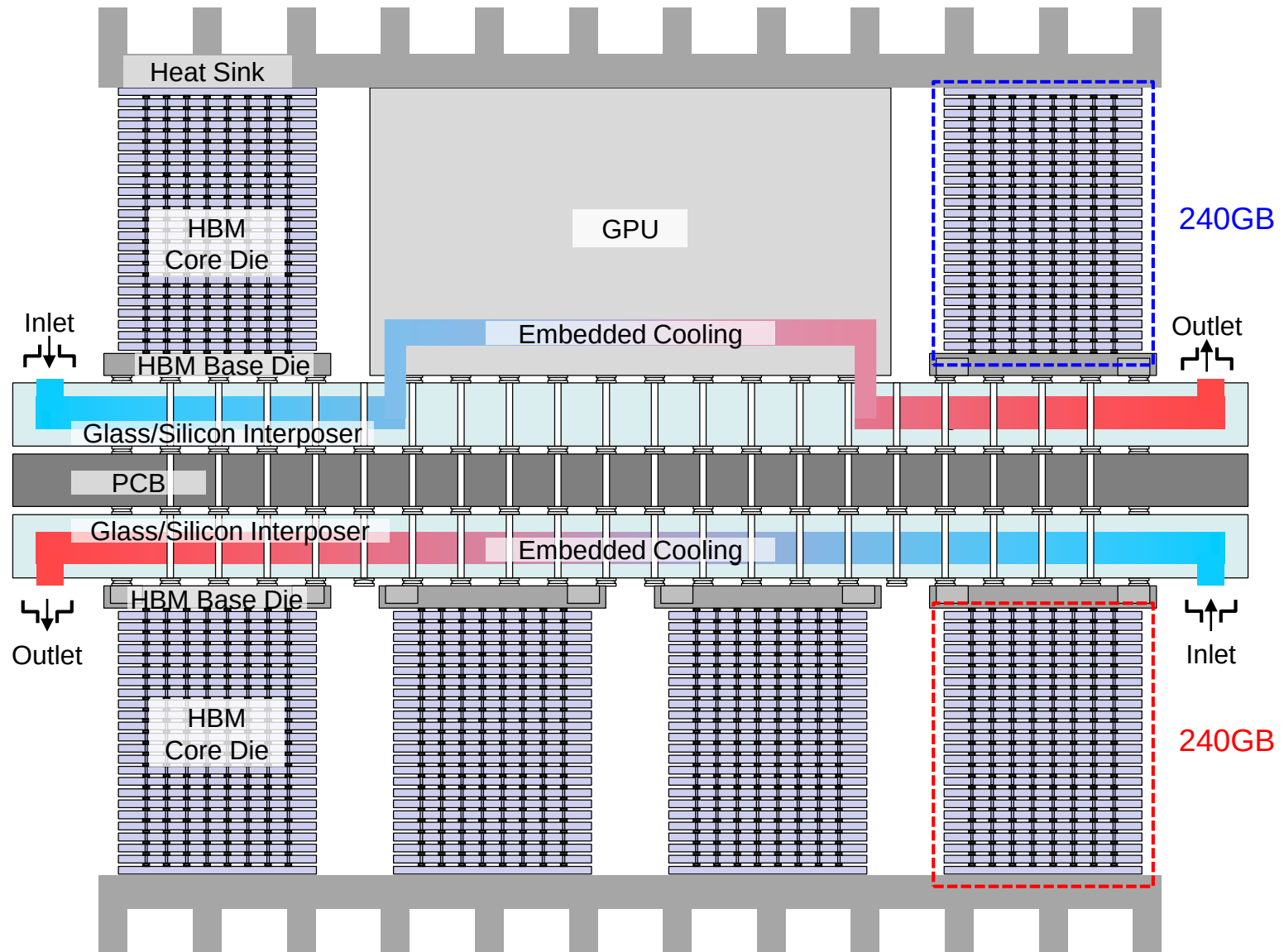


- ① Command decoder
- ② Data (Input x , Input h , Weight w_{ij}) from HBM tower or DIMM → copy to → HBM
- ③ Data (Input x , Input h , Weight w_{ij}) from HBM to SRAM in base die
- ④ Matrix calculation
- ⑤ Save it to SRAM in base die
- ⑥ Save it to DIMM

→ With no **CPU** and **GPU** in the memory transfer path, *delays are greatly reduced*, and there are *very few interconnection steps with a much shorter length*.

3D Memory Expansion Architecture with Embedded Cooling Structure for HBM8 with Double-Sided Interposer [1/3]: GPU-HBM-HBM

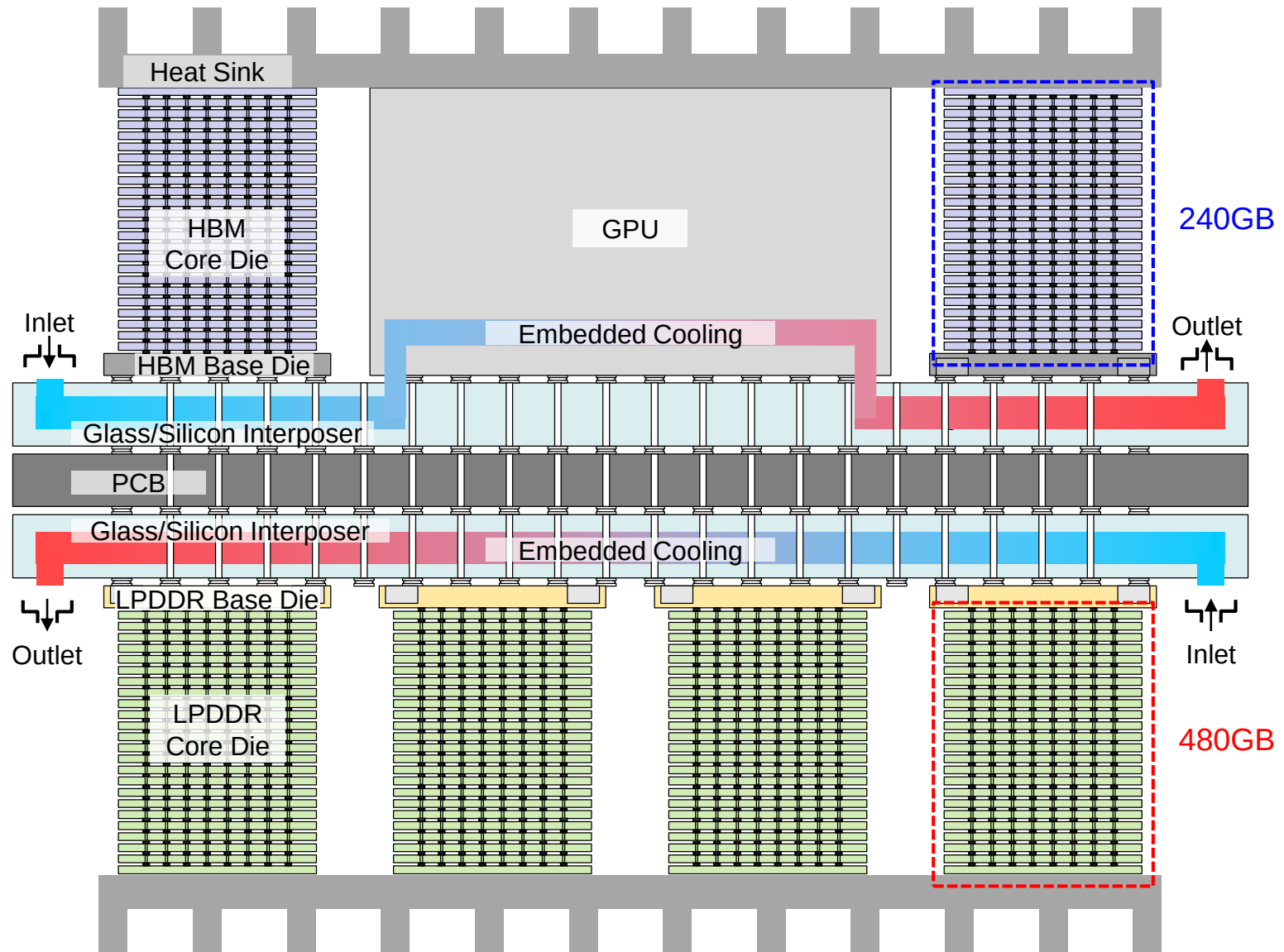
HBM8
Architecture



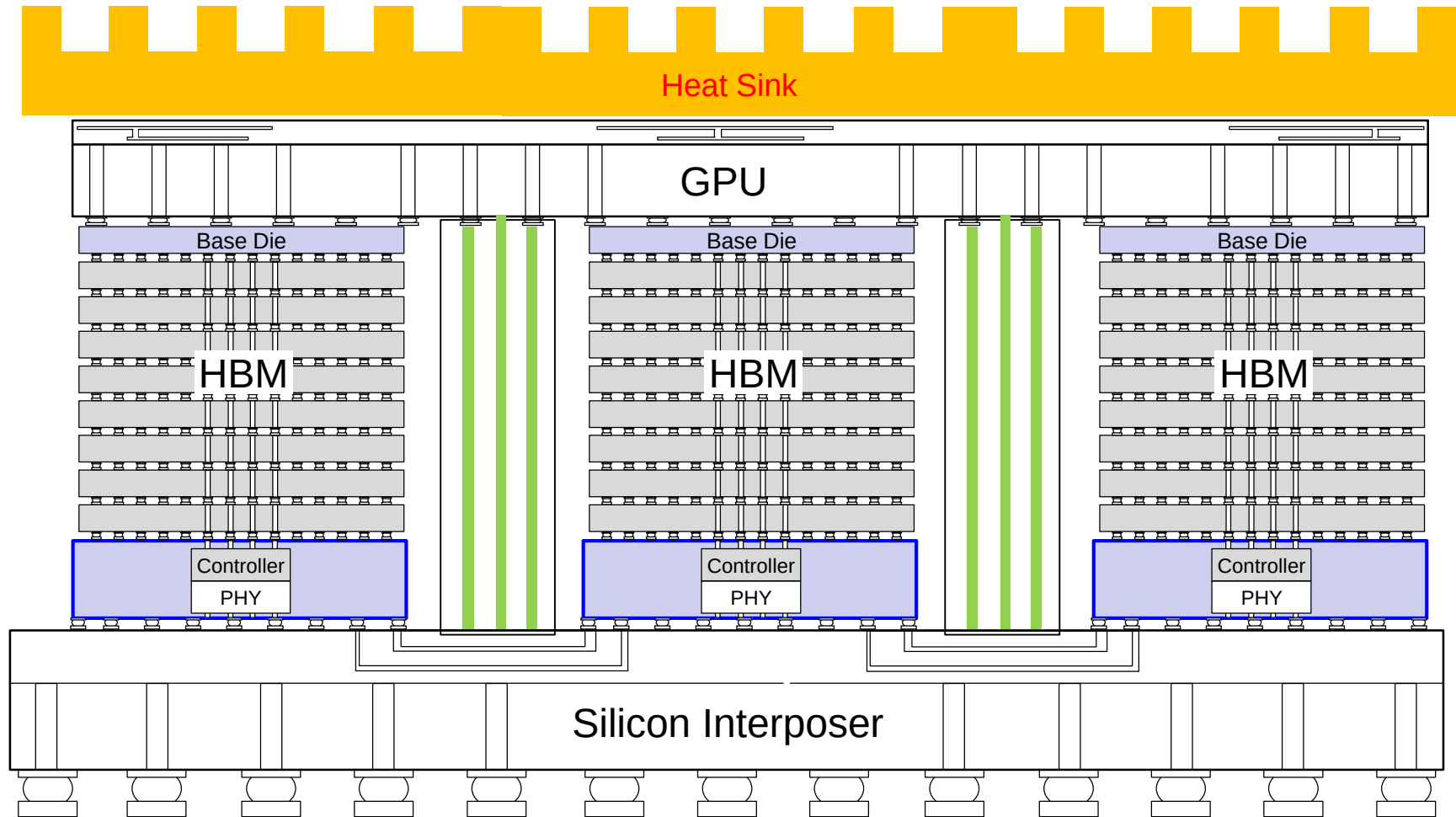
< 3D Memory Expansion Architecture with Embedded Cooling for HBM8 with Double-Sided Interposer using HBM >

3D Memory Expansion Architecture with Embedded Cooling Structure for HBM8 with Double-Sided Interposer [3/3]: GPU-HBM-LPDDR

HBM8
Architecture



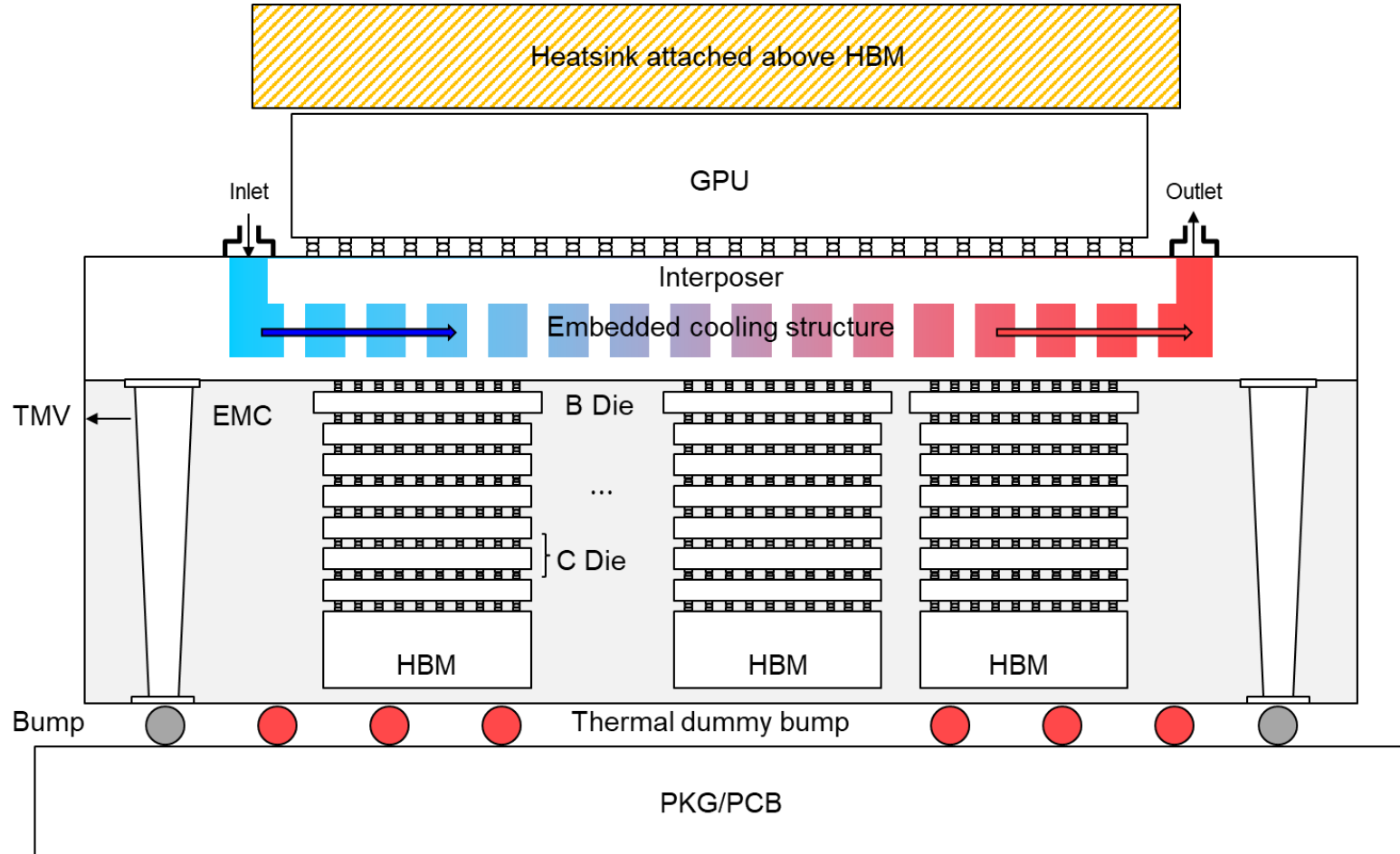
< 3D Memory Expansion Architecture with Embedded Cooling for HBM8 with Double-Sided Interposer using LPDDR >



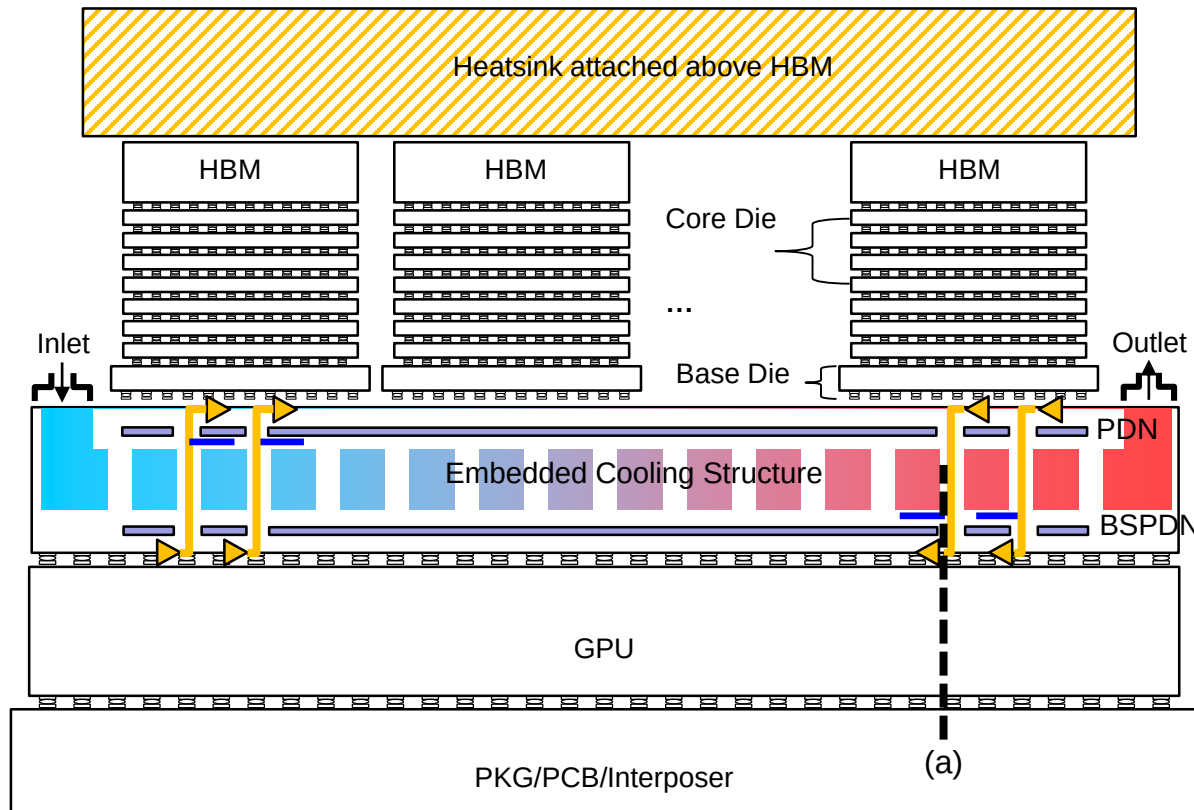
- GPU is implemented at top layer of memory stack for heat dissipation. (Thermal Issue ↓)
- Additional silicon interconnect pillar die is embedded between HBMs to support power to GPU.

3D Package-on-Interposer (PoI) Architecture Using a Double-Sided Interposer with Embedded Cooling Structure for HBM8

HBM8
Architecture

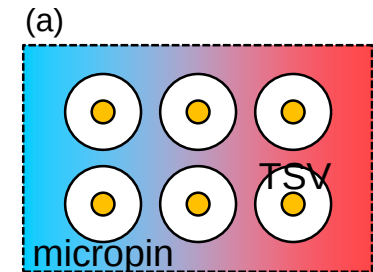


<Proposed 3D Package-on-Interposer (PoI) Architecture Using a Double-Sided Interposer with Embedded Cooling Structure for HBM8 >

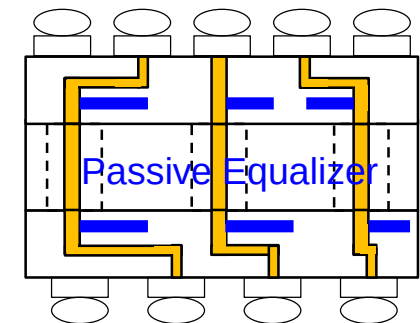


<Proposed Double-Side Interposer Architecture with Embedded Cooling and Top Heatsink for 3D GPU-HBM>

- Both the top heatsink and the double-side interposer are designed with inlet and outlet channels to enable efficient liquid cooling for the entire 3D GPU-HBM stack.



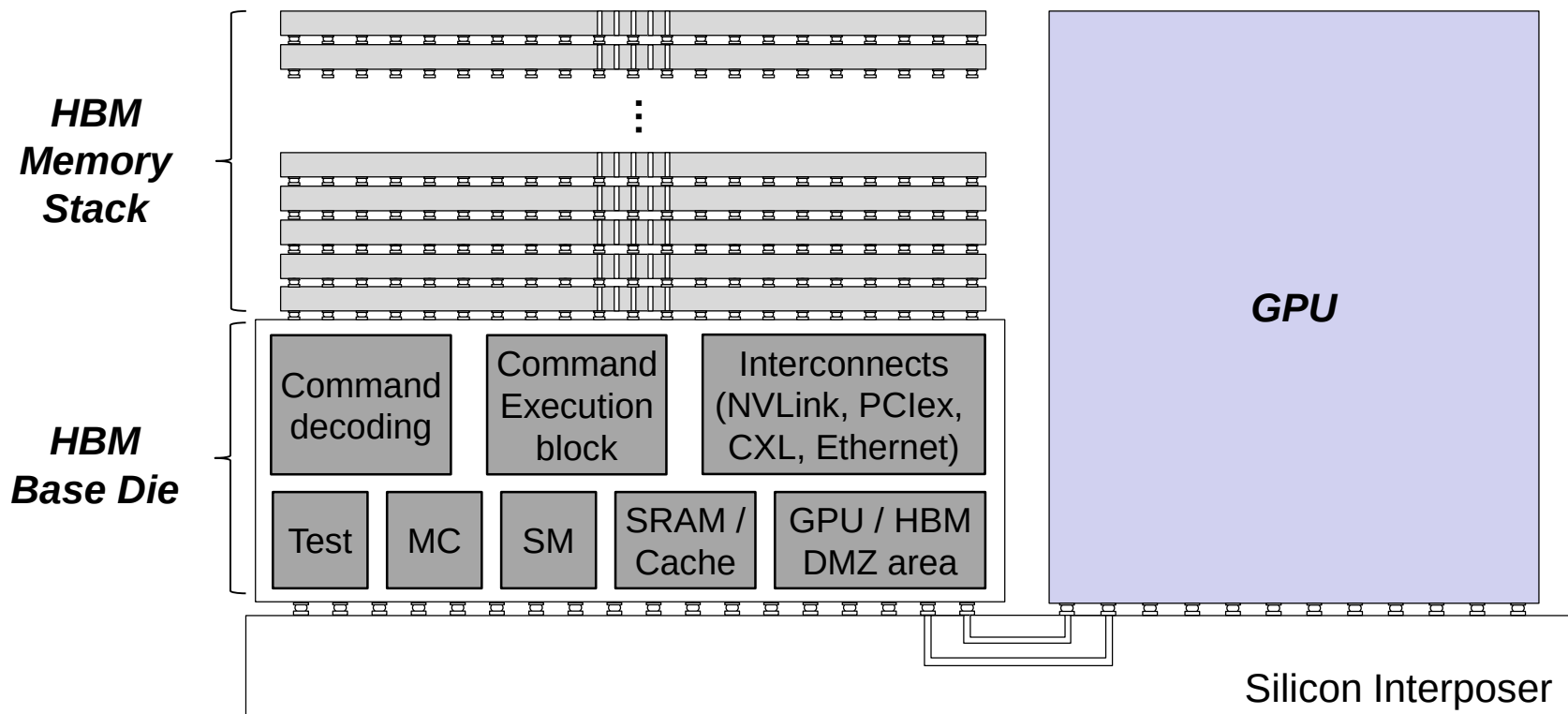
<Embedded Cooling Structure with micropin-fin and tsv>



<Passive Equalizer in double-side interposer>

Part8: HBM Centric Computing Era

HBM Centric Computing (HCC) Structure



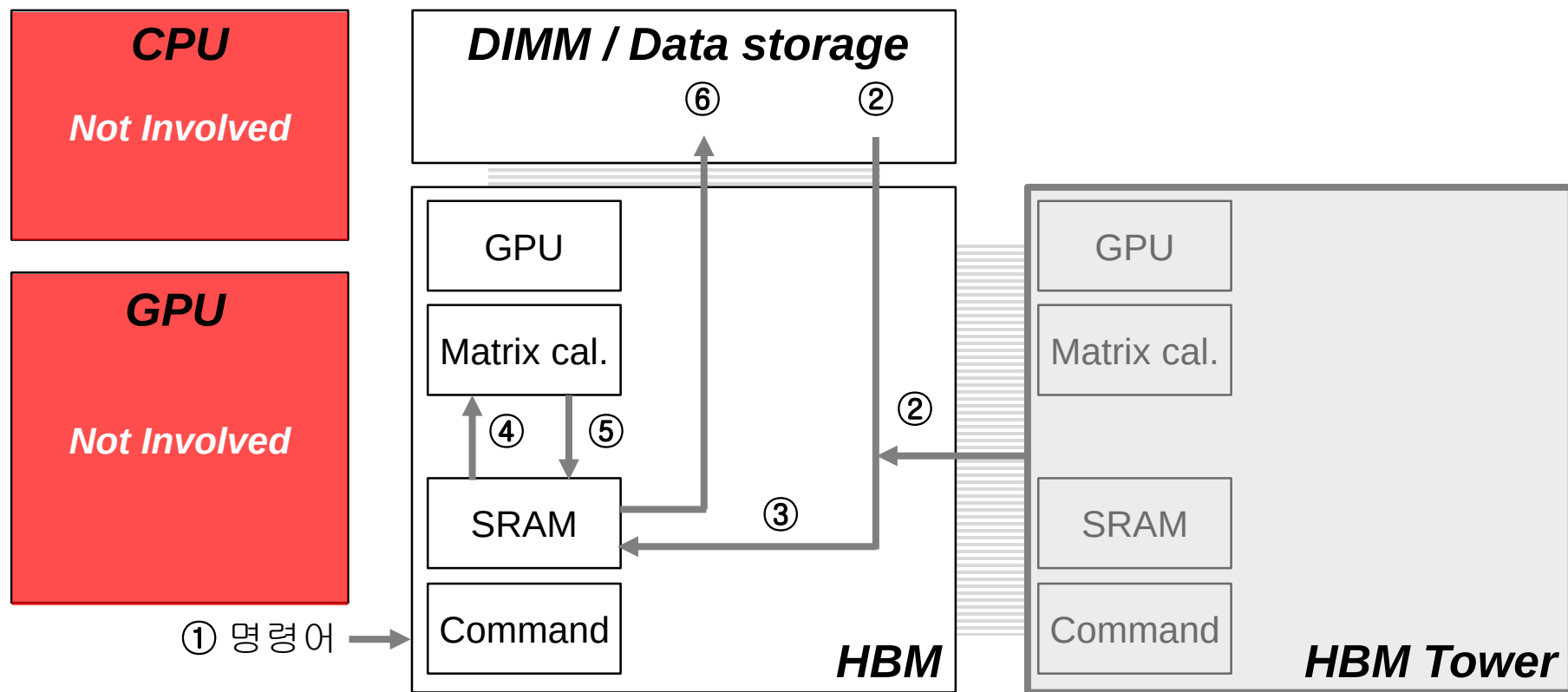
< HBM Centric Computing (HCC) Structure >

- HBM Base Die

- ✓ ① Test ② Memory Controller (MC) ③ Streaming Multiprocessor (SM)
④ SRAM / Cache ⑤ Interconnects (NVLinks, PCIe, CXL, Ethernet)
⑥ Command decoding ⑦ Command execution block ⑧ GPU / HBM DMZ area

Data & Command Flow in HCC [3]:

New Data Flow in HCC with HBM Multi Towers

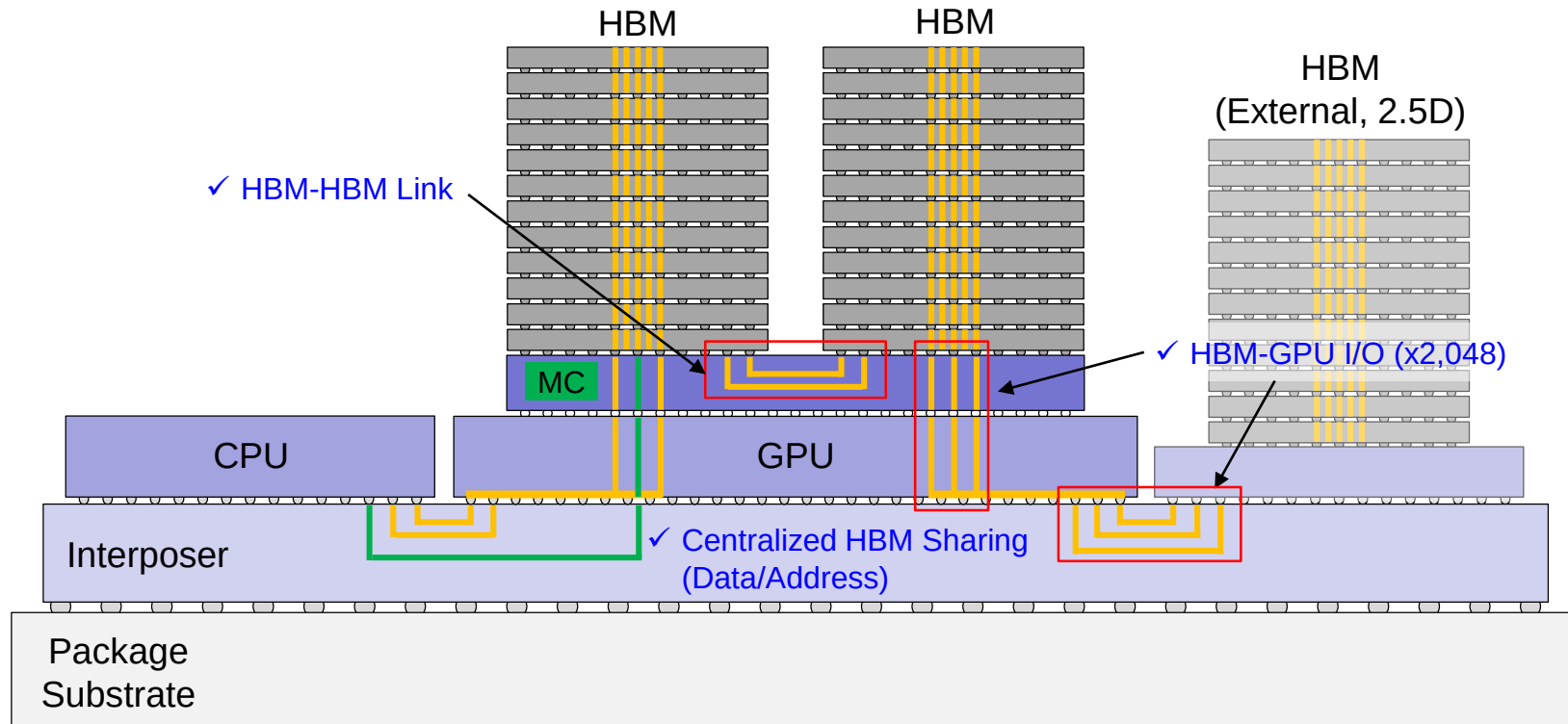


- ① Command decoder
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- ⑥ Save it to DIMM

→ With no CPU and GPU in the memory transfer path, delays are greatly reduced, and there are very few interconnection steps with a much shorter length.

HCC Architecture Roadmap (1)

High-Speed Interconnect Design for HCC Architecture

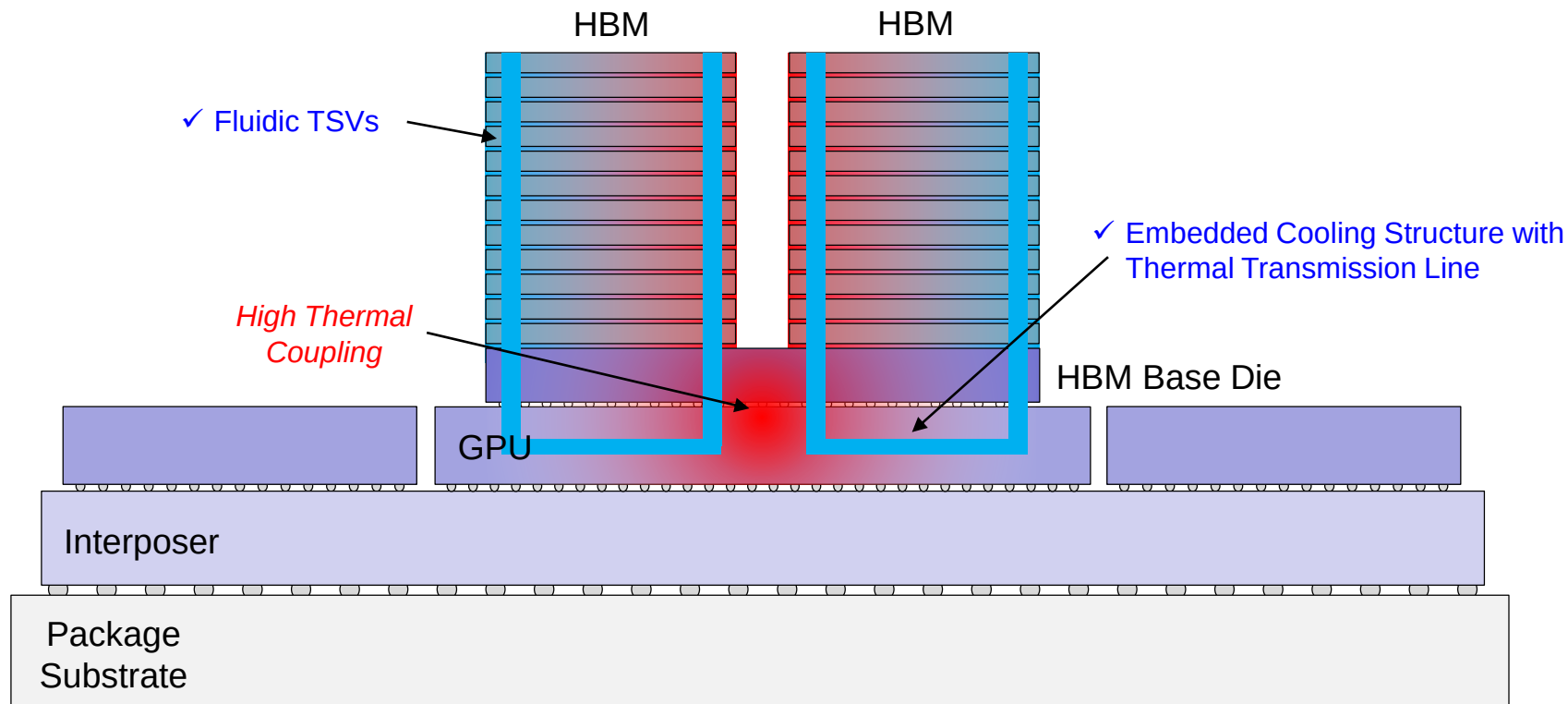


< High Speed Interconnect for HBM Centric Computing (HCC) Architecture >

- High-speed interconnect required for HBM Centric Computing(HCC) architecture
 - ✓ HBM-HBM : Inter-HBM Link for direct memory access (NVLink-like)
 - ✓ HBM-GPU : Wide I/O TSV providing high bandwidth (1,024 → 2,048)
 - ✓ CPU-GPU : Memory Coherent Interconnect (NVLink C2C, Infinity Fabric AP, CXL)

HCC Architecture Roadmap (3)

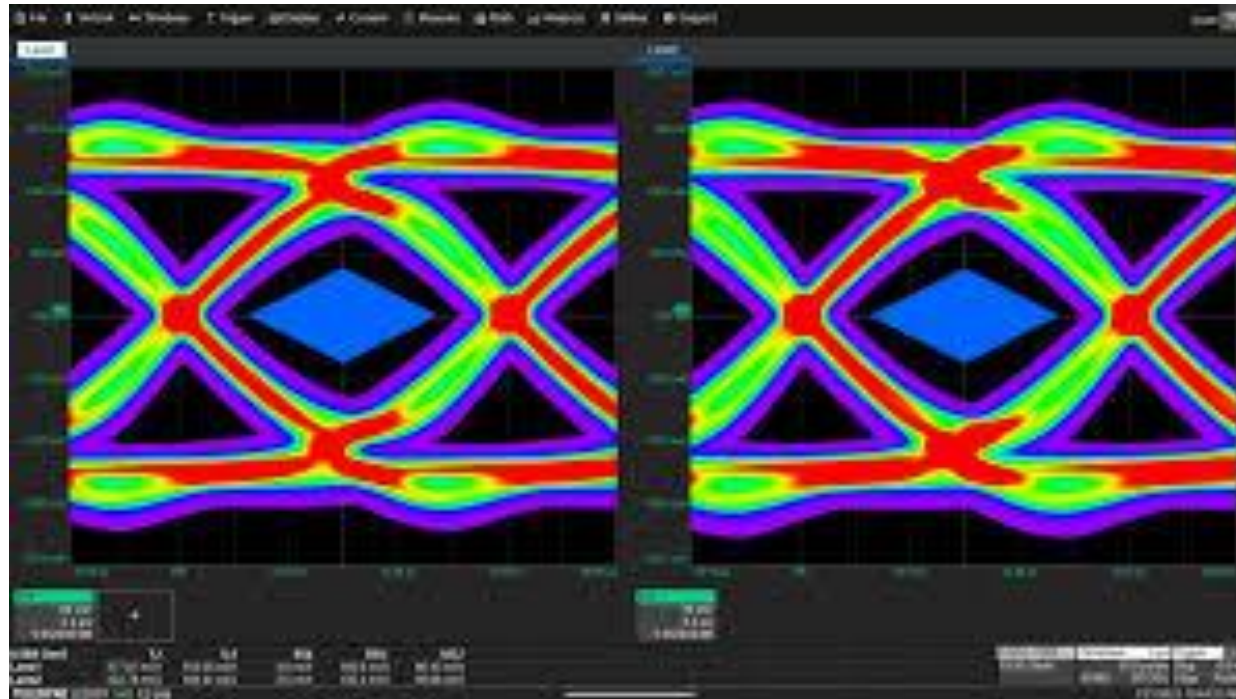
Thermal Integrity Analysis for HCC Architecture



< Embedded Cooling Structure for HBM Centric Computing (HCC) Architecture >

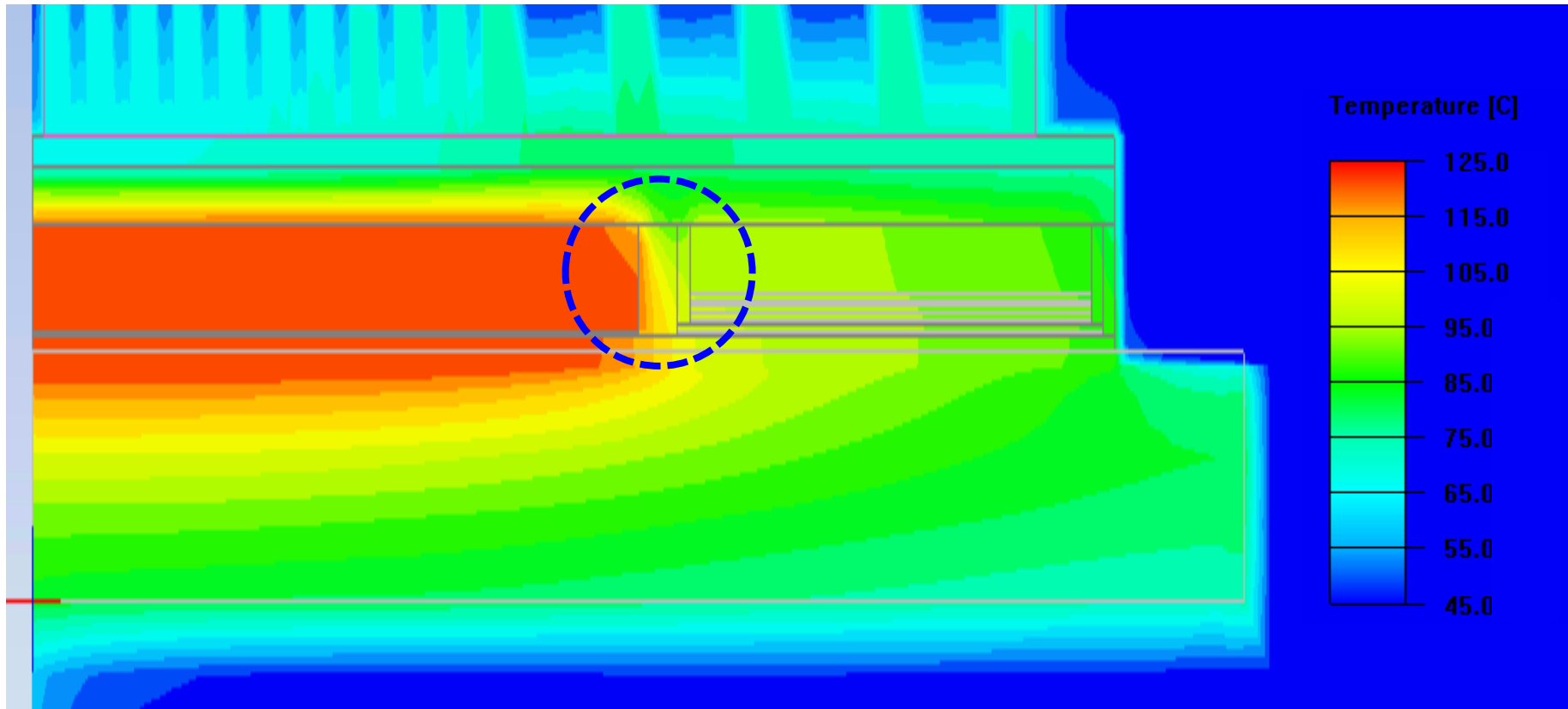
- To minimize thermal coupling within 3D HCC architecture, cooling structure design is required to maintain thermal integrity.
 - ✓ Embedded cooling structure with fluidic via structure & thermal transmission line
 - ✓ Fluidic & Immersion cooling techniques

Ultimate bandwidth limit



Deterministic Jitter
Random jitter
ISI jitter
PDN jitter
PSIJ

Thermal Coupling on Operating HBM System



<Thermal distributions of operating HBM system>

- GPU's heat thermal coupling to HBM through molding compound, heat sink, interposer and etc.
- GPU's heat is dominantly affected to HBM's thermal gradient.
- Only molding compound part can be customized for reducing thermal coupling effects.

Interconnection Future(10): Design Innovations: 5 Is

Design and Analysis for High-performance System

Co-Simul. Co-Design

System on Chip
(On-Chip Level)

System in Package
(On-Package Level)

PCB, Cable, Module
(System Level)



Signal **I**ntegrity
(SI)

Power
Integrity
(PI)

Thermal
Integrity
(TI)

5I
for High-
Performance
System

Electromagnetic
Interference
(EMI)

Artificial
Intelligence
(AI)



Improvement of System Integrity, Performance, Reliability, Cost, ...

Thank You!

HBM